FRONTEX

STUDY ON

Emerging Training Technologies and Concepts

for the European Border and Coast Guard (EBCG)

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Table of abbreviations

Abbreviation	Brief
AGI	Artificial General Intelligence
AR	Augmented Reality
ARES	Augmented REality Sandtable
CCC	Common Core Curriculum
DHS	Department of Homeland Security
EBCG	European Border and Coast Guard
EMIS	Education Management Information Systems
ERP	Enterprise Resource Planning
EU	European Union
FLETC	Federal Law Enforcement Training Centres
GDPR	General Data Protection Regulation
HR	Human Resources
ICT	Information and Communication Technology
INTERPOL	The International Criminal Police Organization
IoT	Internet of Things
L&D	Learning and Development
LMS	Learning Management System
LXP	Learning Experience Platform
ML	Machine Learning
MR	Mixed Reality
QR	Quick response
SCORM	Sharable Content Object Reference Model
SOLO	Structured Observation of Learning Outcomes
SSO	Single Sign-On
TO EIBM	Technical and Operational Strategy for European Integrated Border Management
UK	United Kingdom
US	United States
USBP	United States Border Patrol
VR	Virtual Reality
WSJ	Wall Street Journal
XR	Extended Reality

Executive summary

Background of the study

This study is situated within the evolving context of European border and coast guard management. Aligned with the Technical and Operational Strategy for European Integrated Border Management (TO EIBM), it emphasises the role of well-trained personnel and advanced technical tools in effective law enforcement. Supported by key policy frameworks such as the Digital Education Action Plan and the Osnabruck Declaration, the study aims to contribute to enhancing the existing capabilities for education and training within the European Border and Coast Guard (EBCG). By researching emerging training technologies, it seeks to empower the EBCG community to meet the increasing demands for efficiency and effectiveness through immersive and adaptive educational solutions.

Aim of the study

The objective is to explore emerging technologies and innovative educational concepts that have the potential to enhance training delivery within the EBCG community. This involves examining how these solutions can make training more adaptable, accessible and responsive to the dynamic challenges of border security. Additionally, the study seeks to identify gaps that such technologies can address and evaluate their potential benefits and drawbacks. Ultimately, the goal is to facilitate informed decision-making, driving strategic investments and improvements in training effectiveness within the EBCG community.

Study methodology

The research methods employed in this study include extensive desk research and stakeholder consultations. The **desk research** involved a systematic review of existing literature pertinent to emerging training and education technologies and concepts. This foundational groundwork provided a robust understanding of the current state-of-the-art practices and technologies in border and coast guard training and education management.

Stakeholder feedback was gathered through qualitative methods (several interviews and two workshops), and quantitative methods like questionnaires. More specifically, workshops were organised to ideate on the study's main use cases and on the concept of a technology-enabled academy. Furthermore, interviews were conducted with subject matter experts to understand the different emerging and innovative training and education concepts.

A questionnaire was used during the **Frontex Conference on Future Training Technologies for EBCG** to collect and identify common themes, trends and priorities among the EBCG community on emerging and innovative training and education concepts. During the conference, qualitative feedback was collected through interactions with stakeholders, including discussions with conference attendees and experts in the field. These interactions provided valuable insights into the needs, preferences and expectations of the community regarding training technologies, which were used to refine the study's findings. This qualitative and quantitative feedback helped in identifying common themes, trends, and priorities.

The outcomes of the desk research and stakeholder consultations were pivotal in shaping a comprehensive understanding of current emerging training and education technologies and concepts. These insights were also leveraged to develop a tailored taxonomy that categorises emerging technologies and training concepts and capabilities within the EBCG. Additionally, informed by desk research and stakeholder inputs, the concept of a technology-enabled EBCG academy was formulated, aiming to centralise and enhance training capabilities across various EBCG organisations.

Scope of the study

Research to identify successful initiatives/projects: The research demonstrates the successful implementation of advanced initiatives and projects around the world for training border and coast guard personnel at a wide range of organisations. More specifically, it presents trends, best practices and innovative approaches adopted globally to address the evolving needs of law enforcement education and training. Several organisations and implementations were studied. The United States Border Patrol deployed augmented and mixed reality solutions to train their workforce more effectively, and Network Rail in the UK adopted a learning management system technology to train their dispersed workforce. Interpol's global initiative incorporated metaverse technology, and Deloitte leveraged artificial intelligence (AI) and gamification. BSH in Germany integrated augmented reality and digital twin technology.

Analysis of current and emerging technologies in the EBCG space: The major technologies considered in this study include artificial intelligence, metaverse, gamification, 5G/6G, extended reality (augmented, virtual and mixed reality) and learning management system technology. These technologies were selected for their potential to significantly enhance training effectiveness

and operational efficiency within the EBCG. The research outcomes highlight the technologies' diverse potentials and challenges. LMS technology serves as a centralised platform for developing and delivering standardised training programmes, facilitating centralised resource management, structured learning content and adaptable learning experiences. It supports online training for dispersed personnel and ongoing compliance efforts, with integration opportunities spanning AI, XR, 5G/6G, gamification, microlearning, application programming interfaces (APIs), blockchain and data analytics.

Challenges include content accuracy, user experience optimisation, ecosystem compatibility, scalability, security, stakeholder alignment, resource management, technological readiness and ethical/legal considerations. 5G/6G technology enhances connectivity, IoT use, remote and immersive learning, personalised education and situational awareness, yet faces challenges including security, interoperability, rapid technological change, regulatory compliance, data privacy and ethical concerns. Augmented reality (AR), virtual reality (VR) and mixed reality (MR) technologies create interactive and immersive learning environments, enhancing engagement and flexibility. They integrate with 5G/6G for real-time training experiences, offering simulations for decision-making and teamwork in realistic scenarios, bridging theory and practice. Artificial intelligence automates personalised learning paths, improves information access and enhances decision-making through simulations, but requires addressing bias, legal compliance, user behaviour adaptation and ethical use. Gamification engages learners with game mechanics for goal clarity, collaboration, competition, problem-solving and decision-making improvements. Metaverse technology offers virtual shared spaces for collaborative, immersive training across domains, benefiting from AI, LMS, 5G/6G, XR and gamification integration.

Additionally, this study includes the design for an EBCG technology enabled academy, which is informed by industry standards, desk research, and discussions with the EBCG community at the Frontex Conference on Future Training Technologies. The architecture includes role-based access rights, interconnected building blocks, underlying infrastructure and the involvement of internal and external users. It offers a glimpse into what the academy could potentially become, creates a shared understanding, stimulates and structures discussions and supports the understanding of various use cases explored and detailed in the report. The architecture is designed to provide a cohesive learning experience for students and to bolster operational efficiency and strategic decision-making for administrative personnel, ultimately driving strategic growth initiatives within the academy.

High-level technology taxonomy for training and education: The study also includes a bespoke taxonomy for training and education in the EBCG community. This holistic framework links critical elements (access management, academia, industry partnerships), key management areas (course and curriculum, student and alumni relationships), and content to the highlighted use cases, showcasing how they are being implemented across the industry. The taxonomy enables a common understanding of training technology concepts, use cases and their relationship to one another across different stakeholder groups.

Proposal of relevant research topics: Based on the study's outcomes, several research avenues are proposed to inform the piloting, adoption and further research of innovative market solutions for the EBCG. The research topics are divided into two categories: innovative solutions, and underlying methodologies and frameworks. Each represents a diverse spectrum of research topics worth considering, as they are crucial to the successful integration and application of emerging training technologies within the EBCG community. The proposed topics focus on interoperability, standardisation, futureproofing, scalability, modernisation, clarity and security. These research topics aim to provide a flexible framework to guide the definition and implementation of tailored initiatives from 2023 to 2027.

Conclusions and considerations: This study underscores the growing importance of learning management system technology in EBCG training and education, facilitating collaborative and interactive learning through dynamic interaction, asset sharing, centralised data management and real-time feedback. AI integration offers promising avenues for boosting efficiency, personalising learning experiences, and enhancing decision-making, despite challenges related to high-risk applications and ensuring data privacy. Collaboration emerges as a cornerstone for advancing learning outcomes and embracing innovation within the EBCG community. Future research could focus on practical applications through proofs of concept while ensuring the scalability and interoperability of technology-enabled EBCG academies. Establishing a collaborative framework with inclusive governance could effectively guide strategic direction, resource allocation and adaptation to emerging technologies. Keeping abreast of regulatory developments as technologies evolve (such as those outlined in the AI Act)[1] could be valuable to effectively train EBCG organisations on regulations and ensure training and education compliance. Overall, this study – bolstered by desk research and stakeholder input – provides a robust foundation for informed decision-making to drive strategic investments and make training more effective.

[1] https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32024R1689

1. Introduction

1.1 Context and purpose of the study

This study is framed within the evolving context of border management, emphasising the need to equip border management officials with essential skills and competencies. **Situated within the broader framework of the Technical and Operational Strategy for European Integrated Border Management (TO EIBM), the study acknowledges the critical role of well-trained personnel and cutting-edge technical tools in ensuring effective law enforcement**. Aligned with key policy frameworks such as the Digital Education Action Plan and the Osnabruck Declaration on vocational education and training, the study represents a collaborative effort between the Frontex Research and Innovation Unit (INNOVATE) and expert stakeholders. It aims to steer the European Boarder and Coast Guard (EBCG) community towards greater agility, resilience and effectiveness by leveraging emerging training technologies and concepts.

This initiative seeks to empower border management officials to meet the evolving demands of 21st-century border security. Recognising the pivotal role of research and innovation, Frontex launched this study to explore emerging technologies and concepts relevant to border management training. The goal is to identify specific gaps or challenges that these technologies can address and to assess these technologies' advantages and drawbacks in doing so. With digitalisation and advances in ICT shaping learning and teaching methodologies, there is a notable shift towards immersive and adaptive education. The study underscores the importance of adopting advanced and sustainable methods and tools for training design and delivery within the EBCG education system.

The study explores emerging technologies and innovative educational concepts to achieve several goals. It seeks to examine how these solutions can enhance training delivery, making it more adaptable, accessible and responsive to the dynamic challenges of border security. Additionally, the study emphasises the need for a comprehensive educational approach that fosters continual learning and adaptability among border management officials.

Central to the study's objectives is facilitating informed decision-making. It acknowledges the importance of evaluating the integration of emerging technologies to ensure they yield tangible efficiency gains and align with the overarching goals of the EBCG. Through meticulous research and analysis, decision-makers will gain insights into the latest technological advances and their applicability within border management training, thus driving strategic investments and making training more effective.

1.2 Design of the study

The study takes a multifaceted approach to investigate emerging training technologies and concepts for EBCG training and education. It involved extensive **desk research:** systematically reviewing and analysing existing literature pertinent to emerging training technologies and concepts. This served as the foundation for a robust understanding of the current state-of-the-art practices and technologies in border management training.

Next, the study leveraged qualitative and quantitative feedback from a diverse array of EBCG stakeholders to enrich its insights. The stakeholders included border management officials, training institutions, industry experts and policymakers. **Qualitative methods** such as interviews and workshops were employed to solicit in-depth perspectives, experiences and recommendations regarding the integration of emerging technologies into EBCG training and education. These insights offer nuanced understandings of the challenges, needs and opportunities within the EBCG landscape.

Quantitative data collection methods such as questionnaires gathered structured feedback from stakeholders, providing valuable insights into prevailing use cases and priorities within the EBCG. The data complements the qualitative findings with empirical endorsements and nuances, grounding the study's conclusions empirically and practically to support informed decision-making.

Significantly, the study elaborates on the methodology in each chapter, providing a comprehensive account of how both the desk research and feedback-gathering processes were conducted, analysed and integrated. This includes a detailed breakdown of the data collection steps, the analytical techniques applied to interpret findings, and the synthesis methods employed to distil key insights.

1.3 Structure of the report

The report is structured to provide a comprehensive analysis of the selected key technologies and concepts and identifies areas for potential future research initiatives to inform the research, piloting and adoption of innovative market solutions for the EBCG training and education. In its last Chapter the report concludes with key takeaways.

- **Executive summary:** encapsulating the key points of the research.
- **Chapter 1 – Introduction:** setting the stage by elucidating the study purpose and design, followed by an outline of the report's structure to guide the reader through its contents.
- **Chapter 2 - Research and data collection to identify successful initiatives/projects:** including an introduction to the methods employed, a detailed account of the results and summaries of initiatives and projects from around the globe.
- **Chapter 3 - Analysis of current and emerging technologies in the EBCG space:** outlining the methodologies used, the results and the technology fiches elaborated for key technologies, solutions and concepts. A summary of key technologies is also provided, incorporating feedback from the conference along with high-level use cases.
- **Chapter 4 - High-level technology taxonomy for training and education:** detailing a taxonomy to categorise and organise the various technologies relevant to EBCG training and education.
- **Chapter 5 – Proposal of relevant research topics:** a practical starting point for potential future research avenues to inform the piloting, adoption and further research of innovative market solutions for the EBCG.
- **Chapter 6 – Conclusion and discussion:** a comprehensive conclusion synthesising the study's findings and implications.

The study is accompanied by a technical annex on existing taxonomies in the field of training and education (technology) and the set-up of the workshop for brainstorming on the main use cases of emerging training technologies for the EBCG community. There is a seamless progression from research to operationalisation, and onward to identifying future research directions within the study. The sub-chapters are interconnected as follows.

- **Research:** Chapters 2 and 3 involve a thorough examination of initiatives and projects worldwide, culminating in the presentation and promotion of main use cases identified for the EBCG community at the Conference on Future Training Technologies. This research serves as the foundation for subsequent chapters, providing insights into emerging trends and methodologies aimed at bolstering training effectiveness and readiness within law enforcement operations.
- **Operationalisation:** Chapters 2.3.1, 3.3.4 and 4 translate research insights into tangible strategies to harness the potential of emerging training technologies and concepts for the EBCG. This includes summarising key technologies, elaborating technology fiches for identified solutions, and brainstorming high-level use cases specifically tailored for the EBCG community. This section also proposes a technology-enabled academy to implement practical solutions and leverage innovative approaches to make training more effective and enhance operational efficiency.
- **Looking ahead:** Chapter 6 extends beyond operationalisation to suggest future research directions. These aim to further explore the research questions uncovered in the current study, ultimately expanding understanding and generating additional insights. This ensures a continuous cycle of inquiry and innovation, driving ongoing improvement and adaptation within the EBCG community. The following page illustrates the chapters' interplay.

STUDY CHAPTERS

A GUIDE TO READING THE STUDY

RESEARCH

The research component entails a comprehensive examination of the latest advancements in training technologies and innovative concepts specifically tailored to enhance the operational capabilities of the European Border and Coast Guard. This involves examining emerging trends, evaluating cutting-edge methodologies, and identifying promising tools aimed at bolstering training effectiveness and readiness within the context of law enforcement operations.

Technology fiches elaborated for the key technologies/ solutions/ concepts identified

Summary of the workshop for brainstorming on the main use cases

Stall at the Conference for Future Training Technologies to present and promote the main use cases identified for the EBCG community

Summary of the key technologies investigated incorporating the knowledge and feedback collected duting the Conference on Future Training Technologies

Scan of relevant initiatives and/or projects from around the world

High-level use cases identified for the EBCG community

OPERATIONALISATION

The operationalisation component includes the translation of research insights into tangible strategies aimed at harnessing the potential of emerging training technologies and concepts for the European Border and Coast Guard, to implement practical solutions and leverage innovative approaches to enhance training effectiveness and operational efficiency.

High-level groupings and sub-categories identified and listed

High-level technology taxonomy for training and education

LOOKING AHEAD

Looking ahead components includes future research directions, such as suggesting new studies to further explore the research questions uncovered in the current study, aiming to expand understanding and generate additional insights in the field of emerging training technologies and concepts for the European Border and Coast Guard.

Proposal of relevant research topics

2. Research and data collection to identify successful initiatives and projects

2.1 Introduction

In the dynamic landscape of law enforcement education and training, the integration of new and emerging technologies plays a pivotal role in driving innovation and enhancing operational effectiveness. This chapter explores successful initiatives and projects worldwide, highlighting the key findings. A comprehensive review **identifies trends, best practices and innovative approaches adopted globally to address the evolving needs of law enforcement education and training**. Additionally, this chapter provides insights gained at the Conference on Future Training Technologies, where the main use cases identified for the EBCG community were presented and promoted. It also summarises the post-conference data collection exercise. Elucidating the current landscape of initiatives and projects, the chapter serves as a foundation for further exploration and discussion on the potential applications and implications of emerging technologies for the EBCG community.

2.2 Methods

The research methods for scanning initiatives and projects were supported by comprehensive secondary desk research.[1] This allowed us to gather and analyse information, reports, case studies, academic papers and other literature related to the implementation of new technologies in law enforcement and defence education and training. Information from different sources was used to gather case studies, map them based on their relevance for the EBCG community and select the most relevant based on the client requirements and the analyst's subjective assessment.

From a shortlist of 13 case studies, five were chosen as the most relevant for this project, in consultation with the client and internal SMEs. To extract the most relevant use cases we leveraged publicly available sources (open web, vendor websites, blogs, whitepapers etc.), as well as third-party research tools and databases (Factiva, Gartner, EMIS, WSJ, Deloitte Marketplace intelligence reports etc.).

To identify and analyse successful initiatives and projects and introduce new training technologies across various types of organisations, the analysis followed a top-down approach. It is structured in four layers with different granularity levels.

- Border management ecosystem
- Global defence, security & justice sector
- Large public sector
- Private sector

[1] Secondary desk research refers to scanning data from publicly available sources only. The team did not conduct any primary research, such as interviews or surveys, to gain information around the case studies.

2.3 Results

2.3.1 Scan of relevant initiatives and/or projects from around the world

Case study 1: United States Border Patrol

Box 1. Key details of the case study at the United States Border Patrol
Development status: Implemented
Key stakeholders: US Border Patrol, Department of Homeland Security (DHS) Science and Technology Directorate, Federal Law Enforcement Training Centres (FLETC)
Technology: Augmented reality, mixed reality, 2D/3D virtual reality system, night vision
Solutions used: Sign Cutting and Tracking Solution, Augmented REality Sandtable (ARES), Microsoft Kinetic Sensor
Region: United States
Implementation year: 2019[2]
Sector: Border management ecosystem

Context: The United States Border Patrol (USBP) was looking to improve the training process for its agents, to help them maximise tracking performance using innovation and emerging technologies.

Challenge: The United States faces regular border crossing attempts by terrorists, smugglers, human traffickers and undocumented immigrants. Safely and efficiently identifying and tracking them is made more difficult by the complexity of the terrain.

Objective: The USBP wanted to update training processes for patrol agents by integrating innovative tools and emerging technologies, expanding the agents' skillset and capabilities to maximise detection and tracking performance. It also wanted to give them easy access to the experience of expert trackers, professionally trained in tracking/detection/apprehension of undocumented immigrants. As part of this initiative, the USBP operationalised and integrated this solution as part of its training programme.

Partners involved: DHS[3] Science and Technology Directorate (S&T) and Federal Law Enforcement Training Centres (FLETC).

Product details
Sign Cutting and Tracking Training – The system includes cutting-edge, commercially available augmented reality technology, night vision technology, a 90-minute computer-based training course and instructional movies. The content of the agent training course (recognising visual signals of human activity across varied topography, locations and times of day) was developed using 2D and 3D video. To improve situational awareness and the ability to identify and monitor the movement of people and groups along the border (sometimes referred to as 'sign cutting'), the system also uses night vision, augmented reality and virtual reality technologies.

[2] https://www.dhs.gov/science-and-technology/news/2019/11/05/snapshot-us-border-patrol-agents-leverage-emerging-st-tech

[3] Department of Homeland Security

Product details
Augmented REality Sandtable (ARES) – This low-cost, commercial, off-the-shelf solution provides an innovative, multi-modal interaction and visualisation.[4] With the ARES Tactical Planner app for Android mobile devices, users can add 2525C symbols and visuals to base maps created from several sources. They can use the Tactical Planner software with or without a real sand table. To deliver a user-defined common operating image at the point of need, the Army Research Lab is working to enhance battlespace visualisation. One component of this endeavour is ARES.
Microsoft Kinect Sensor – This 3D motion-sensing input device, detecting players and their environment, is integrated within ARES as part of the final solution to provide innovative visualisation options through a tangible user interface.[5]

Solution development: The USBP partnered with DHS S&T and FLETC to develop and implement an advanced technology solution using augmented and virtual reality, combined with input sensing devices for improving the tracking skills of border patrol agents. The Sign Cutting Tracking Training solution was originally developed by the U.S. Army (Army Research Laboratory). It was repurposed by S&T to implement virtual and augmented reality as part of their agent training and development.

"The new and improved Sign Cutting and Tracking Training solution, ARES, and the night vision technology are just a couple of examples of solutions being delivered by S&T to enhance national security and are resulting in measurable performance impacts." - Darren P. Wilson, Programme Manager, DHS S&T

Implementation: The USBP implemented the Sign Cutting and Tracking Training solution, integrated with ARES technology, to improve patrol agent training, mission planning/briefing, resource deployment, real-time decision-making and risk mitigation in field operations.

Uniqueness: Agents using the solution showed better knowledge retention and skill levels than agents using the traditional training method of 'tracking boxes.

- Improved visualisation: The solution gave agents a better visual understanding of the terrain, which helped during their training and exercises.
- Real-life simulation: Augmented reality allowed agents to experience on-field challenges in a simulated environment and learn from their mistakes without major adverse consequences.

Note: Implementation details on parameters such as deployment environment, type of user access, number of users, implementation phases and duration of implementation were not available in secondary sources.

Outcomes: Achievements for USBP with the Sign Cutting and Tracking Training solution:

- Virtual environment: Extending the ARES technology into a virtual reality environment allows staff to operate without interruption, enabling coordinated network coverage in remote areas where the physical sand table cannot be accessed.
- Integrated training environment: The solution offers multiple integration avenues, such as 2-D/3-D technology, augmented reality technology and motion sensors, to offer an interactive training environment.
- Improved operational efficiency: The solution helps agents respond significantly faster and more safely during operations, due to better situational awareness of the terrain.

Impact: Key benefits for USBP from the S&T ARES solution are stated below.

- Agent tracking proficiency improved by 63% compared to training by legacy methods.
- Increased situational awareness for agents due to better visualisation and identification capabilities. The solution allows agents to explore the various types of terrain using overlay maps through 3D hologram projection.
- An extended virtual environment allows patrol agents in remote locations to leverage terrain-mapping capabilities without the physical components of the solution.
- The networking capabilities between various solutions at multiple locations enable greater coordination between patrol agents across sites.

Status: USBP deployed the solution in 2019 at its sectoral headquarters in Yuma, Arizona, with two ARES sand tables and S&T solution implementation for agent training.

When combined with 3-D video capabilities and computer-based training course content, these technologies enable USBP and FLETC to maximise

[4] https://www.youtube.com/watch?v=DSstuycfJag&list=PLPGdJVRp7dNjfmrpqj_20meCRakLNgCbq&index=3

[5] https://apps.dtic.mil/sti/pdfs/ADA622471.pdf

the knowledge, skills, and abilities of agents and officers who are tasked with identifying and tracking activity along the border, enforcing immigration laws, countering drug and human trafficking, pursuing fugitives and conducting search and rescue operations.

Darren P. Wilson,
Programme Manager,
DHS S&T

Future potential: The agency is exploring the application of this solution in other areas. Some options are:

- DHS S&T, along with FLETC, is planning to deploy the solution across its training centres in Glynco/Brunswick, Georgia, and Artesia, New Mexico.
- The capability of this innovative solution to integrate multiple commercially available technologies provides flexibility to adapt it to similar scenarios outside border patrol. DHS S&T is exploring the applicability of the solution across drug trafficking, capturing fugitives, and search-and-rescue operations.

Relevance for EBCG community: Key takeaways

- Applicability of extended reality (XR) solutions: These technologies are exceptionally well-suited to training in the EBCG sector, offering significant advantages. The immersive nature of virtual reality, augmented reality and mixed reality devices enhances the effectiveness of training, providing a dynamic and interactive learning experience for individuals undergoing preparation for diverse tasks and responsibilities. It also leads to significant cost benefits by eliminating the need for physical equipment, the mobilisation of employees at a single place, and associated travel and administrative costs.
- Improved engagement/learning: Agents training with the Sign Cutting and Tracking Training solution show better engagement levels and visualisation capabilities compared to legacy training environments. This offers a good reference point for the EBCG community to understand the relevance and applicability of its border management processes and explore phased adoption within its training model. This is expected to provide flexibility to organisations deploying young recruits in field operations using such solutions to improve tracking of undocumented migrants across the border areas.
- Cost savings: The seamless integration of commercially available technologies enabled substantial cost savings for USBP. This is a key focus area for EBCG when considering large-scale implementation, making it sustainable in the long run and justifying its investments. Ensuring the availability of equipment across multiple training centres would help to set up an effective training delivery model.
- Increased efficiency: The solution can be used to train multiple employees simultaneously, leading to higher practical efficiency without needing physical trainers or seminar sessions, and reducing costs.

Case study 2: Network Rail

Box 2. Key details of the case study at Network Rail
Development status: Implemented
Key stakeholders: Titus, Moodle
Technology: Learning management solution
Solutions used: Moodle LMS
Region: UK
Implementation year: 2021[6]
Sector: Public sector

Context: The learning and development (L&D) team at Network Rail is responsible for driving the training ecosystem across its various entities. These include approximately 800 partner agencies and sponsor organisations at different levels of the supply chain that support Network Rail and its sub-entities in their operations, maintenance and upgrading. Network Rail wanted to optimise the monitoring and tracking of users across its different learning environments.

Challenge: Building an integrated e-learning platform for learning and development across the organisation. The L&D team wanted to distribute its content across the organisation in a more time-efficient way. In addition, Network Rail had no way to identify the number of users on its platform, track their progress or shift users between the separate virtual environments. The training team had to obtain data across multiple unconnected learning environments. And any shift of users from one environment to another led to the loss of their learning records. The organisation also depended on multiple in-house systems to manage its workforce, onboard new hires and manage employees leaving the organisation.

Objective: Network Rail wanted to implement a digital solution to bring together users and content across its various virtual environments. This would offer an optimised, integrated learning solution using advanced technology. It also wanted to explore commercialisation opportunities through a digital training platform for external stakeholders. This would recover some of the cost of creating eLearning products, which was indirectly borne by taxpayers.

Partners: Titus, Moodle

Product details
Moodle[7] is an open-source learning management system (LMS) offering an easily scalable platform for enterprises and integration with third-party applications**.**
The solution provides significant flexibility for enterprise customers, making it a popular choice across workplaces, schools and universities.
Moodle LMS implementation requires the organisation to have dedicated ICT support or high technical expertise.
Network Rail leveraged the services of Titus, a Moodle implementation partner. Its Moodle migration[8] solution enables the client to create, customise, and integrate the Moodle learning management system within their learning platform.

Solution development: Network Rail chose Titus as its technology partner to develop the LMS as a customised, integrated platform with centralised control. To ensure easy scalability options, they worked together to build

6 https://moodle.com/news/network-rail-moodle-workplace/
7 https://moodle.com/solutions/lms/
8 https://www.tituslearning.com/moodle-migration-services/

a custom server within the cloud service. The solution used a multi-tenancy model: the client could create a sub-platform for each sub-entity, with centralised control from a single installation. This allowed Network Rail to create a sub-platform for internal staff, in addition to those created for sub-entities. It made content deployment easier, and synchronised updates across all platforms.

Implementation: The Moodle solution allows clients to mirror their organisational structure within the LMS platform, improving visibility for course administrators and providing better real-time visibility into learners' progress. The solution includes built-in reporting tools that ensure an efficient learning journey and has multiple integration options with third-party enterprise software solutions. The deployment environment was Cloud, and the number of users was approximately 100,000.

Uniqueness: Unique attributes include:

- The LMS-enabled solution offers course administrators significant flexibility in managing multiple learning initiatives, enabling efficient content distribution across the organisation. It mirrors the structure of the implementing organisation, giving team leads and administrators a clear view on employees' learning journey and guiding them by analysing progress and performance.
- The interactive dashboard also provides clear visibility into the number of users accessing the system and the courses being followed, which allows the organisation to maintain regulatory compliance and set up career pathways for each user.

Outcomes: Achievements for Network Rail using Moodle LMS platform

- Streamlined digital learning solution - Reporting tools allow data-driven decision-making on potential uses of the platform.
- Integration with third-party applications - Integration with Network Rail's ERP software automatically updates staff movement from the organisation data set within the LMS platform solution.
- Multi-tenancy workplace - A learning management system enables Network Rail to create sub-platforms for each organisation in their supply chain, all centrally managed from a single installation.

Impact: Key benefits for Network Rail from the LMS solution

- Streamlined content management and distribution across sub-entities within the organisation, creating operational efficiencies for both L&D team and users.
- Multi-tenancy structure for course administrators allows the L&D team to create multiple sub-platforms for each entity or partner within its supply chain network while managing it centrally.
- Real-time visibility for trainers/course administrators into learners' course progress, enrolment and completion status.
- Diverse learning paths, aligned to career progression goals, offer employees flexibility to drive their career growth and seek out relevant content from the digital platform.
- Aligned auto-enrolment for learners enables the organisation to achieve its learning goals while ensuring all users comply with the organisation's regulatory requirements.

Status: The Moodle LMS deployed by Network Rail is currently used by over 100,000 people across its organisation, which highlights the wide reach of the solution and its scalability. The organisation is exploring options to monetise some components of the platform as a long-term opportunity.

Future potential: Network Rail plans to explore integration with advanced and emerging technologies and wants to optimise the solution to include the delegation of administrative tasks.

Relevance for the EBCG community:

- Multi-tenancy functionality: A multi-tenancy structure within the LMS platform can be based on an ecosystem. This enables efficient control over organisation learning goals and processes, as well as the building of futureproof structures. EBCG organisations can inherit from centralised processes or content, but also push towards other content/training/information.
- Integration options: The LMS platform can be integrated with third-party enterprise software solutions (such as ERP) and with extended reality training (accessed via the LMS solution). The centralised platform can gather all user and system data, including training data and other information from third-party applications. It offers the EBCG community a window to explore commercialisation opportunities.
- Reporting dashboard: Along with the solution's reporting functionality, the platform includes a custom reporting tool from its technology partner. This can be combined with artificial intelligence (AI) to develop learning content, activate it, and more.

Case study 3: Interpol

Box 3. Key details of the case study at Interpol
Development status: Implemented
Key stakeholders: Microsoft, Meta, Mastercard, Animoca Brands, Decentraland
Technology: Metaverse
Solutions used: NA
Region: Global
Implementation year: 2022[9]
Sector: Global defence, security & justice

Context: The International Criminal Police Organisation (Interpol) wanted to explore the potential of the metaverse and build awareness around it across the organisation. It also wanted to develop employees' skills to oversee future cases concerning the metaverse virtual environment.

Challenge: Exploring the metaverse to understand, evaluate, and prepare for potential criminal activity within the virtual environment. Participation in metaverse virtual environments is growing. Yet Interpol, as a law enforcement agency, is not fully aware of the distinctions between criminal activity in the real world and the virtual world. The agency lacks the expertise to address potential criminal activity within metaverse virtual environments.

Objective: Interpol wanted to understand potential criminal activity within the metaverse, and the applicability of criminal law there. To do so, it wanted to build employees' expertise around activities in the metaverse, along with improving collaboration and learning across its global locations.

Partners: Microsoft, Meta, Mastercard, Animoca Brands, Decentraland

Products used: Information on the open web is lacking about the specific products used by Interpol for the metaverse digital academy.

Solution development: Interpol leveraged metaverse technology to develop a virtual digital academy for training its agents across various locations in separate countries. Boosting collaboration and the exchange of ideas, it enables team members dispersed across the globe to connect, enhance skills in such areas as virtual border management, and help one another solve cases. Interpol has presented its digital academy at global conferences of security agencies, highlighting another way of spreading awareness about innovative technology implementation within the industry. In addition to offering an environment for team members to collaborate, the platform has huge future potential for integrating other technologies and tools, such as gamified modules, LMS, and others.

Implementation: Exposing Interpol agents to the metaverse academy helped the organisation get a better idea of risks in that virtual environment, the regulatory frameworks required for improved governance, and the potential for criminal activities. It also enables agents to train in virtual scenarios, such as virtual border screening in an airport environment; simulated forensic investigations; collecting and preserving crime-scene evidence; and monitoring digital criminal activities. It is thus a highly efficient way to prepare agents before deployment.

Note: Implementation details on parameters such as deployment environment, type of user access, number of users, implementation phases and duration of implementation were not available in secondary sources.

Uniqueness: A metaverse-based learning and development environment for employees across the world enables Interpol to leverage senior staff members' experience and knowledge remotely. It also boosts collaboration, allowing employees to share knowledge and tap into their peer network in a time-efficient manner. The relatively lower maturity level of law enforcement organisations in terms of digitalisation makes the collaboration aspect significant. To fight against cyber-criminality in virtual worlds, these organisations must collaborate with peers and/or competing organisations. They also need to develop metaverse skills, otherwise it is not possible to understand this virtual environment. Additionally, the metaverse is an effective tool for law enforcement that is gaining credibility due to its holistic approach to multi-stakeholder engagements and cross-border collaboration, covering multiple jurisdictions, dimensions and organisations.

[9] https://www.interpol.int/en/News-and-Events/News/2022/INTERPOL-launches-first-global-police-Metaverse

Achievements: Major accomplishments for Interpol through the metaverse platform:

- Inter-organisation collaboration: It has enabled interaction between agents across various Interpol offices, significantly improving collaboration and allowing agents to tap into the expertise of their peers.
- Future readiness: Interpol agents can proactively identify risk in the metaverse and work with stakeholders to formulate policy or set up governance frameworks.
- Remote training: The metaverse platform gives Interpol significant flexibility in conducting agent training across locations, resulting in significant cost savings and operational efficiency.

Impact: Key benefits for Interpol from the metaverse solution:

- Improved collaboration between agents using the digital academy platform for learning, allowing them to leverage the expertise and skills of colleagues across the globe in solving cases without the need to be physically present.
- More efficient training of new hires on topics such as document verification and passenger screening, increasing efficiency in the field.
- Virtual 3D avatars can be customised, which makes exploring the metaverse digital academy more interesting and interactive.
- Self-paced training modules allow agents to learn at their own speed, leading to higher interest levels and course completion rates.
- Remote learning allows agents to access the virtual academy from any location, facilitating learning as well as collaboration with peers across locations worldwide.

Status: The metaverse digital academy was deployed by Interpol in 2022. There is not enough data on how the organisation leveraged the solution, what benefits it achieved or its plans for the next steps.

Future potential: Interpol plans to work with government and industry stakeholders on metaverse security.[10] Utilising the metaverse for agents' learning and development helped Interpol understand the nuances of virtual environments and the implications for crime. Interpol established an expert panel focused on ensuring that innovative virtual environments follow a 'secure by design' approach during their development stage.

Relevance for EBCG community: Key takeaways

- Real-time knowledge sharing: The metaverse's capacity for real-time knowledge transfer, fostering collaboration, and enhancing user interactivity has the potential to greatly support the EBCG community in cultivating a modern learning culture, especially at a time when individuals must navigate constant shifts in digital information.
- Proactive approach: Interpol decided to deploy a metaverse solution to better understand virtual worlds and specific environments and improve collaboration for its users in these spaces. This highlights the significance of using agile methodology in such scenarios, rather than waiting for the development of the perfect solution.
- Integration capabilities: The metaverse learning environment offers multiple integration options. The EBCG community can mandate a centralised, technology enabled academy to integrate with this type of technology and enable learners to access this training from a central point.

[10] https://www.interpol.int/content/download/20828/file/Metaverse%20-%20a%20law%20enforcement%20perspective.pdf

Case study 4: Deloitte

Box 4. Key details of the case study at Deloitte
Development status: Implemented with ongoing expansion/development
Key stakeholders: Deloitte UK, Deloitte Belgium, Totara
Technology: Enterprise LMS, gamification, digital analytics, artificial intelligence, microlearning
Solutions used: Totara Learn (eEnterprise LMS platform)
Region: UK
Implementation year: 2022
Sector: Private sector

Context: Deloitte UK wanted to implement a digital onboarding solution and capability development journey for its new pool of employees: experts brought in to address global transformation using a multidisciplinary approach, aligned with input from multiple experts. The firm's wide array of service offerings and scale of operations posed a series of challenges. An enterprise LMS-powered digital academy enabled the firm to enhance employees' onboarding experience and improve user compliance and provided flexibility to undertake digital transformations through major disruptions.

Challenge: Leveraging technology within the learning and development process to address the growing complexity and dynamic nature of business. It was crucial to familiarise employees with the overall story and approach and ensure alignment with various concepts and topics to improve workforce collaboration. The firm decided to provide this critical onboarding initiative through a virtual academy with a highly interactive communication programme, along with a customised learning pathway. The digital academy used gamification and interactive modules to offer a unique learning experience.

Objective: Develop a digital onboarding and learning pathway solution integrating diverse skills and expertise on a common platform. The firm aimed to deliver a captivating onboarding experience for the Major Programmes division, providing learning through concise, bite-sized content structures. It consolidated its diverse expertise and offerings portfolio, creating an academy-focused programme offering a learning journey using emerging technologies such as gamification, and customised learning analytics. In Phase 2 of the programme, the firm will implement an artificial intelligence (AI) layer in the proof-of-concept stage to direct capability development for users in a more streamlined manner.

> *The exam question I set the Belgian team was how can you help me with an L&D solution that is scalable, educational, and engaging? It needed to land the knowledge needed across a very diverse range of people in my business from all different industries. It also needed to use innovation and cutting-edge technology.*
>
> Richard Sedgwick,
> Chief Transformation Architect,
> Major Programmes at Deloitte

Partners involved: Deloitte UK, Deloitte Belgium, Totara

Products used: Product details are listed in the table below.

Product details
Totara LMS technology includes advanced features for course creation, learner tracking, assessments and reporting. The platform supports various learning styles and ensures that organisations can tailor their training programs to meet specific needs, whether for compliance, skill development or staff onboarding. A valuable feature is that learners can engage with peers and instructors through forums, discussions and collaborative projects. The platform offers customisable playlists, enabling learners to curate and organise content that aligns with their personal learning paths and goals. This flexibility ensures that users can access information when they need it, promoting a more personalised learning journey.
The implementing organisations can set performance goals, track progress, and provide feedback within the same system, creating a cohesive and efficient approach to workforce development.

Solution development: Create a hybrid learning journey to help employees build capabilities efficiently. Deloitte UK collaborated with its Belgian counterpart to conduct a learning needs analysis, ensuring alignment with the learner audience and promoting engagement across different course levels. The development team utilised user feedback to create easily digestible content in formats such as word clouds, podcasts, quizzes, reflection logs and videos. The platform provided flexibility in content and design, facilitated automated interactions with learners and administrators, generated analytics for monitoring performance and demonstrated scalability. The academy will be deployed across other Deloitte firms worldwide.

Implementation: Implemented through the enterprise LMS solution, 'Major Programmes' is a digital academy that hosts a programme consisting of five online modules. It is complemented by a one-day in-person workshop leading to certification. Users can track their progress through reflection logs, engage with self-paced online modules, and receive regular updates via automated emails. Additionally, the solution incorporates a personal action plan to apply the knowledge acquired in real-world scenarios. For deeper engagement, learners are prompted to maintain private reflection logs during courses to record thoughts, ideas and responses to questions. With one click, learners can revisit their answers throughout the learning journey. Automated emails, tailored to progress status, help learners stay on track with mandatory activities. Alongside online learning, participants attend in-person workshops at the end of each journey segment to apply knowledge and exchange ideas with peers. At the end of onboarding, learners create a personal action plan, committing to three goals: short-term (30 days), medium-term (60 days), and long-term (90 days). Automated reminders track progress and reinforce goal commitment.

Implementation details: The solution was deployed in a cloud environment to ensure scalability, flexibility, and accessibility. Users accessed the system through Single Sign-On (SSO) access, streamlining the login process and enhancing security measures. With over 1,000 users expected to use the platform, robust infrastructure and user management protocols were paramount. The implementation was divided into two phases. The first was successfully completed, and Phase 2 is currently in preparation. This approach allows for efficient deployment and gradual integration of features, ensuring a smooth transition and optimal functionality.

Uniqueness: Unique attributes include:

- A thorough analysis at the start of the learning journey helps analyse learners' needs, helping the team curate learning models.
- Bite-sized content delivery allows learners to consume their learning with ease, reducing time commitment and enhancing effectiveness.
- Maximised flexibility allows users to learn whenever and wherever they want.
- Personalised learning modules, reflection logs and progress emails drive deeper engagement and higher compliance rates.
- The hybrid learning model allows users to leverage both online mentorship and in-person workshops.
- A gamified learning module leads to higher user engagement with their learning journey, an interactive learning environment, and better knowledge retention.

Outcomes: Achievements for Deloitte UK through the enterprise LMS solution include:

- High scalability: The solution is easy to implement and provides multiple customisation options, including extension across global locations in one LMS platform.
- Automated interaction: It generates and uses data based on performance monitoring to offer insights for users and administrators, improving the experience for both.
- Mobile-friendly: Continuing the onboarding journey via mobile devices extends the experience and provides users with flexibility to continue their learning regardless of location.

Impact: Key benefits from the solution include the following.

- Growing scalability: A high approval rating from users highlighted the platform's simple-to-use structure and effective impact on the learning journey. Within the first 12 months of implementation, over 500 practitioners across Deloitte UK had completed the programme, marking over 85% in satisfaction ratings. It generated an additional 250 registrations for 2024. The programme is being expanded to the Netherlands, Belgium, the US and the Middle East and could be deployed to other teams/divisions across Deloitte member firms. The implementation time will depend on any further development.
- Enhanced flexibility and utilisation: Flexible, self-paced training modules empower learners to manage their time. The learning journey includes interactive e-learning modules to gauge comprehension. Choosing concise content allows learners to better progress at their own pace.
- Increased visibility: Enhanced visibility for course administrators, supported by a strong communication plan to learners, streamlines operational processes. Instructors and administrators can effectively monitor and guide learners by leveraging insights from learning needs analysis, personal action plans, progress status and other inputs.

- Learning platform reporting dashboard: This is the basic tool that allows course administrators to understand and have real-time visibility on the learning journey and progress levels, through achievement activities and logs.
- Effective analysis: Learning analytics is achieved through a combination of survey data, learning platform data and HR data. Analysed at the individual level, it enables the personalisation of the learning journey based on user data and showcases clear differences in the learning patterns of individuals and/or groups.

Status: The initiative received broad approval across the organisation, with over 1,000 practitioners completing it within the first year and reporting an impressive 85% satisfaction rating. An extra 250 participants enrolled to finalise their learning journeys by summer 2024. Testimonials from both customers and managers have attested to the programme's substantial impact, resulting in the creation of numerous new sales opportunities since inception.

Future potential: The favourable feedback from users and the solution's role in generating new sales opportunities have inspired Deloitte UK to explore extending its implementation to member firms in other countries, including the Netherlands, Belgium, the US, and the Middle East. The successful launch shows how major enterprises can build new business cultures, specifically on emerging topics with the potential for large-scale impact.

Relevance for EBCG community: Key takeaways

- Potential of programme enhancement: Using the solution's analytics feature, further interpretation of the data and feedback collected from participants could lead to significant enhancements and development. There is also scope for exploring artificial intelligence (AI) integration within the solution to make it more robust and enrich the user experience. Safeguards for privacy and the protection of fundamental rights need to be in place from the very beginning of the process when considering AI integration.
- High flexibility: This solution showcases the ability of LMS tools to be flexible and futureproof by using open-source software, along with offering a great user experience through a highly visual interface.
- Scalability potential: This study showcases the significant upscaling possibility of the solution to quickly extend to an enterprise level, leading to benefits for employees across remote locations.
- Focus on bite-sized content: Due to the short, easy-to-consume content form, users reported higher engagement during their learning journey.
- Integrated gamification: The LMS platform offers a customisation option to provide learners with gamified modules, as per the client enterprise requirement.
- Assessment and certification module: The platform provides enterprises with features that enable instant grading, showcase learner results, run online tests and generate certificates upon course completion.
- Changing user behaviour: The platform can use the technology to alter behaviour, upskill and reskill people via personalised learning pathways and change user mindsets around sensitive topics.

Case study 5: BSH

Box 5.Key details of the case study at BSH
Development status: Implemented
Key stakeholders: BSH Hausgeräte GmbH,[11] BSH Startup Kitchen,[12] Staufen Akademie,[13] rooom AG[14]
Technology: Augmented reality, virtual reality, artificial intelligence, digital twin
Solutions used: rooomSpaces, rooom360
Region: Germany
Implementation year: 2023
Sector: Private sector

Context: BSH Hausgeräte GmbH, a major player in the home appliances industry, wanted to explore ways to create accessible online training for its global workforce, minimise travel costs for in-person sessions and address engagement issues in online courses. BSH aimed to establish an in-house digital learning platform for employees, ensuring easy access without requiring additional device installations. The case study highlights how BSH developed and implemented the platform by using XR, artificial intelligence (AI) and digital twin technologies.

Challenge: Although online formats offer convenience, they come with drawbacks like limited interaction, insufficient personal exchange and increased distraction, all of which dilute their effectiveness. Despite advancements in e-learning tools, the absence of social aspects negatively impacts learner motivation, emphasising the need for in-house training and education for employees.

Objective: To promote virtual upskilling and bridge the significant communication and collaboration gaps among global teams, BSH recognised the need for a unique approach to digital training, emphasising gamified and interactive elements. The firm's goal was to create a scalable, cloud-based in-house learning platform with secure data handling. To engage younger, digitally savvy employees, BSH aimed to reinvent smart learning using innovative technologies like augmented and virtual reality. The project prioritised individualised virtual training sessions, allowing global reach and minimising travel expenses.

Partners involved: Staufen Akademie, rooom AG.

Products utilised: Product details are listed in the table below.

Product details
rooomSpaces is an interactive platform for creating engaging virtual spaces. It enables the development of digital twins for physical locations like campuses, with features such as customisable avatars, expandable whiteboards and seamless learning content integration. Ideal for dynamic virtual rooms, it supports multiple learning modules and interactive discussions
rooom360 offers a suite of tools designed for virtual rooms, providing a comprehensive and interactive digital environment. These tools include interactive whiteboards, 3D models for product visualisation, augmented reality features for detailed examinations, on-demand learning paths, **artificial intelligence (AI)** -driven tutorials, and smart learning environments for collaborative workshops
BSH Virtual Campus is an innovative digital learning platform designed to enhance training experiences. It features a Digital Twin of the physical BSH Campus Traunreut, allowing users to create personal avatars and explore different 3D models of products, view individual objects in augmented reality, access on-demand learning paths, and engage in collaborative live workshops set in various virtual environments

[11] BSH Hausgeräte GmbH: Technology Implementor

[12] BSH Startup Kitchen: Technology Consultant (corporate venture division of BSH Hausgeräte GmbH)

[13] Staufen Akademie: Technology Consultant (Digital training course developer)

[14] rooom AG: Software developer (Virtual platform) – platform provider for virtual rooms

Solution development: In developing an internal learning platform that would be scalable and cloud-based while guaranteeing secure management of sensitive data, BSH aimed to elevate user experiences through technologies such as augmented reality and virtual reality. The project team asked trainers to design their virtual training modules individually to minimise in-person travel expenses. Requirements included swift data transfers and access to the learning platform on a range of devices without installations. To leverage gamification and other interactive elements for an enhanced digital learning experience, the firm, in collaboration with BSH Startup Kitchen, identified rooom as a leading provider of smart learning platforms. Offering high-quality depictions of virtual environments and seamless data integration, rooom's advanced visual implementation techniques were used to generate superior 3D models and augmented reality objects in a smart-learning collaboration environment.

BSH relied on the expertise of Staufen Akademie to develop the content for digital training courses using artificial intelligence (AI), and then created the course module. This was aimed at helping employees and managers learn to apply theoretical concepts in real-world scenarios. Various experts in lean management and higher education helped the project team prepare the learning content. The team used the rooomSpace platform to develop the Digital Twin of BSH Campus Traunret, with personalised avatar options for learners. rooom360 allowed the team to create a learning path with multiple stations within the 360-degree factory tour. The stations featured interactive elements offering learning content on value-stream workflows at the BSH factory in Traunreut. Users could engage with an expandable online whiteboard for notetaking and recording key insights in real time. Each station also included questions to reinforce earlier learning. The rooom system made it easy to bring in content from current courses, improving access and keeping the learning process smooth. The e-learning platform thus brought together 3D models, augmented reality objects and optional virtual reality headset integration, to provide a unique experience for learners.

Implementation: Learning content was curated in collaboration with education experts to make courses easily available to BSH employees across the globe. The metaverse platform on which the solution was developed is highly ranked by users, due to the high-quality visual environment. The team developed the 'Value Stream Organisation' e-learning journey for around 400 global managers responsible for organisational transformation, ensuring consistent principles and methods across various locations using virtual support. The team also implemented innovative training via virtual stream mapping, with virtual factory tours to document information, material flows, and processes. The data was further visualised on a value stream map, facilitating the application of theoretical knowledge in practical scenarios for managers and employees. The metaverse platform allowed easier integration of existing data within the solution.

Uniqueness: The BSH Virtual Campus platform gives users access to various enhanced learning methods.

- Differentiated 3D models of products with explanatory annotations for better understanding
- Augmented reality for viewing individual objects
- On-demand learning paths covering a range of subject matter
- Learning videos and tutorials developed using AI for an enhanced learning experience
- Smart learning environments to facilitate collaborative live workshops in various virtual settings
- Optional access to virtual reality headsets

The platform allows learners to experience a 360-degree tour of the BSH museum within the Digital Twin, allowing them to explore historical home appliances in 3D and see product history. It also provides virtual tours of various production halls in Traunreut, offering insights into processes and workflows.

Note: Implementation details on parameters such as deployment environment, type of user access, number of users, implementation phases and duration of implementation were not available in secondary sources.

Achievements: The initiative enabled BSH to accomplish the following objectives:

- Enhanced virtual learning experience: The platform introduced an innovative tool for users to leverage virtual learning in a new and effective manner. It also provides them with a wide range of learning possibilities, facilitating both synchronous and asynchronous learning, individually or as a group.
- Long-term cost reduction: In-house virtual training will allow the firm to significantly reduce long-term costs. These include infrastructure costs, such as classrooms, libraries, and laboratories; maintenance and operational costs; and commute costs for in-person sessions.
- Extended accessibility and reach: The firm has considerably expanded its reach by making training available across the globe at the user's convenience. The 360-degree tours and 3D spaces, museum, factory, and campus in Traunreut are easy for employees to access. Managers from around the world can engage with each other through avatars in virtual lounges on their 'Smart Learning Island'.

- Smart learning interfaces: Standards such as SCORM[15] and xAPI[16] help ensure compatibility between learning platforms. Modules with these standards allow easy transfer between platforms, reducing costs through content reuse. Single sign-on authentication across multiple systems facilitates data integration. The flexible rooom platform supports diverse interfaces compatible with various systems, allowing smoother interoperability among learning management systems.

Impact: The development of the platform brought the following benefits:

- Increased interest in e-learning: Personalised interactions, immersion and the integration of AR, VR and AI have increased interest in learning about the 'Value Stream Organisation' e-learning journey. Further impact came from improved access to training courses for global employees, with increased avenues to engage with peers from other locations.
- Added flexibility in learning options: Employees decide for themselves when and where they attend training, adding flexibility and convenience. Courses can be held online or hybrid (partly digital, partly in-person). This choice further encourages adoption of the tool.
- Less global travel has considerably lowered the cost of workforce learning and development requirements.

"The virtual campus is a great opportunity for BSH and modern, cutting-edge learning for the future."

Kai Sponer,
responsible for the BSH
Global Learning Campus

Status: BSH has implemented the solution across its organisation, with a special focus on middle to senior managers. The team is working with colleagues to integrate additional learning content and production sites into the virtual campus.

Enhancement and scalability: BSH will widen the reach of the digital platform to increase engagement between employees in its Germany and India offices, while making its courses accessible to employees across the globe.

- BSH aims to establish the platform as a fundamental tool within the corporation to promote a culture of learning and enhance staff efficiency.
- The team plans livestream directly from the BSH campus in Traunreut to share learning content internationally and make the course content available on demand.
- The firm is exploring the potential of avatars in combination with AI to further enhance the platform.

Relevance for EBCG community: Key takeaways

- Centralised platform: This enables significant interoperability, seamlessly integrating various technologies and systems to provide users with a diverse array of learning content, video, audio, text, blended formats and more. The platform empowers users with the flexibility to engage in self-paced learning from any location, free from the constraints of physical seminars and time restrictions. Additionally, it serves as a hub for gathering learning data, including certificates, pass/grades and compliance tracking, which can be combined with human resources (HR) data and seamlessly integrated with other technologies. Such a digital-enabled 'academy' functions as the backbone for all training administration, covering aspects such as subscriptions, certifications, recertification and training materials. Centralisation not only brings standardisation opportunities but can also accommodate Member States' preferences for personalised learning. Generic learning paths meet the needs of border management personnel EU-wide, covering a wide range of topics from interviews and passport checks, through detecting suspicious individuals, to operational skills like patrolling. Additionally, Member States' learning administrators can augment generic courses with tailored content to meet their national requirements. Authorised individuals gain access to specialised training, facilitating rapid adaptation to new contexts and benefitting both national personnel and international officials, who can transition between roles or countries with a simple account adjustment.
- Metaverse environment: The use of the metaverse to create a virtual campus offers an innovative learning environment leading to increased collaboration between learners across different business units, and higher interest in the course.
- Using third-party expertise: rooomSpaces showcases the benefits for EBCG, such as significant user customisation, system interoperability and third-party integration options across vendors, which enables platform connectivity. The plug-and-play digital

[15] SCORM (Sharable Content Object Reference Model) principles revolve around creating a standard for e-learning content interoperability. It ensures that learning content adheres to a set of guidelines, allowing it to be easily shared and reused across different learning management systems. SCORM-compliant content follows specific packaging and sequencing standards, enabling seamless integration, tracking and reporting of learners' progress.

[16] xAPI, or Experience API, principles are centered on enhancing the tracking and analysis of learning experiences in a flexible and interoperable manner. Unlike traditional e-learning standards, xAPI allows the collection of a wide range of learning activities and experiences, including offline and non-traditional interactions.

platform shows that for the user, a single place to access learning is a best practice.
- Cost savings: A metaverse-based solution can save costs by eliminating the travel associated with traditional, in-person training.
- Increased efficiency: Self-paced learning saves significant amounts of employees' time.

2.3.2 Stall at the Conference on Future Training Technologies to present and promote the main use cases identified for the EBCG community

The objectives of the stall were twofold: to raise awareness about the high-level use cases and technology fiches being promoted for the EBCG community, and to collect feedback. To achieve these, the stall was designed in line with very specific principles. Brand consistency was paramount, with prominent incorporation of Frontex corporate identity elements and alignment with the visual identity of the conference. Visual quality was prioritised, with meticulous attention to detail in the design of elements to create an appealing and well-organised display.

Interactive elements were integrated into the stall to actively engage attendees, including messaging to convey key information concisely and interactive zones designed to facilitate discussions with participants. The role of facilitators was crucial: managing the stall during the conference, ensuring continuous engagement with visitors, collecting audience feedback, distributing technology fiches and providing information about the objectives of the study.

Key design features included curved walls, which elegantly framed the stand to prominently display Frontex branding, the definition of the tech-enabled academy, statements on assumptions about the tech-enabled academy and the study's definition and objectives. Dual vertical screens strategically placed on the walls showcased dynamic dioramas highlighting three use cases and presenting detailed information on six technology fiches. The screens were large enough to attract participants' attention and encourage them to stop by.

Additionally, QR codes were strategically placed on the outer limit of the booth to enable participants to engage with specific questions, such as readiness assessments for certain technologies. A designated discussion space with standing tables and a comfortable seating area provided an ideal environment for discussions with participants and short interviews. Overall, the conference stall served as an interactive platform to promote key use cases and technology fiches, while actively engaging with attendees to gather feedback and insights.

3. Analysis of current and emerging training and education technologies relevant to the EBCG domain

3.1 Introduction

Expanding on the preceding chapter, which explored global trends, best practices and innovative methodologies in education and training around the world, this chapter directs its attention to technologies. **Here, the focus lies on the analysis of current and emerging training and education technologies relevant to the EBCG domain.** A comprehensive examination will discern their present applications and future potential within the specific context of the EBCG. This analysis will help identify pathways for enhancing vocational training and education, addressing the evolving needs and complexities encountered by law enforcement personnel. The goal is to facilitate the integration of state-of-the-art technologies that can drive innovation and efficacy in border security training and education.

The chapter outlines the methods employed to assess current and emerging training and education technologies within the EBCG domain. From data collection to analysis frameworks, the methodology provides a robust foundation for understanding this landscape. Subsequently, the chapter presents the results of this analysis, encompassing a summary of key technologies investigated, technology fiches elaborated for identified solutions, and insights from workshops brainstorming main use cases. This is followed by high-level use cases identified for the EBCG community that offer actionable insights into the practical implementation of emerging technologies in border management training. Additionally, the chapter explores the concept of academy architecture, laying the groundwork for designing innovative educational frameworks tailored to the needs of border management personnel.

3.2 Methods

The research approach involved comprehensive secondary desk research, which allowed us to gather and analyse reports, case studies, academic papers and other relevant literature related to the implementation of new technologies in law enforcement and defence education and training. Various sources were used to gather data, map them based on their relevance for the EBCG community and include the most relevant pointers. We identified experts in the field of emerging technologies and conducted multiple rounds of interviews (in-person and virtual) to explore their expertise, experiences and recommendations for enhancing training and education capabilities for border management.

The analysis followed a top-down approach structured into four distinct layers, with different levels of granularity, to comprehensively identify and analyse successful initiatives and projects introducing new technologies for workforce training.

- Border management ecosystem
- Global law enforcement, defence & justice
- Public sector
- Private sector

Leveraging the desk research, as well as a workshop between the different research parties, this section conceptualises a technology-enabled EBCG academy.

3.3 Results

3.3.1 *Summary of the key technologies investigated, incorporating the knowledge and feedback collected during the Conference on Future Training Technologies*

This section highlights various projects and initiatives by organisations incorporating new and emerging technologies into their training, learning and development processes. It presents information about initiatives and projects (case studies) worldwide, along with preliminary conclusions regarding their relevance to EBCG. Importantly, the report highlights the transformational potential of each technology for workforce education and training through sustained innovation and digitalisation processes.

As part of this section, technology fiches have been developed for each of the key technologies: Metaverse, AR/VR, Gamification, Learning Management System, Artificial Intelligence. Each technology fiche is a two-page document summarising the key technology, its use-cases and relevance for EBCG. These fiches were printed and shared during the Frontex Conference on Future Training Technologies to share key research outcomes, spark conversations and collect feedback on training technology concepts and use-cases. An example of a technology fiche is included below.





Technology 1: Artificial intelligence (AI)

AI for training and education, powering various learning solutions.

Training technology overview

Artificial Intelligence (AI) has transformed various industries in recent years, fundamentally changing the way businesses and organisations operate. From revolutionising healthcare and finance to optimising manufacturing processes, AI has ushered in a new era of innovation and efficiency, reshaping entire industries and setting new standards for productivity, decision-making, and technological capabilities.

AI has roots in the early 20th century. The term 'robot' originates in a 1920 Czech novel[1] and in 1950, the famous Turing test[2] was first proposed to the public. The term 'artificial intelligence' was coined by Stanford Professor John McCarthy in 1955;[3] he defined it as the science and engineering of creating smart machines. Over the years, AI has evolved from machines following pre-set instructions to systems that can learn from data or experience, a bit like how we learn. Some focus on specific tasks, while others with general intelligence aim for human-like capacities. The idea is not just about machines doing

1 R.U.R. and the Vision of Artificial Life (2004) https://doi.org/10.7551/mitpress/14136.001.0001

2 Turing, A. (1950). I.—COMPUTING MACHINERY AND INTELLIGENCE. Mind, LIX (236), 433–460. https://doi.org/10.1093/mind/lix.236.433

3 https://hai.stanford.edu/sites/default/files/2023-03/AI-Key-Terms-Glossary-Definition.pdf

things on their own, but about creating technology that helps humans and meets the needs of society.

Integrating AI within learning and development initiatives has the potential to drive significant advances, enhancing efficiency and improving learner connectivity and engagement across border management organisations.

AI has developed significantly in recent years, catapulting it to the centre of the world's attention. With the breakout and rapid advance of generative AI, about 40%[4] of senior executives expect their organisations to invest more in the technology. Around 20%[5] are already using AI, while a majority are either undertaking pilot projects or exploring the use of these tools.[6]

AI is already enabling the creation of personalised content, providing efficient content distribution and management, and acting as a conversational partner to quickly find relevant answers and aid learning professionals in making decisions based on data. The process of translating the power of AI into the realm of learning is an essential strategic endeavour and given today's trends it seems inevitable.

Main benefits

Integrating artificial intelligence into EBCG learning and development can offer a range of benefits, such as introducing innovative training approaches, improving efficiency and enabling adaptability across the workforce.

Automate personalised learning paths: AI could facilitate personalised learning paths and adaptive modules, tailoring training content to individual needs and optimising the learning experience. It helps in connecting training resources across different entities and platforms, along with encouraging global collaboration.

Efficient access to information for learners: AI-driven analytics provide in-depth quantitative insights into staff study performance, allowing learners to make informed decisions about their training and learning pathway.

Reduced load on learning administrators: The automation of curriculum creation and curation, as well as advanced insights into staff performance, reduce the burden on training administrators, allowing them to focus on higher-level strategic aspects of workforce development.

Realistic simulations and scenario-based training: These enhance decision-making skills for border and coast guard operations. The use of AI would support the alignment of skill-driven workforce training with broader organisational goals.

Increased decision-making capacity: Access to a wide range of data sets and analytics makes the decision-making process quicker and more efficient. Insights on skills allow organisations to be more strategic with workforce planning and development, and to optimise training.

Automated content creation for training: AI-driven tools accelerate training-material creation, automating processes faster than traditional methods. Algorithms enhance relevance, recommending supplementary materials based on progress and preferences, fostering quicker, deeper staff understanding.

Improved language inclusivity: Cross-border organisations can break down language barriers by using AI-powered tools. These include real-time translation of training materials, interacting in multiple languages through chatbots/virtual assistants and customising user interfaces with the learner's language.

These benefits are expected to enable the strategic realignment of the organisation's operations and processes, keeping them current in the digital age. The integration of AI technologies should not be looked at as an exercise in adopting the latest digital solutions. Instead, it should be a major component to help EBCG organisations become efficient, future-ready and globally connected.

Implementation insights

This section explores best practices to help EBCG organisations deploy emerging technologies within their learning and development processes, and the impacts on workforce training and learning requirements.

4 https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai-in-2023-generative-ais-breakout-year

5 https://www.gartner.com/en/newsroom/press-releases/2023-07-05-gartner-survey-finds-79-percent-of-corporate-strategists-see-ai-and-analytics-as-critical-to-their-success-over-the-next-two-years

6 About 51% of executive strategists exploring the use of AI are focused on machine learning, while 45% are looking at predictive analytics.

Training content personalisation: artificial intelligence (AI) plays a pivotal role in streamlining and expediting content development for training, offering transformative benefits. Automated content creation tools harness AI-driven algorithms, revolutionising traditional methods by extracting key insights from diverse sources like reports, publications and videos. These algorithms not only automate processes faster, but also enhance the training material's depth and relevance. By sifting through vast amounts of data, AI-driven tools recommend supplementary materials based on individual progress and preferences, fostering quicker and deeper staff understanding.

Use of AI to generate video content allows for content creation, distribution and updates on an automated cycle based on pre-specified instructions. This eliminates the need to repeatedly seek stakeholder approval and leads to operational efficiencies. It also allows the instructional design team to focus on understanding learners' preferences and customising content, accordingly, eliminating the manual effort of content curation across formats. AI-driven language assistance optimises the learning landscape by offering automated, one-click translation of training resources, including video and audio elements. This innovative approach ensures multilingual accessibility and a seamless, real-time, user-centric experience. Accommodating diverse linguistic preferences and offering dynamic interactions, the adaptable platform fosters inclusivity and global collaboration, a transformative advance.

Automated personalised learning: artificial intelligence (AI) is expected to drive the modernisation of training across the European Border and Coast Guard community. Today, Frontex and the Member States work in a decentralised manner, using separate learning resources; there is a recognised need to synchronise these resources, possibly by means of a technology-enabled academy. AI can play a pivotal role in this process by analysing various learning resources, including courses and content created by different community members, while considering aspects such as translation.

AI analysis can identify strengths and weaknesses within these learning resources, aiding in the optimisation of the learning environment. This optimisation does not extend to creating personalised curriculums, as this approach is not currently under consideration due to associated risks. Instead, AI technologies can streamline the management of individual learning paths. One key area where AI can contribute is identifying mandatory courses necessary for achieving qualifications or skill levels. This approach ensures that employees are guided through a structured learning journey tailored to their needs. Leveraging AI technologies within the EBCG community facilitates the synchronisation of learning resources and the optimisation of individual learning paths, ultimately making the learning ecosystem more effective without delving into the complexities of personalised curriculum creation.

For border management, where precision is paramount and to-and-fro between physical and virtual/digital training is required, the integration of AI-driven competence learning is expected to show remarkable results. Border management personnel would benefit from targeted training that addresses their specific areas of improvement, fostering a more skilled and adaptive workforce. This could not only enhance overall team competence, but also contribute to a more agile response to dynamic border security challenges. The use of AI-driven competence learning could reduce onboarding time by analysing historical and performance data to identify effective training courses or sequences. This could lead to greater efficiency and higher proficiency levels. The use of AI to personalise language training can improve communication skills, positively impacting interactions with users and enhancing overall operational efficiency. In a cross-border management task force scenario, AI-driven tactical skills training can enhance skill development by assessing performance in tactical exercises and providing tailored recommendations for improvement. This would foster better team cohesion and coordination during real-world operations.

Efficient access to information for learners: AI-driven analytics provide in-depth quantitative insights into learner performance. This includes tracking progress, providing learning action items and recommending upcoming courses transparently and responsibly that comply with the AI Act. With AI-driven analytics integrated into a learning and development framework, users experience a transformative shift in how they interact with information. Unlike traditional methods, an AI system ensures efficient access to information while prioritising user data protection and transparency. Users see a comprehensive and nuanced overview of their progress, learning action items and upcoming courses, allowing them to gauge their performance with precision and empowering them to make informed decisions about their training and learning pathway based on personalised recommendations. A key differentiator of AI solutions lies in their ability to adhere to the stringent data and governance requirements of the AI Act[30]

Learning analytics for an efficient digital learning platform: Artificial intelligence is a transformative force in education, prioritising learning analytics over traditional business intelligence. Predictive analytics anticipates group performance trends, allowing for personalised interventions at a collective level. Adaptive assessments dynamically tailor evaluations, providing real-time feedback for a customised learning environment and continuous improvement, scrutinising group performance metrics, knowledge gaps and skill deficiencies. Educators gain actionable insights to refine curriculums and optimise teaching methodologies, ensuring a dynamic and effective educational experience. It is important to note that the application of AI in individual performance assessment is carried out in strict adherence to the AI Act, with the necessary safeguards for high-risk AI applications, protecting individual rights and ensuring transparency.

Dynamic learning management systems (LMS): Incorporating AI into a dynamic LMS yields a multifaceted approach to elevate the entire learning experience. An advanced LMS seamlessly integrated with AI capabilities becomes the cornerstone of transformative education, which could offer innovative solutions such as regional modification of content based on users' learning preferences. The real-time potential of AI is to enable users in their learning journey, supporting them in their workflow (improving collaboration, making daily operational activities efficient, etc.)

The content authoring included in the LMS is either built in or an external tool, either powered by or using AI. The LMS's increased decision-making capacity is complemented by accelerated and automated content creation for training. AI-driven tools expedite the development of materials, ensuring that content remains engaging, informative and compliant with evolving regulation.

Personalised learning paths and adaptive modules are made possible, ensuring the adequacy of the individual training map. The LMS is the platform where multiple technologies come into play, to create efficient access to information which is streamlined and allows learners to navigate a huge pool of resources efficiently. This not only fosters engagement but also enhances comprehension, providing recommendations for the content and its visual structure, as well as the learning path.

The AI-powered LMS goes beyond benefiting learners: it significantly reduces the load on learning administrators. Learners benefit from AI-driven analytics beyond basic tracking, offering nuanced and real-time progress data, a departure from traditional static insights. The personalised nature of AI analytics provides actionable information tailored to learners' needs, offering recommendations for optimal training pathways based on individual performance. With AI, learners access a data-driven dashboard, empowering them with real-time insights for informed decision-making, ensuring a more dynamic and responsive learning experience compared to traditional methods. An AI-powered LMS supports learners through features such as multi-lingual ecosystems, providing a user-friendly learning experience.

Moreover, the real-time translation features break down language barriers and foster a more inclusive learning environment. The integration of AI enables continuous improvement and customisation of training content, ensuring it evolves in tandem with educational needs.

Future-proof architecture ensures the LMS can adapt to the ever-changing landscape of educational technology, including the seamless integration of new AI features. The AI-driven LMS thus becomes not just a tool but a catalyst for educational transformation, catering to diverse learning needs, enhancing administrative efficiency and aligning with the future of education.

Box 6. Case studies on the adoption of AI technology in education

The virtual campus developed by BSH[7] leveraged artificial intelligence (AI) in the development of learning videos and tutorials for users, significantly improving the efficiency of overall project implementation and showcasing an innovative method of using AI within learning systems.

CrossEngage, a German company,[8] deployed an AI-enabled LMS to automate feedback collection from the training process, enabling the organisation to improve user experience, optimise operational efficiency and reduce user effort during a learning path.

The Workforce Transformation division of Deloitte Belgium has used AI for various learning-related purposes, including content creation, updating and tagging, as well as optimising the work of learning administrators. The team mapped training resources onto skills and skill levels, developing an engine for analysing skills and content gaps. Additionally, they used learning analytics to create personalised learning journeys for employees.

Integration opportunities

This section explores the opportunities to integrate artificial intelligence, so that organisations can harness its full capabilities to enhance efficiency and drive innovation.

Initial steps to integrate AI within the EBCG's digital academy involve assessing the current learning ecosystem and defining clear objectives. Stakeholder engagement is essential to ensure buy-in and alignment across departments. Evaluation of AI technologies is crucial, followed by the establishment of robust data infrastructure and governance frameworks to support AI integration securely. Building internal capabilities through training programmes and iterative implementation of AI solutions helps drive success. Effective change management and communication strategies ensure smooth adoption, while continuous improvement and optimisation foster a culture of innovation and adaptation. This collaborative strategy enhances learning experiences, ensuring a comprehensive and effective approach for diverse industries and organisations.

Metaverse integration: Metaverse technologies can be integrated with artificial intelligence (AI) to create immersive training environments. AI-driven simulations within the metaverse can replicate real-world scenarios, giving border security personnel realistic training experiences and enhancing decision-making skills and operational readiness.

Augmented reality (AR) and virtual reality (VR) integration: AI can enhance augmented reality and virtual reality training modules by dynamically adapting scenarios based on user interactions. AI algorithms can personalise simulations, track performance and provide real-time feedback, creating highly realistic and tailored training experiences for border security personnel.

Mixed reality (MR) integration: The integration of AI with MR enables seamless blending of physical and digital elements in training. AI algorithms can enhance the contextual relevance of digital overlays in the real-world environment during mixed reality training sessions, providing a more immersive and effective learning experience.

Integration with LMS: Artificial intelligence can transform traditional LMS deployment by offering personalised learning pathways, adaptive modules and real-time analytics. AI-driven LMS can categorise, tag and manage content automatically, reducing administrative burdens and providing a more efficient and customised training experience.[9]

Integrating with 5G/6G connectivity: The high-speed and low-latency capabilities of 5G and future 6G networks facilitate seamless connectivity for AI-powered applications. This ensures real-time data exchange, enabling border security personnel to access AI-driven training resources, simulations and collaborative tools with minimal latency for enhanced learning experiences.

Gamification as an integrated module: Artificial intelligence (AI) can be integrated into gamified training platforms to adapt to challenges based on individual progress and learning styles. By analysing performance data, AI algorithms can dynamically adjust game elements, providing engaging and targeted training experiences that align with the specific needs of border security personnel.[10]

7 https://www.rooom.com/blog/employee-development-in-the-metaverse

8 https://www.zavvy.io/case-studies/crossengage

9 An example of AI-enabled learning management system solution is the Zavvy learning management system (https://www.zavvy.io/learning-management-system)

10 An example from College of Policing (UK) (https://www.capita.com/our-work/using-gamification-upskill-employees-improve-learner-retention-and-combat-crime)

Data analytics and predictive modelling: Integrating AI with advanced data analytics and predictive modelling enhances the understanding of trends and potential threats. AI algorithms can analyse vast datasets to identify patterns, allowing organisations to proactively address emerging challenges and tailor training programmes and hiring to ensure they have access to the right skills.

Integration of conversational artificial intelligence for improved interactivity: Incorporating conversational AI into training modules fosters interactive learning. AI-driven chatbots or virtual assistants can respond to queries, provide instant feedback and offer guidance, creating a more engaging and learner-centric training environment.

By leveraging AI-enabled learning solutions, organisations can streamline learning administration, align content with skill gaps and boost learners' motivation. Automatic content tagging facilitated by AI ensures a personalised learning experience. Additionally, administrators benefit from AI tools such as virtual assistants, enhancing execution speed and optimising processes. The overall result is a learning organisation that is not only more efficient and personalised but also future-proofed, ready to adapt to evolving educational landscapes and technological advancements. The strategic integration of AI becomes a cornerstone for sustained relevance and effectiveness in organisational learning and development.

Main challenges

Integration and implementation of AI in the training and education of EBCG personnel will require various gaps and challenges to be addressed.

Addressing the artificial intelligence bias: The success of an AI model relies on the quality, accuracy and relevance of the training data. Inaccurate or biased data can lead to skewed models, affecting their ability to make accurate predictions or decisions. It is imperative to ensure biases are checked to avoid any operational and/or ethical risks. EBCG organisations must align their security policies, incorporate ample datasets for model training and prioritise diversity to prevent biases and ensure model generalisability.

Ensuring regulatory compliance: The dynamic EU technology regulatory framework poses challenges for the EBCG community. Rapid changes in regulations related to data privacy, security and ethical AI use may require continuous adaptation of AI-driven training protocols. Ensuring compliance with regulations addressing bias, transparency, accountability and data protection becomes a complex task. In addition, ensuring human supervision will be compulsory in the case of high-risk AI systems. Striking a balance between innovation and compliance is essential for the adoption of AI in training.

Evaluating learning and development teams: To implement AI in workforce learning, organisations must train their learning and development teams in emerging technologies, ensuring they have the skills to understand and use AI effectively. They also need a clear understanding of AI's impact on learning outcomes. By empowering their teams and investing in training, organisations can optimise AI integration in the digital academy, enhancing the learning experience and achieving better outcomes for employees.

AI framework: Developing a monitoring and evaluation framework is a key component of leveraging AI within workforce learning and development. The aim is to balance automation with human-centric learning.

Impact of user behaviour and consumption pattern: Rapid changes in user behaviour and consumption patterns pose challenges in anticipating evolving preferences and adapting AI-driven training solutions to them. The dynamic nature of user engagement may necessitate continuous adjustments to AI algorithms and platforms to align with the needs of EBCG personnel's ethical and cultural challenges.

Rapid development of emerging technologies: The fast development of new digital solutions and technologies introduces challenges in ensuring compatibility and integration with existing AI-driven training systems. The EBCG community may face the need for frequent updates and modifications to keep up with emerging tools, leading to potential disruptions in the learning process. Implementation cycles are longer in public organisations due to higher procedural requirements, necessitating an agile approach to make sure this technology is deployed effectively.

High dependency on technologies: Overreliance on AI technologies for training may create dependency issues. Technical glitches, system failures or cybersecurity vulnerabilities could temporarily disrupt training programmes. Striking the right balance between leveraging AI and maintaining alternative methods is crucial to mitigate the risks associated with technology dependency.

Impact on commercial adoption: Regulatory uncertainties and evolving standards can impact the adoption of AI-driven training solutions. The EBCG community may face challenges in making long-term investments and commitments due to the evolving regulatory landscape. Clear and consistent rules and regulations are crucial to fostering confidence in AI adoption.

Ethical considerations: Ethical considerations in deploying AI for training, like algorithmic bias and fairness, pose challenges. Impact assessments on privacy and fundamental rights are crucial. This ensures compliance with regulatory limitations for an EBCG tech-enabled academy. Continuous vigilance is needed to prevent AI algorithms from perpetuating bias or discrimination.

Data quality, security and sovereignty: Concerns related to data quality, security and sovereignty, especially in cross-border collaborations, can impact the deployment of AI technologies. Striking a balance between leveraging global AI capabilities and ensuring control over sensitive data becomes a complex challenge for organisations operating in the border security domain.

Box 7. Insights on the ethical and legal implications of adopting AI technology

General Data Protection Regulation (GDPR) and European digital rights: In the EU, the GDPR sets stringent standards for data protection and privacy. The EBCG community, operating within the EU, must adhere to GDPR principles when implementing AI for training. This includes ensuring transparent data processing, obtaining explicit consent and providing individuals with control over their data. Additionally, considerations related to European Digital Rights and the Principles Declaration underscore the importance of respecting individuals' rights and autonomy in the digital realm.

Ethical guidelines for educators: Educators within the EBCG community must adhere to ethical guidelines concerning AI and data use in teaching and learning. This involves transparent communication with learners about the use of AI, ensuring fairness in assessments, and addressing potential biases in algorithms. Ethical considerations extend to the responsible handling of learner data, emphasising confidentiality and the right to privacy.

Health and safety issues: The use of AI in training may involve immersive technologies, such as virtual reality simulations. Ethical considerations extend to health and safety issues, ensuring that training environments are designed to minimise any potential physical or psychological harm. Regular risk assessments, user feedback mechanisms, and adherence to industry safety standards are critical components of ethical AI implementation.

Data protection and fundamental rights: The deployment of AI in training necessitates a thorough assessment of data protection and fundamental rights implications. It is crucial to strike a balance between leveraging AI for effective training and safeguarding fundamental rights, such as the right to privacy and non-discrimination. Transparency in data processing, clear consent mechanisms, and robust security measures help uphold these rights.

Assessment of data use in teaching and learning: Ethical guidelines emphasise the importance of fair and transparent assessments in AI-driven learning environments. Educators should critically assess the data used in teaching and learning to ensure that algorithms do not perpetuate biases or discriminate against individuals or groups. Regular reviews of assessment methodologies and outcomes contribute to ethical AI adoption.

Principle of accountability: The principle of accountability is central to ethical AI implementation. The EBCG community must establish clear lines of responsibility for AI-driven training programmes, ensuring accountability for algorithmic decisions and their impact on individuals. This includes mechanisms for redress in the event of errors or unintended consequences.

Human-centric approach: Adopting a human-centric approach is foundational to navigating the ethical and legal landscape of AI in training. This involves prioritising human values, respecting human rights and actively involving individuals in decision-making processes related to the use of AI in their training and development.

The ethical and legal implications of AI in workforce training for the EBCG community demand a comprehensive and proactive approach. By aligning with GDPR and the AI Act, and prioritising human-centric values, EBCG can foster a culture of responsible AI use that respects individual rights, upholds ethical standards, and ensures the health and safety of personnel undergoing AI-enabled training. Continuous ethical evaluation and adaptation are crucial in navigating the evolving landscape of AI ethics and regulations.

Outcome/future potential

Currently at about 15%, in the next two years an estimated 50%[11] of strategic planning and execution within

[11] https://www2.deloitte.com/ch/en/pages/press-releases/articles/ai-study-almost-half-of-all-employees-are-worried-about-losing-their-jobs.html

organisations is expected to be automated using artificial intelligence tools. The creation of so-called artificial general intelligence (AGI), able to consider any manual or intellectual task holistically and respond accordingly, is expected to significantly alter the way businesses operate and function. The anticipation is that AI will revolutionise current job roles and tasks, enhancing human abilities. This progression implies that job descriptions will undergo changes, and the workforce will be required to learn new skills, upgrade current ones, redefine job parameters and collaborate with AI as the industry continues to adapt.

By analysing vast datasets in near real-time, AI will empower personnel to make better-informed decisions, making them more efficient and effective. Additionally, AI's role extends beyond decision support to the creation of dynamic learning environments. Personalised and adaptive training modules will continuously evolve based on individual performance, ensuring that border security personnel are equipped with the most relevant skills and knowledge. Emphasising learning as part of the workflow, micro-learning aspects will be highlighted, fostering synergy between learning materials and operational situations. This entails the seamless integration of bite-sized learning modules within the work context, enabling prompt access to information and its direct application to tasks. Such dynamic interplay ensures that AI-driven technology makes learning experiences more effective, facilitating continuous and contextualised learning processes aligned with the demands of the dynamic work environment.

Moreover, AI is expected to seamlessly integrate with Internet of Things (IoT) devices, enhancing situational awareness by collecting and analysing data from sensors, cameras and other connected devices along the border. This integration will facilitate the creation of more comprehensive training scenarios. Looking ahead, the potential collaboration between AI and unmanned aerial systems holds promise for improving coordination and response strategies through the simulation and analysis of interactions between security personnel and autonomous devices. Furthermore, the evolution of AI technology suggests continuous advances, including improved natural language processing, enhanced pattern recognition and increased adaptability. Such progress will equip border management and security organisations with more accurate and efficient training solutions, bolstering the overall efficiency of security operations by swiftly identifying potential threats or anomalies.

In anticipation of emerging threats, AI-driven training will prioritise agility, enabling rapid adjustments to training programmes through continuous learning from evolving scenarios. For instance, AI learning tools could integrate with other security applications, facilitating comprehensive understanding and adaptation to dynamic security challenges across different platforms. Integration with blockchain technology holds promise for enhancing data security, leveraging its decentralised and tamper-resistant nature to secure sensitive training data, thereby ensuring its integrity and confidentiality. Emphasising augmented intelligence, the future of AI in border management training underscores human-AI collaboration to optimise decision-making processes. AI systems will complement human capabilities by providing valuable insights and support, mandating a comprehensive understanding of AI's broader applications in border management. Additionally, as cyber threats evolve, AI will play a crucial role in cybersecurity training, preparing personnel to build resilience against potential cyber-attacks through simulated scenarios.

Ethical considerations: Future implementations of AI in training must prioritise ethical practices, especially in border management. Organisations will emphasise fairness, transparency and accountability in AI algorithms to ensure responsible use of the technology. This includes stringent adherence to ethical standards in data collection, processing and use, with a focus on safeguarding individuals' rights and privacy. Moreover, special attention will be given to addressing potential biases in AI systems to uphold fairness and equity in training outcomes.

Human considerations: AI-driven training programmes for border management personnel will extend beyond technical skills to prioritise learners' mental health and well-being. Incorporating features to evaluate and support mental resilience in high-stress environments will be crucial. These programmes may include AI-powered tools for stress management, coping mechanisms and personalised support to enhance learners' overall well-being. Additionally, ensuring user experience considerations are integrated transparently will enhance the effectiveness and acceptance of these tools among learners.

Public perception and acceptance: Clear communication and transparency regarding the use of AI in border management training will be essential to gaining public trust and acceptance. Educating the public on the benefits, safeguards and ethical considerations associated with AI in training and education is paramount. Efforts to demystify AI technologies and address misconceptions will facilitate broader acceptance and foster trust among stakeholders. Furthermore, data considerations, aligned with the principles outlined in the AI Act, will emphasise group-based data evaluation to minimise privacy concerns and ensure responsible data-handling practices.

Technology 2: The Metaverse

Metaverse for training and education shaping interactive learning

Training technology overview

The Metaverse is defined as a virtual shared space, enabled by 3D digital reality. This space can help break down geographical barriers by enabling people across locations to connect and have immersive virtual experiences. Although metaverse technology is expected to reach maturity only by 2030,[12] it is still foreseen that about 25%[13] of the global population will use the metaverse by 2026 for activities such as jobs, trades, education and entertainment. Much like virtual reality (VR), the metaverse offers themed virtual environments. Its primary emphasis lies in creating a collaborative, shared learning experience that breaks free from the confines of physical spaces. This evolution is set to redefine the landscape of learning technology, opening doors to innovative educational possibilities. As we embark on this metaverse era, it challenges our conventional understanding of reality: it presents a transformative vision for a virtual world where interactions, economies, and experiences are boundless and interconnected, forging a new frontier in human engagement and exploration.

The metaverse can be used as an immersive online collaborative platform to gather and learn together. While extended reality (XR) technologies provide immersive experiences, the metaverse goes beyond by creating a comprehensive and interconnected virtual universe. Metaverse training encompasses a broader spectrum of experiences, including social interactions, persistent environments and interconnected simulations, surpassing the capabilities of standalone extended reality solutions.

Main benefits

The ability of metaverse technology to enable experiences in 3D virtual reality has significant implications for training purposes. The metaverse provides access to training resources across multiple domains and environments, maximising their availability for the EBCG community. Additionally, metaverse training platforms are more cost-effective and sustainable compared to in-person training setups. These provide the flexibility to quickly switch between training situations and environments, reducing the time and cost associated with traditional and physical training setups. The availability of metaverse training platforms also ensures that border management-related training can happen more frequently, since it is not reliant on physical setups. Organisations can also create personalised learning paths on metaverse platforms to ensure a high degree of relevance and applicability.

Enables collaboration: For large organisations with a distributed workforce, the technology also adds value by transcending the limitations of conventional online training in terms of physical space and hands-on team interactions. Implementation of a metaverse in the training of border management officials is an innovative and novel approach, expected to generate better collaborative training spaces by enabling immersive virtual environments to simulate border operations. Learners can get immersed in a 3D shared experience where they can explore virtual worlds (replicating operational environments) via their avatars alongside co-workers.

Increased understanding: Training within the metaverse not only equips border management professionals with essential skills but also empowers them to gain a deeper understanding of how individuals navigate and use this emerging public realm. This insight is crucial for formulating strategies to mitigate emerging risks.

Implementation insights

This section explores implementation aspects that could help border management organisations better understand and decide on the deployment of emerging technologies within their learning and development process. It provides key insights into different areas affected when implementing metaverse technologies for training and education.

[12] https://www.thecuneiform.com/insights/the-future-of-metaverse-in-terms-of-product-development/
[13] https://www.thecuneiform.com/insights/the-future-of-metaverse-in-terms-of-product-development/

Full-body tracking for 1:1 representation of physical movement: Full-body tracking in the metaverse allows for a one-to-one representation of the learner's physical movements. This enables immersive and realistic training experiences that simulate physical actions, such as tactical manoeuvres, weapon handling and coordination exercises.

3D assets to visualise the dimensions and scale of objects: The metaverse, with its 3D assets, allows learners to explore and navigate through different landscapes, enhancing their spatial awareness and understanding of the physical surroundings they may encounter during border management operations.

Virtual avatars for presence in a virtual environment: Virtual avatars offer a level of immersion and interpersonal interaction that is not easily achievable in traditional online training. Learners can visually perceive the presence and actions of their peers and instructors, fostering a collaborative environment that simulates real-world team dynamics.

Blockchain technology for transparent record-keeping: Blockchain technology in the metaverse ensures secure and tamper-proof record-keeping. Learner achievements, certifications and other data can be securely stored on the blockchain, providing a transparent and verifiable record of individual and collective training accomplishments. This feature is particularly valuable for maintaining accurate personnel records and facilitating seamless audits.

Use of cloud: Using cloud-based technology to create a repository of learner data can ensure accurate evaluation of progress to provide real-time feedback to learners. It can also help ensure continuous improvements in the training content.

Awareness of the platform: Educating trainers and facilitators about the various benefits of using the metaverse platform for training could help reduce the barrier to quick and seamless adoption within the EBCG training community. Refresher sessions on evolving metaverse features can help ensure that trainers are updated about the technology's capabilities, ensuring maximum use of the technology. Consequently, trainers can provide relevant and timely inputs to continuously develop the training content by using the latest capabilities.

Integration of the metaverse with existing learning management systems and organisational workflows: This can ensure that the metaverse complements the learning curriculum when implemented by the wider EBCG community. The metaverse twin seamlessly integrates with existing learning systems, making the curriculum more effective. This integration benefits learner awareness, fostering a cohesive learning environment within the digital realm.

Global collaboration: The metaverse breaks down geographical barriers, fostering global collaboration. Teams and individuals can seamlessly connect and work together irrespective of physical distances, enhancing communication and cooperation on a global scale.

Immersive training infrastructure: Serving as a comprehensive infrastructure for immersive training, the metaverse enables organisations to create realistic and dynamic simulations, beyond the capabilities of hands-on experiences, enhancing the effectiveness of training programmes, especially in fields like border security where practical skills are paramount.

Enhanced brand recognition: Organisations leveraging the metaverse observe increased engagement by learners, which enables improved employer branding, particularly among younger generations. The immersive and engaging nature of metaverse experiences captures the attention of younger generations, establishing a unique and innovative brand presence that resonates with this tech-savvy category of staff.

Adaptability to industry changes: The metaverse empowers organisations with increased adaptability to industry changes. As industries evolve, the metaverse provides a versatile platform for organisations to pivot and adjust their strategies, ensuring they remain agile and responsive to emerging trends and challenges.

Engaging new generations: For new generations, the metaverse serves as a captivating public touchpoint. Organisations can engage with this demographic in refreshing and stimulating ways, aligning with their digital preferences and creating a meaningful connection. This not only enhances outreach but also establishes a forward-thinking image that resonates with the younger audience.

Main integration opportunities

This section explores the integration opportunities offered by the metaverse. By identifying key areas where the technology can be seamlessly integrated, organisations can harness its full capabilities to enhance efficiency and drive innovation.

Learning and development in border management organisations is poised for transformation through the integration of the metaverse with cutting-edge technologies. This convergence includes AI for personalised experiences, seamless connectivity with 5G/6G, streamlined management through a learning management system (LMS), enriched experiences with extended reality (XR) and heightened engagement via gamification. Together, these integrations create a powerful and adaptive learning ecosystem, enhancing the effectiveness and efficiency of training programmes for border management personnel.

AI integration: AI integration with the metaverse enables personalised learning experiences. AI algorithms can analyse learner interactions, adapting content to individual needs. In border management training, AI-driven simulations can respond dynamically to learners' decisions, enhancing real-world scenario simulations.

LMS integration: Integrating the metaverse with LMS streamlines content delivery and tracking. This ensures a cohesive learning experience, allowing border management organisations to centralise training materials, monitor progress, and provide real-time feedback within the immersive metaverse environment.

5G/6G connectivity: Leveraging 5G/6G connectivity in the metaverse offers seamless, high-speed interactions. This is crucial for border management training, where real-time collaboration, immersive simulations and data exchange are enhanced. Faster connectivity ensures a more responsive and immersive learning experience.

Extended reality integration: Combining the metaverse with extended reality technologies, such as virtual reality (VR) and augmented reality (AR), enriches the training environment. Border management personnel can experience augmented information overlays in real-world scenarios or immerse themselves in virtual border simulations, enhancing hands-on skills and situational awareness.

Gamification elements: Incorporating gamification into the metaverse enhances engagement and motivation in border management training. Game mechanics, rewards and challenges can be integrated, making the learning experience more enjoyable and fostering a competitive and goal-oriented approach to skill development.

These integrations offer a holistic approach to learning and development in border management organisations. They combine the immersive nature of the metaverse with the adaptive capabilities of artificial intelligence, centralised management through LMS, enhanced connectivity with 5G/6G, enriched experiences through extended reality and increased engagement through gamification. Together, these technologies create a synergistic learning ecosystem aligned with the evolving needs and challenges of border management personnel.

Box 8. Case studies on the adoption of metaverse technology in education

Interpol[14] adopted metaverse technology to train its learners in travel document screening and passenger screening, using Interpol capabilities in a virtual classroom. Following this session, learners were able to apply their learning at a virtual border point in a metaverse airport to ensure high applicability from their learning session.

The Indian Army[15] announced that it would be developing a metaverse-based training platform for its soldiers. The platform will help soldiers learn and adapt to modern warfare patterns, across multiple environments, while ensuring minimal casualties. The Army will provide its data from the field to the platform to add a high degree of relevance to the training exercises. The Indian Army has collaborated with local universities and Tech Mahindra to design and implement this training laboratory.

Main challenges

Integration and implementation of metaverse technology into the training and education of EBCG personnel will require various gaps and challenges to be addressed.

Metaverse integration issue: Integrating metaverse experiences with existing LMSs and software presents compatibility challenges. As metaverse technology evolves rapidly, ensuring seamless connectivity with dynamic LMSs becomes pivotal. Harmonising these

[14] https://www.interpol.int/en/News-and-Events/News/2022/INTERPOL-launches-first-global-police-Metaverse

[15] https://defencedirecteducation.com/2022/05/27/ai-powered-metaverse-training-for-indian-army-troops/

components in a constantly evolving landscape demands strategic planning and potential system updates. Moreover, metaverse training must be available in operational locations or physical training facilities, requiring dedicated equipment with varying standards. Achieving seamless integration and availability amidst evolving technologies requires careful consideration and adaptation.

Technical complexity in the metaverse: Metaverse participation faces challenges due to technical complexity, especially in creating demanding 3D content. Advances in generational AI are vital for realising the metaverse vision. Assessing its actionability for organisations and the readiness of learning and development teams is key for effective integration and utilisation.

Metaverse versus physical training: While virtual environments offer a standardised version of the equipment for training, the reality in the field might be significantly different. This may require additional training on the use of equipment for daily operations. Metaverse training will never entirely replace physical training, so hybrid programmes are needed.

Data privacy and security: Ensuring high data privacy and security for the integration of the metaverse into learning systems, especially when confidential datasets are being used, continues to be a cost- and time-intensive challenge, as sensitive information needs to be safeguarded. GDPR requires stringent data protection measures to secure personal data used in metaverse training, necessitating robust cybersecurity measures and adherence to data processing principles. Additionally, the Rights to Erasure and Rectification under GDPR require mechanisms allowing participants to exercise these rights for data used in metaverse training. The policies outlined by the EU under Digital Rights Principles emphasise informed consent, transparency and ethical use of technology, guiding the development of metaverse scenarios for training purposes.

Compliance with dynamic regulatory standards: Challenges could arise in ensuring that metaverse technologies adhere to changing security standards. Ensuring continuous compliance with dynamic national and international security regulations will become crucial.

Ethical usage: Metaverse technologies need to ensure adherence to ethical usage guidelines. This includes addressing potential biases in simulations, to prevent any unintended consequences of immersive training scenarios.

Mindset shift: Educating the EBCG community about the benefits of metaverse-enabled training set-ups will be crucial to ensure the success of the platforms. It plays a crucial role in fostering a mindset of continuous learning among both trainers and learners. This educational initiative is essential to overcome adoption barriers, ensuring that the EBCG community embraces metaverse technologies for skill enhancement and stays adept in an ever-evolving educational landscape.

Cross-border collaboration: As border management often involves collaboration between multiple agencies, regulatory frameworks need to facilitate seamless cross-border metaverse training initiatives. Ensuring that regulations enable international cooperation in virtual training environments is essential for addressing shared security challenges and concerns.

Collaboration between developers and educators: Aligning metaverse training experience with curriculum standards and educational goals requires collaboration between technology developers and educators to ensure effective learning environments. This also involves educating the trainers about the benefits of integrating the metaverse into training curriculums.

Accessibility and inclusivity: Bridging the technology gap for diverse user demographics and languages demands careful consideration to ensure accessibility and inclusivity for all learners.

Outcome/future potential

The future of the metaverse will bring an important feature: the standardisation of virtual interactions. Similarly to what today's internet is for websites, the metaverse will provide the infrastructure for the 3D interactive layer of the future internet. The metaverse is reshaping every industry: in manufacturing, virtual prototyping is enabled with improved product understanding; virtual labs in technology accelerate hands-on practice, advancing technical skills; healthcare leverages the metaverse for lifelike medical simulations, enhancing surgical expertise. These examples signify a transformative shift in workforce training, fostering efficiency and proficiency across industries. Metaverse technology holds the potential to transform border management training, but it prompts consideration of how it integrates with extended reality platforms and technologies. The seamless integration of such elements within the metaverse is essential for creating a unified and cohesive user experience. The technology needs to advance rapidly to enable mainstream adoption. High device costs, bulky hardware and lack of cross-device/platform integration limit the reach of this technology.

Interoperability across platforms: Future development of metaverse technologies will likely prioritise interoperability across devices and technologies, allowing seamless integration of metaverse training platforms

with existing systems, thus ensuring a comprehensive training experience.

Integration with AI, generative AI and machine learning (ML): Metaverse technology is expected to incorporate advanced AI and ML algorithms to create dynamic and adaptive training scenarios, helping simulate diverse and evolving training environments for learners.

Personalised training modules: The evolution of metaverse technology will enable the creation of personalised training modules, tailoring scenarios to individual skill levels, roles and specific challenges and scenarios.

Global collaboration and joint exercises: The cross-border capabilities of metaverse technology could lead to collaborative metaverse training initiatives involving multiple countries. Joint exercises in virtual environments can help enhance cross-border collaboration and preparedness.

Regulatory adaptation and standardisation: Expected changes in regulatory frameworks will lead to further adaptation and standardisation of metaverse technology in border security training, with regulations evolving to address emerging challenges and ensure responsible use.

Stakeholder endorsement: Endorsement and sponsorship from key stakeholders and leadership groups within the organisation will ensure effective implementation and adoption of metaverse technology for training purposes.

User-friendly interface design: Creating a user-friendly interface for metaverse technology is crucial to ensuring maximum adoption and accessibility. Designing intuitive and easy-to-navigate platforms can help ensure seamless integration and adoption into training functions.

Training for diverse skill levels: Metaverse training platforms should provide training and support for individuals with varying levels of technological proficiency. Customising training programmes to accommodate beginners and advanced users can help foster inclusivity and enhance adoption.

Regular accessibility audits: Conducting regular accessibility audits and assessments can help identify areas for improvement. This ongoing evaluation ensures that metaverse technology remains inclusive and accessible to all learners.

Adaptive and inclusive features: Incorporating adaptive features, such as language preferences, enhances the accessibility of metaverse training platforms.

Technology 3: Gamification

Gamification for training and education, from nice-to-have to must-have

Training technology overview

Gamification is the process of integrating game design elements and principles, such as badges, leaderboards, points and rewards, into non-game contexts to enhance engagement and motivate learners.[16] It involves the strategic application of gaming features to routine tasks, aiming to improve adoption, engagement and success rates. Gamification is used in various fields, including education, marketing, staff training, health, wellness and customer engagement. It is essential in contemporary learning and development approaches and will remain vital in the future. Gamification's capacity to boost learner engagement and motivation, enhance learning and retention, foster social interaction and collaboration among employees, and positively impact workplace culture makes it indispensable.[17]

Over the last 3-5 years, organisations have been increasingly adopting gamification for workforce training to enhance engagement and improve learning experiences. Initially linked to entertainment, its surge in adoption across sectors is driven by proven effectiveness, technological advances and the broader trend of digital transformation. The adoption of gamification as part of learning and development has become critical due to its ability to enhance learner engagement, improve skill retention (because of its interactive and immersive nature), simulate real-world scenarios for training and adapt across various learning styles and solutions. Though not a technology, gamification yields a 99% increase in learner confidence, a 97% improvement in course utility, and demonstrates resourcefulness, even aiding in case resolutions.[18] Customisation options further empower organisations to tailor courses to specific needs.

16 https://www.biworldwide.com/gamification/what-is-gamification/

17 https://moodle.com/news/why-you-should-incorporate-gamification-into-your-l-and-d-programs/

18 https://www.capita.com/our-work/using-gamification-upskill-employees-improve-learner-retention-and-combat-crime

Main benefits

In addressing the complexities faced in training the border management workforce, integrating gamification emerges as a transformative approach. The simplicity of implementation, incorporating key elements into existing platforms, caters for immediate improvements. Gamification taps into intrinsic motivation, fostering self-driven learning and energising the learners, showcasing a commitment to excellence. Techniques like repetition, challenges and rewards enhance knowledge retention for effective real-world application. Embracing a tech-savvy learning approach reflects the EBCG's dedication to staying current for long-term workforce competence. Beyond immediate impact, gamification strategically supports workforce retention by fostering job satisfaction, continuous learning, skill development and a positive work culture. As the EBCG community seeks a comprehensive training experience, gamification emerges as a holistic solution, catering to diverse learning requirements.[19]

Enhanced engagement and motivation: Gamification can make training more engaging, productive and enjoyable for employees, creating a unique user experience across multiple platforms such as mobile and tablets. It can reframe the way the workforce looks at learning, putting the learner at the centre of the design process. This approach can boost intrinsic motivation, giving learners more control over their learning and illustrating their competence through scores and public recognition. Organisations will thus not only ensure the upskilling of learners but also enhance their personal experience simply by having fun.

Improved learning outcomes: Gamification enhances learning outcomes by providing proactive directives and feedback through game mechanics, making goals clear and easy to follow. Learners can see progress in performance, receive immediate feedback on accomplishments and connect with colleagues through collaboration and competition**.**

Real-world connection/scenarios for training: Gamification can create a real-world connection, helping learners improve problem-solving and decision-making skills. This connection can make training more relevant and meaningful, which can enhance collaboration, understanding and application of knowledge**.**

Promotes social interaction and collaboration: Gamified learning encourages social ties and interactions, giving learners a reason to collaborate with and get to know their colleagues. This can facilitate relationship-building, which is crucial in a team-oriented environment like border management.

Measurable results: Gamification provides data and reporting to help see what's working and where improvements can be made. This can help in continuously improving the training programme.

Workforce retention: Gamification supports workforce retention by fostering job satisfaction, providing continuous learning opportunities, facilitating skill development and creating a positive work culture that values and recognises employees' contributions.

Embracing gamification is therefore crucial for EBCG workforce training. By catering to engagement needs and delivering an immersive user experience akin to popular consumer apps, it ensures training aligns with the digital preferences of today's workforce. The dynamic and interactive nature of gamified training particularly resonates with younger professionals, making careers in border security more appealing. This approach not only increases participation but also contributes to a positive perception of the sector. Gamification serves as a strategic tool, preparing personnel for real-world challenges and attracting a diverse pool of talent, thereby shaping the future of training in this critical sector.[20]

Implementation insights

This section explores implementation aspects that could help border management organisations better understand and decide on the deployment of emerging technologies within their learning and development process. It provides key insights into different areas affected when implementing gamification technologies for training and education.

[19] https://www.sumtotalsystems.com/blog/the-role-of-gamification-in-learning

[20] https://medium.com/

Tapping into intrinsic motivation: People naturally desire a sense of achievement and recognition. Tapping into this will motivate learners.

Reward and progression system: Implementing a point or levelling-up system within gamification serves as a tangible acknowledgement of individual and team achievements. This not only reinforces positive behaviours but also introduces an element of healthy competition, driving continuous improvement in the workforce's skills and knowledge.

Narrative flow of the content: Gamified content is strategically structured to unfold narratively, creating a compelling storyline for learners. This storytelling approach adds an immersive layer to the learning experience, increasing interest and motivation for learners to complete content modules within the context of their roles in border management.

Portioning of the learning content: Gamification leverages on the concept of 'learning nuggets', bite-sized and easily consumable portions of content. This approach mirrors the format of language learning apps, allowing border management personnel to access short, focused content at their own pace. This flexibility accommodates learning within the flow of their work, ensuring minimal disruption while maximising knowledge acquisition.

Box 9. Case studies on the adoption of gamification technology in education

The College of Policing, North Yorkshire – UK, 2023:[21] The college implemented gamification in its workforce training and development initiatives by creating a gamified, 'point-and-click' digital series called Op Modify. This programme consisted of 10 modules, each focusing on a different type of crime, demonstrating how digital resources can be used to combat crime. The highly visual nature of the game and its engaging design aimed to upskill officers, staff and volunteers in the digital sphere, increasing their expertise in combating crime. The initiative was part of the college's commitment to the professional development of its workforce in the digital domain, particularly to prepare for innovative crimes and the increasing digital expertise expected of new officers joining the police force.

The Dutch Police Academy, Netherlands, 2023:[22] The academy has embraced virtual reality (VR) to enhance training efficiency and address evolving learning needs. Virtual reality technology, including scenario-based training, provides a realistic and engaging experience, overcoming limitations in time, budget and safety associated with traditional methods. The academy has introduced multiplayer virtual reality courses for forensic investigation, proving effective in improving knowledge, resilience and discussions about policing quality. This adoption reflects the academy's commitment to leveraging innovative technologies for effective training and meeting the evolving needs of its workforce.

Deloitte Major Programmes, UK, 2021: Deloitte has inaugurated a new sector dedicated to managing significant transformations, leveraging the collective expertise of specialists across various domains to assist clients in accomplishing major changes. To integrate professionals into this innovative programme, the Belgium Workforce Transformation team established a virtual academy with a comprehensive learning pathway. Interactive modules have been strategically incorporated into the learning solution, seamlessly linking offline classroom activities with the personalised aspect of online learning.

LAW-GAME European Project, 2021:[23] The LAW-GAME project was a European initiative aimed at developing an advanced learning experience to train law enforcement agencies (LEAs) and measure their proficiency in conducting forensic examination, effective questioning and intelligence crime analysis. The project aimed to elevate experiential training for police officers through gamification technologies, including serious games, virtual reality and AI-assisted procedures. The project was supported by the EU's Horizon 2020 Programme and brought together 19 partners from 11 European countries. It helped enhance LEAs' training experience by recreating real-world scenarios through introducing an attractive approach to developing core competencies required for intelligence crime analysis.

21 https://www.capita.com/our-work/using-gamification-upskill-employees-improve-learner-retention-and-combat-crime

22 https://re-lion.com/how-the-dutch-police-academy-uses-vr/

23 https://www.simavi.ro/en/node/101

Main integration opportunities

This section explores the integration opportunities offered by gamification. By identifying key areas where the technology can be seamlessly integrated, organisations can harness its full capabilities to enhance efficiency and drive innovation.

Integrating various evolving technologies such as artificial intelligence, mixed reality, augmented reality and virtual reality with gamification can considerably boost the effectiveness of learning and development capabilities. By leveraging these technologies, law enforcement and military forces can make their training programmes more effective, improve knowledge retention, and foster a competitive spirit among their workforce, ultimately equipping them with the skills and knowledge necessary to address evolving needs and combat crime more effectively. The strategic use of gamification in workforce learning and development for border management organisations goes beyond surface-level engagement. It taps into intrinsic motivation, introduces reward structures, employs engaging narrative techniques and facilitates learning through easily digestible portions — all tailored to enhance the effectiveness and appeal of training initiatives in the unique context of border management forces.

Serious games and gamification:[24] Serious games are purposeful learning environments that target key curriculum areas for explicit learning, while gamification involves the use of game design elements in non-game contexts. Integrating serious games with gamification enhances learning by increasing motivation, deepening engagement, improving outcomes and boosting attendance. This combination offers immersive and tailored experiences, addressing evolving workforce learning needs.

Interactive training platforms: Interactive gamified training platforms can enhance problem-solving skills and create opportunities for learners to face real-world challenges, fostering a competitive spirit and improving knowledge retention.

AI-assisted procedures: AI elevates gamification through adaptive feedback, personalisation and analytics. It tailors dynamic scenarios, challenges and rewards based on the learner's profile, performance and preferences. AI offers personalised feedback and guidance aligned with the learner's needs, goals and emotions. Furthermore, it analyses gamified data to offer insights and recommendations for improvement.

Game-based training:[25] Game-based training blends elements of gamification and serious gaming by incorporating game mechanics into specific modules or courses. It offers a dynamic and effective approach for security forces to enhance their skills, improve decision-making abilities and prepare for a variety of complex and challenging scenarios they may encounter in their duties.

Metaverse and gamification:[26] The metaverse, an interoperable virtual space, revolutionises content ownership and trading with gaming elements. It offers interactive gamification, leveraging in-game features to maintain learner engagement. This approach, applicable across industries like learning and development, fosters customer loyalty, incentivises behaviours and drives attractive promotional campaigns. Embracing gamification in the metaverse enables the creation of intuitive, immersive games with social features, ensuring engaging and meaningful experiences.

Main challenges

Integration and implementation of gamification technologies into the training and education of EBCG personnel will require various gaps and challenges to be addressed.

Staying true to the concept: Gamification goes beyond the implementation of a few tactics. It first requires good content quality and learning methodology, on top of which the gamification strategy reinforces the storytelling and creates a coherent user experience.

Ensuring the right perspective: Gamification should be viewed as a layer that sits on top of the qualitative content to implement the concept and create a balance between the fun and the educational aspects of the application.

Customisation: The gamification approach should be tailored to the target audience, considering parameters such as workforce diversity and age groups**.**

Resistance to change: Some individuals may resist game-based initiatives, perceiving them as forced or manipulative. This highlights the need to communicate the benefits and value of gamification to overcome this resistance.

Integration with existing systems: Integrating gamification across various learning management systems (LMS) or diverse training platforms becomes complicated when dealing with a hierarchical structure. Rather than attempting to connect gamification frameworks across

[24] https://moodle.com/news/why-you-should-incorporate-gamification-into-your-l-and-d-programs/
[25] https://moodle.com/news/why-you-should-incorporate-gamification-into-your-l-and-d-programs/
[26] https://landvault.io/blog/brands-gamification-metaverse

disparate LMS levels, a more effective approach is to establish a common, collective higher infrastructure, like a technology-enabled academy. This centralised structure defines principles and allows the community to inherit and implement them. In the current landscape, the challenge arises as individual entities maintain their LMS, limiting gamification to lower levels due to the lack of a unified framework. Therefore, ensuring compatibility and interconnectivity of different academies and LMSs is key to preventing inconsistency in gamification and isolated systems.

Content development challenges: The creation of compelling and impactful gamified content demands specialised skills and resources. This underscores the importance of investing in training the digital academy's administrative staff, enabling them to produce, distribute and manage content that effectively balances educational objectives with gaming elements, and trains L&D teams to use gamification in a more strategic way for learning pathways.

Accessibility and inclusivity: Certain features of gamification may not be accessible to individuals with disabilities, potentially violating accessibility regulations. In these instances, emphasising inclusive design principles and ensuring that gamified content adheres to accessibility standards becomes crucial, particularly within the EBCG community.

Intellectual property rights: Employing game elements or content from third-party sources may give rise to concerns regarding intellectual property rights. Therefore, obtaining appropriate permissions or generating original content is essential to steer clear of legal issues associated with copyright and intellectual property.

Fairness and equality: Ensuring that gamification methods do not unfairly favour certain individuals or groups, promoting equal opportunities for all training participants.

Transparency: Communicating the rules, objectives and rewards of gamified learning to maintain transparency and build trust among participants.

Informed consent: Obtaining informed consent from participants before incorporating gamification, ensuring they are aware of how their data will be used and the nature of the gaming elements.

Avoiding exploitation: Ensuring that gamification does not exploit the competitive nature of individuals, leading to negative consequences or unhealthy competition among the team members, is important to retain the purpose and motive of the task.

Legal implications for the EBCG community: When adopting gamification technology, it is essential to consider regulations concerning data protection and privacy, securing intellectual property rights, complying with staff-related regulations, safeguarding fundamental rights, clearly defining the terms within contractual agreements (if applicable) and ensuring overall regulatory compliance.

Outcome/future potential

Though gamification is mature in learning technology, it is not a standard approach. Influenced by the gaming industry's successful tactics in marketing and consumer experiences, the learning sector is expected to adopt and elevate gamification to meet the expectations of digitally nurtured generations. The increasing popularity of virtual reality/augmented reality/mixed reality, the metaverse and gaming, along with affordable hardware, challenges learning to embrace blended formats and virtual worlds. What is currently a mobile learning application where the learner gathers badges and levels to progress could transform into a 3D role-playing game with deeper game mechanics based on the learning content tomorrow.

Integration with emerging technologies: Gamification is likely to integrate with emerging technologies such as VR, AR and AI to create more immersive and personalised learning experiences.

Metaverse learning environments: The rise of the metaverse could lead to creation of innovative gamified learning environments, where learners engage in collaborative and interactive experiences across virtual spaces.

Adaptive and personalised learning: Gamification is expected to evolve towards more adaptive and personalised learning experiences, tailoring content and challenges to individual learner profiles and preferences.

Integration of data analytics: Gamification platforms will increasingly leverage data analytics to provide detailed insights into learner performance, allowing organisations to make informed decisions about training effectiveness and areas for improvement.

Mobile and microlearning integration: As mobile and microlearning gain prominence, gamification strategies will likely adapt to these trends, offering bite-sized, mobile-friendly gaming experiences that align with the preferences of today's learners.

Gamified certification and credentialing: The use of gamification for certifications and credentialing is expected to increase, providing a more engaging and

verifiable way for individuals to showcase their skills and achievements.

Inclusion of social and collaborative elements: Social and collaborative elements within gamified learning, such as multiplayer scenarios and team-based challenges, are likely to become more prominent for fostering teamwork and communication skills.

Realism and relevance: Enhancing the relevance and effectiveness of gamified training involves crafting lifelike scenarios and challenges that closely replicate real-world situations encountered by military and police personnel.

Adaptive learning paths: Customising gamified learning paths according to individual skill levels and learning styles ensures adaptive training that caters to learners' diverse needs.

Risk-free simulation: Offering a consequence-free setting for practising skills and decision-making through virtual simulations enables law enforcement practitioners to learn from their errors without facing real-world consequences.

Integration with actual tools and equipment: Incorporating actual tools, equipment and procedures used by security forces into gamified training scenarios enhances the applicability and transferability of learned skills to real-world situations.

Scalability and consistency: Creating gamified training programmes that are scalable for large groups and consistently delivered ensures uniformity in training experiences across various units and locations.

Alignment with training objectives: Ensuring that gamified elements directly align with the training objectives and desired learning outcomes helps maintain the focus on essential skills and knowledge for the sustained development of similar technologies for learning and development initiatives.

Technology 4: 5G/6G

5G/6G offers training and education anywhere, anytime

Training technology overview

With every generation of communications technology, the focus of the network changes. The 2G and 3G eras centred on human-to-human communication through voice and text. 4G heralded a fundamental shift to the massive consumption of data, while the 5G era has turned the focus to enabling the Internet of Things (IoT) and industrial automation systems. 5G and emerging 6G technologies represent significant milestones in the evolution of wireless communication.

5G, the 5th generation mobile network, is a global standard that enables a new kind of network designed to connect virtually everyone and everything, including machines and devices. Already deployed in many regions, 5G technology is meant to deliver higher multi-Gbps peak data speeds, ultra-low latency, more reliability, massive network capacity, increased availability and a more uniform user experience to more users. Higher performance and improved efficiency empower new user experiences and connect new industries,[27] enabling innovations such as augmented and virtual reality, autonomous vehicles and IoT applications. It empowers smart cities, remote healthcare and immersive experiences.

6G, the anticipated sixth-generation technology, is envisioned to be the next, evolutionary step in wireless communication. While still in the conceptual stages, 6G is expected to offer even higher data rates, lower latency and advanced features with near-instantaneous communication and ubiquitous connectivity. Potential technologies for 6G include the use of terahertz frequencies, AI integration, tactile internet and holographic communication. The goal is to provide an infrastructure that supports unprecedented levels of connectivity, enabling futuristic applications such as immersive augmented reality, seamless AI integration and efficient communication in challenging environments.

Main benefits

The integration of 5G and the forthcoming evolution to 6G holds immense potential for revolutionising learning and development processes. These advanced telecommunication technologies offer real-time immersive training experiences through increased connectivity and decreased latency. The integration of Internet of Things (IoT) devices and artificial intelligence (AI) ensures personalised learning journeys and responsive control in training scenarios. The implications for learning technology are profound.

[27] https://www.qualcomm.com/5g/what-is-5g

Enhanced connectivity: With faster data transfer speeds, 5G/6G facilitates increased connectivity, enabling seamless access to training materials and simulations, and providing an uninterrupted learning experience for security personnel.

Low latency for responsive control: Decreased latency enables new scenarios of asset usage such as responsive UAVcontrol or remote unmanned vehicle control. Training exercises involving responsive control benefit from minimal delays, providing security personnel with realistic simulations and hands-on experiences, enhancing their operational readiness.

Maximising IoT sensor usage: Sensors can be deployed alongside borders, fences and in terrain to detect breaches, general movement, heat signatures, vibrations and similar. By maximising IoT sensor usage, training programmes can simulate real-world scenarios where sensors detect breaches, monitor movement, identify heat signatures and capture vibrations. This hands-on experience enhances the practical skills of security personnel, preparing them for the diverse challenges they may encounter in the field.

Remote education: 5G enhances remote education with high-quality video streaming and real-time collaboration. It also fosters coordination, overcoming connectivity challenges in remote areas. 5G can transform how employees acquire knowledge, breaking down traditional barriers to create an accessible, adaptive and interactive learning environment. Leveraging high connectivity, EBCG organisations can extend training to challenging terrain that was previously inaccessible. This ensures comprehensive training for personnel across diverse landscapes, enhancing their adaptability to real-world scenarios.

Immersive learning experiences: The high bandwidth and low latency of 5G and 6G allow immersive technologies like virtual reality and augmented reality to be integrated into education. Learners can explore complex concepts through immersive simulations, increasing comprehension and retention.

Personalised education: 5G and 6G can support advanced AI algorithms that personalise learning experiences. These technologies analyse individual learner data to tailor content, pacing and assessments, ensuring that each learner's needs are met.

Efficient resource management: Educational institutions can benefit from the efficiency of 5G and 6G in managing resources, optimising network usage and reducing operational costs.

Expanded reach in challenging terrain: The extended connectivity in challenging terrain provided by 5G becomes even more robust with 6G, comprehensively addressing coverage challenges. The ability of 6G to sense the environment enhances situational awareness, a critical aspect of border management and security training. With a strong focus on security measures and privacy considerations, the journey from 5G to 6G promises to redefine training paradigms, providing a technological foundation for the future of border security and management preparedness.

Implementation insights

This section explores implementation aspects that could help border management organisations better understand and decide on the deployment of emerging technologies within their learning and development process. It provides key insights into various areas affected when implementing 5G/6G technology for training and education.

Remote education: Providing high-quality video streaming and enabling seamless real-time collaboration translates into more engaging and interactive online learning experiences, bridging geographical gaps and making education accessible to a wider audience.

Global access to experts: 6G's promises of global connectivity at unprecedented speeds means that learners and educators can easily tap into a worldwide network of experts, creating a borderless exchange of knowledge and expertise. This not only enriches the learning experience but also facilitates cross-cultural understanding and collaboration.

Mobile learning with augmented reality/virtual reality integration: The implementation of 5G/6G allows seamless integration of augmented reality/virtual reality solutions into training scenarios. This facilitates mobile learning, allowing personnel to connect UAVs, mobile devices and other hardware for simulation exercises. This immersive, on-the-go training enhances skill acquisition in diverse environments.

Immersive training simulations: Implementing high-quality immersive simulations enabled by 5G/6G could make training more realistic. This would allow instructors to create dynamic scenarios, improving the preparedness of the workforce for border and security challenges. The high bandwidth and low latency of 5G and 6G enable the integration of immersive technologies like virtual reality and augmented reality) into educational programmes. Border management personnel can explore complex concepts through immersive simulations, significantly enhancing comprehension and retention. 6G may revolutionise learning by introducing immersive virtual classrooms where learners can participate in lifelike environments, enhancing knowledge retention and fostering active engagement. Additionally, 6G's advanced AI capabilities can offer personalised education experiences, tailoring content and pacing to individual learners' needs, optimising the learning process.

Enhanced connectivity for uninterrupted learning: Increased data transfer speeds and connectivity ensure security personnel have seamless access to training materials and simulations. Uninterrupted learning experiences make the training process more efficient and effective. For example, the deployment of 5G technology within the medical ecosystem has showcased improved connectivity, near-zero latency and integration capabilities to accelerate the adoption of advanced technology. It will allow the deployment of capabilities such as virtual reality-supported imaging features and virtual training environments for surgeons to work with simulated conditions and augment the growth of emerging technologies such as artificial intelligence.

Enhanced brand recognition: Organisations leveraging the metaverse observe increased engagement by learners, which enables improved employer branding, particularly among younger generations. The immersive and engaging nature of metaverse experiences captures the attention of younger generations, establishing a unique and innovative brand presence that resonates with this tech-savvy category of staff.

IoT integration for practical scenarios: Integrating IoT devices in training scenarios through 5G/6G enhances practical learning. Educators can deploy sensors to simulate real-world border control situations, allowing learners to analyse data and respond to detected breaches, movement, heat signatures and vibrations. Hands-on experience with IoT sensors enhances the practical skills of security personnel, preparing them for diverse challenges in the field.

Efficient resource management and cost reduction: Educational institutions benefit from the efficiency of 5G and 6G in managing resources and optimising network usage. Reduced operational costs contribute to the overall efficiency of the digital learning academy.

Main integration opportunities

This section explores the integration opportunities offered by 5G/6G. By identifying key areas where the technology can be seamlessly integrated, organisations can harness its full capabilities to enhance efficiency and drive innovation.

Enhanced AI-based training applications: The increased speed and connectivity offered by 5G/6G significantly benefit AI-based training applications. Organisations can leverage AI for adaptive learning modules.

Real-time simulation with AR/VR: Combining 5G/6G with augmented and virtual reality creates realistic training simulations. Border security personnel can engage in real-time scenarios, improving decision-making skills and responsiveness.

Edge computing for instant feedback: Edge computing brings computation and data storage closer to the user, reducing latency. Together with 5G/6G, it facilitates instant feedback in training scenarios. This enhances the learning experience by providing immediate assessment and correction.

IoT-enabled practical scenarios: Integrating IoT devices with 5G/6G allows for the creation of practical scenarios. Sensors deployed in training environments simulate real-world border situations, enhancing hands-on experience.

Secure and encrypted communication: 5G/6G ensures secure communication channels for sensitive training materials. Encryption protocols safeguard against cyber threats, maintaining the confidentiality of training data.

Collaborative learning environments: 5G/6G enables collaborative learning environments for border security personnel. High-speed connectivity fosters teamwork and knowledge-sharing between learners through the metaverse or collaboration in gamification.

Continuous professional development platforms: Establishing continuous professional development platforms supported by 5G/6G ensures ongoing skill enhancement. Access to real-time updates and collaborative tools fosters a culture of continuous learning.

Main challenges

Integration and implementation of 5G/6G technology into the training and education of EBCG personnel will require various gaps and challenges to be addressed.

Maritime challenges to remain: 5G coverage is problematic in maritime areas away from the shore. Internet via satellite is required, which is expected to always struggle in scope and efficiency. Technologies such as satellites, buoy networks and antennas should be combined to effectively address this limitation and improve border management training.

Security concerns: Safeguarding sensitive training data and communications in a high-security environment demands robust cybersecurity measures to prevent potential threats. In the absence of the network slicing[28] features of 5G, robust encryption capabilities and advanced authentication measures, training of law enforcement staff in such an environment will prove to be a major issue.

Device authorisation and authentication: Ensuring that only approved users and devices are accessing a private 5G network is a crucial yet challenging task.

Interoperability: Ensuring seamless interoperability between different generations of technology and legacy systems during the transition phase can be complex.

Behavioural shifts: Anticipating and adapting to changes in user behaviour and consumption patterns is crucial. The rapid evolution of digital solutions demands continuous updates to training content and methods.

Dependency on technology: High dependency on 5G/6G technologies for training poses risks in the case of network failures or cyber threats. Ensuring backup systems and cybersecurity measures is imperative.

Rapid technological development: The fast-paced development of new digital solutions and technologies introduces challenges in aligning training programmes with the latest advances. Regular updates are essential to keep the workforce abreast of technological shifts.

[28] Network slicing is a feature of 5G technology that allows for the creation of multiple virtual networks. Each “slice” can be customised to meet the specific requirements of different applications, services or business models. For example, a slice designed for emergency services can ensure high priority and minimal latency, while another slice for consumer internet might focus on maximising throughput. This level of customisation not only optimises resource use but also improves service quality by isolating and managing traffic according to specific needs. Moreover, network slicing supports the creation of isolated environments, ensuring that performance issues in one slice do not affect others, thereby enhancing security and reliability across the network. By enabling such granular control, network slicing makes it possible for operators to support a wide range of use cases and business models on a single 5G infrastructure, driving innovation and efficiency in ways unattainable with earlier network technologies.

Dynamic regulatory landscape: Technology regulations vary across regions, impacting the uniform adoption of 5G/6G. Harmonising regulatory frameworks is essential for consistent implementation within the EBCG community.

Global focus areas: Major economies and private players have distinct focuses on technology regulation. Collaborative efforts are required to address common concerns and streamline regulatory guidelines for workforce training.

Data ownership and privacy: Educators using 5G/6G must navigate data ownership issues. Establishing clear guidelines on data ownership and privacy protection is essential to maintain ethical standards.

Health and safety concerns: Assessing the health and safety implications of prolonged exposure to 5G/6G technologies is critical. Ensuring compliance with health regulations and guidelines is necessary for the well-being of educators and learners.

Data protection and fundamental rights: The implementation of 5G/6G in training raises questions in these areas. Establishing frameworks that prioritise individual rights while leveraging technology for learning is paramount.

Compliance with ethical guidelines: Adhering to ethical guidelines in the use of emerging technologies is vital. Educators must ensure that training content aligns with ethical principles, avoiding biases and potential harm.

Regulatory policies: Navigating diverse regulatory policies in different regions requires careful consideration. Educators must be well-versed in the legal landscape to ensure compliance with local and international regulations.

Outcome/future potential

In the realm of border management and security training, the forecasted evolution of 5G and anticipated 6G technologies promises a rapid transformation. The general industry outlook envisions higher data rates, reduced latency and improved connectivity, particularly with 6G aiming for ground-breaking advances. This evolution is set to usher in more sophisticated training systems marked by the integration of VR/augmented reality applications, offering immersive and realistic training environments. This facilitates high-quality, real-time streaming, a game-changer for training and development, offering unprecedented opportunities for immersive learning, enhanced skill acquisition and real-time situational awareness in critical sectors like security and defence. The regulatory outlook anticipates international collaboration for standardisation as 6G technologies emerge, emphasising the need for common standards to facilitate seamless integration across borders. The future-proof 6G network will allow the integration of emerging technologies, such as holographic interfaces and communication. This advanced approach enhances personnel skills, offering a more precise and immersive training environment, critical to tackling complex operational challenges.

From a technical perspective, the integration of 5G/6G is poised to introduce advanced features shaping the future of security training. Automation and AI are expected to play pivotal roles, enabling intelligent decision-making in training scenarios, with AI-driven simulations dynamically adapting to evolving threat landscapes. Edge computing, when incorporated into 5G/6G networks, will facilitate faster data processing, enhancing overall training efficiency, especially in real-time analysis, decision-making and response. Biometric technologies, coupled with 5G/6G, enhance security in training applications through robust authentication protocols.

Ensuring secure and reliable networks remains paramount, with continuous improvements in network security features, including network slicing and advanced encryption protocols. Ultra-reliable low-latency communication in 6G is expected to refine connectivity, ensuring sub-millisecond latency for improved real-time communication and coordination during training exercises. AI-driven personalisation within 5G/6G networks emerges as a key feature, adapting training content based on individual performance and needs, promising more effective and tailored learning experiences.

Energy efficiency remains a challenge, with a focus on balancing the demand for high data rates. Optimising energy consumption in 5G and planning sustainable energy models for 6G are deemed crucial for both environmental and economic reasons. Finally, the integration of 5G/6G significantly advances emergency response systems, playing a vital role in public safety and crisis management by enabling faster communication, real-time data sharing and efficient coordination. The implementation of 5G/6G in border management and security training holds the promise of revolutionising systems, enhancing connectivity and preparing organisations for future challenges through the integration of cutting-edge technologies.

Technology 5: Extended reality

Extended reality for training and education, offering immersive learning experiences

Training technology overview

Extended reality (XR) is a comprehensive term encompassing virtual reality (VR), augmented reality (AR) and mixed reality (MR). Innovations to adapt to remote working situations helped democratise XR, making it more affordable and accessible. For training, extended reality is a transformative tool offering immersive simulations by merging physical and digital environments, and a platform for remote learning, training and collaboration.

Extended reality has evolved into a transformative technology across industries. Adoption of XR technologies has accelerated since 2010, supported by periodic innovation. The global pandemic accelerated wider adoption for training, remote collaboration and virtual events. It benefits learners by offering them an enhanced experience, unique simulated environments and improved training efficacy. Originally rooted in aviation, recent advances have democratised XR, making it affordable and accessible. In healthcare, XR aids in medical training and surgical planning. Manufacturing benefits from XR in efficient prototyping, minimising errors. Education leverages the technology for interactive learning, enhancing comprehension. In architecture, XR enables immersive design visualisation. The gaming industry pushes extended reality innovation for realistic experiences. The key driver is XR's democratisation, with increased affordability and accessibility, making it a widely adopted technology with diverse applications across sectors.

Virtual reality (VR): virtual reality creates fully digital environments for learners to interact with, allowing for immersive simulations of hard-to-illustrate situations, such as naval interventions. Headsets, controllers, gaming consoles and motion simulators are examples of virtual reality devices used to immerse learners in these fully digital simulations.

Augmented reality (AR): augmented reality allows for the layering of additional non-interactive elements, such as text descriptions, on top of reality. It is useful for real-time descriptions of the control features of patrol devices, for example. In training, devices such as AR glasses, or smart glasses, overlay contextual information onto the learner's view, facilitating real-time understanding of functionalities.

Mixed reality (MR): mixed reality blends the real-life environment with virtual objects, creating a more dynamic and engaging approach to workforce training. The learner can see reality plus additional interactive elements they can manipulate. MR headsets, like those worn by factory workers, blend digital instructions with real-world equipment for interactive assembly training, combining the best of augmented reality and virtual reality experiences.

By allowing learners to 'learn by doing', this technology improves learning efficiency for both hard and soft skills. In a cost-effective, scalable and realistic way, XR not only enhances the learner's skills and motivation. but also builds confidence to apply them in real life. XR can help learners from the EBCG community across a range of digital learning environments, from crisis event simulation to operation of complex device controls. Border control tasks have undergone a profound transformation, transitioning from traditional reliance on paper documents and individual observations to a modern, multifaceted approach supported by technology. This evolution demands a comprehensive understanding of diverse information and signals by border guards. Integrating extended reality technologies can optimise training for this complex scenario, enhancing situational awareness and providing real-time data overlays, interactive elements and immersive training simulations. The integration of extended reality aligns seamlessly with the evolved border management landscape, offering a technological edge to navigate and process the multifaceted nature of contemporary border control.

Physical field training remains essential for law enforcement practitioners. XR enhances this by simulating realistic scenarios, allowing agents to practice decision-making, threat responses and teamwork. It bridges the gap between theoretical knowledge and real-world application, ensuring agents are well-prepared for dynamic and challenging border control situations. Leveraging extended reality technologies for virtual training can help address standardisation challenges across the EBCG ecosystem. By immersing learners in realistic narratives, XR ensures a unified approach to

security protocols, language proficiency, adherence to policies and familiarity with diverse locations. This virtual training methodology promotes standardised, efficient and adaptable responses, enhancing learners' overall preparedness for real operational environments.

Main benefits

Extended reality is exceptionally well-suited for training in the preparatory phases of products and services, offering significant advantages. The immersive nature makes training more effective, providing a dynamic and interactive experience for individuals preparing for diverse and ever-evolving tasks. Virtual reality creates fully digital environments for learners to interact with. Reality completely disappears, allowing for immersive simulations of hard-to-illustrate scenarios such as search-and-rescue interventions. VR headsets put users into these fully digital simulations. For example, medical learners wear them for surgical simulations, creating a realistic virtual environment for safe and immersive medical training. Other VR devices include controllers, gaming consoles, motion simulators and gloves.

Immersive real-life scenarios: Mixed reality blends the real-life environment and virtual objects, making a more dynamic, engaging and practical approach to workforce training compared to traditional methods. It allows the learner to see reality plus additional virtual elements they can interact with. For instance, factory workers undergoing assembly training wear mixed reality headsets blending digital instructions with real-world equipment, combining the best of augmented reality and virtual reality experiences. The evolving nature of border management operations, having transitioned from paper documents and physical systems to a technology-enabled approach, calls for modern training technologies. XR technologies could give EBCG learners better situational awareness by providing real-time data overlays, interactive elements and immersive training simulations. The ability of XR technology to accurately simulate real-world scenarios by overlaying real-world practical data ensures that learners can sharpen critical skills related to decision-making, threat responses and teamwork — all in a controlled environment. XR technologies could also help standardise training exercises by ensuring a unified approach to security protocols, language proficiency, policy adherence and familiarity with diverse locations.

Standardised virtual training centre: Through XR technologies, law enforcement organisations can internalise knowledge efficiently, standardising training processes while replacing physical training centres with a universally accessible virtual counterpart, ensuring consistency and cost-effectiveness.

Better engagement and assessment: Augmented reality allows the learner to see both reality and non-interactive elements such as text descriptions and explanations. In training, devices such as augmented reality glasses or smart glasses play a crucial role. They overlay contextual information onto the learner's view, facilitating real-time understanding of functionalities, especially in scenarios like explaining the control features of patrol devices. For example, technicians use augmented reality glasses for maintenance, with step-by-step instructions overlaying real-world equipment, enhancing on-the-job learning. Adopting extended reality technologies in training requires mechanisms for learners to provide real-time feedback on the technology and any associated issues, and to receive real-time mixed reality based on their training exercises. These performance analytics can also help instructors identify improvement areas and tailor training to individual needs.

Dedication to learning: Extended reality technologies rule out multitasking, fostering an active learning environment. This ensures full commitment to the training process, optimising the absorption of crucial information and skills.

Cost-effective: Extended reality technologies make training more cost-effective by engaging learners in real-world scenarios without the logistical challenges and risks associated with on-site training exercises. Additionally, they provide scalable and repeatable experiences and training, for consistent skill development.

Implementation insights

This section explores implementation aspects that could help border management organisations better understand and decide on the deployment of emerging technologies within their learning and development process. It provides key insights into various areas affected when implementing extended reality technologies for training and education.

Course curriculum design and scenario development: Developers need to collaborate with educators to ensure that extended reality (XR) courses are aligned with the specific themes and challenges encountered in border management. Additionally, constructing a curriculum that progresses in difficulty levels can facilitate gradual development, allowing learners to build foundational skills and gain confidence over time. This also ensures that learners with varied skill levels and proficiency can use the XR-based curriculum to advance their skill sets.

Technology integration and accessibility: XR hardware and software solutions need to be carefully selected to align with training objectives, and to ensure compatibility and scalability across different devices such as VR headsets and AR glasses. XR training should address accessibility concerns, by accounting for factors such as learner comfort, thus promoting easy and equitable access to all training personnel.

Instructor training and support: Instructors would require comprehensive training on XR technologies and scenario management to boost their proficiency in operating XR platforms, troubleshooting technical issues and guiding learners through simulations. Instructors would also need continuous support about new features and improvements.

Main integration opportunities

This section explores the integration opportunities offered by extended reality. By identifying key areas where the technology can be seamlessly integrated, organisations can harness its full capabilities to enhance efficiency and drive innovation.

Artificial intelligence (AI) integration: AI can enhance extended reality simulations in border management training by providing dynamic, adaptive scenarios. AI algorithms can analyse learner responses, enabling personalised learning paths and realistic AI-driven interactions, creating more immersive and effective training experiences.

LMS integration: Integration with learning management systems allows seamless incorporation of extended reality modules into existing training programmes. This ensures centralised content management, tracking and analytics, streamlining administrative processes and optimising the overall learning experience.

5G/6G connectivity: Leveraging 5G/6G connectivity enhances XR training with low latency and high bandwidth. This facilitates real-time, high-quality interactions, enabling border management personnel to engage in immersive training scenarios without network hindrances, ensuring a smooth and responsive learning experience.

Metaverse integration: Integrating XR technologies with the metaverse expands training possibilities. Learners can engage in collaborative virtual environments, enhancing team-based scenarios. This integration fosters a more interconnected and immersive learning ecosystem, allowing border management personnel to prepare for real-world challenges in a virtual collaborative space.

Gamification elements: Gamification enhances extended reality training by introducing game-like elements such as rewards, challenges and competition. This not only makes learning more enjoyable but also boosts engagement, motivation and skill retention among border management personnel, transforming training into an interactive and competitive experience.

Biometric and facial recognition for realism: AI-powered biometric and facial recognition can add realism to training scenarios. Integrating these technologies into simulations allows border security personnel to practice identity verification and response strategies in lifelike situations, increasing their preparedness for real-world incidents.

Box 10. Case studies on the adoption of XR technology in education

The US Border Patrol used extended reality to improve learning programmes by incorporating virtual terrain-based training, to increase agents' situational awareness and tracking capabilities. To accomplish this, the US Department of Homeland Security's Science and Technology function developed new training and technologies, including Sign Cutting and Tracking, an augmented reality sand table and night vision. The Sign Cutting and Tracking technology uses augmented reality (AR) and virtual reality (VR) to teach agents how to visually identify human activity along the border across different types of terrain and during different times of day. The AR sand table uses both augmented reality and virtual reality to create a 3D map of any terrain and helps improve learners' situational awareness during operations.

The US Air Force employed augmented reality and mixed reality solutions to provide learners with immersive, 3D training and task performance tools. The tools offered a platform that engaged users in all modes of learning. These included visual instructions, aural cues, synthetic voice narratives and multi-media images, as well as realistic, immersive, full-scale, 3D digital representations of aircraft. The tools made learning considerably more effective. By spending just five-to-ten hours on training with them, learners were able to achieve an 8% increase in scores compared to legacy training systems.

The Air Force Special Operations Command recently adopted a comprehensive XR-based training solution to upskill its fighters. The aim is to be able to conduct repetitive readiness training any place, any time through secure and autonomous learning systems for all command disciplines. The solution addresses four distinct areas: qualification for all aircrew support and mission support personnel; incorporating VR into all flying training units; integration of XR into operational units with an emphasis on mission events; and using AR for aircrew and special tactical teams.

Main challenges

Integration and implementation of XR in the training and education of EBCG personnel will require various gaps and challenges to be addressed.

The requirement for custom training applications: Creating realistic scenarios for border management training using XR solutions that accurately simulate real-world situations and environments is crucial. Interoperability issues must also be addressed, ensuring that these custom applications integrate seamlessly with the diverse array of XR devices available. The investments directed by the EBCG community should be driven by common standards, allowing members of the community to split the overall cost.

Prolonged device use: XR training requires extended use, but current hardware may not be designed for this. It poses a challenge for developing training experiences that aligns with the operational duration of law enforcement operations.

User comfort: Ensuring user comfort is paramount, as individuals may not be familiar with XR technology. Designing training experiences that accommodate a diverse range of physical abilities and preferences is essential for existing systems to avoid disruptions in workflow, and issues with compatibility or collaboration with external suppliers.

Adapting ICT roles: Integrating XR in border security requires ICT teams to evolve beyond routine maintenance and embrace new capabilities, ensuring optimal device functionality, cybersecurity and integration. This role shift is crucial for the implementation of XR technologies in organisational training.

Security and privacy: Protecting sensitive information collected during XR training sessions is imperative, requiring robust data security measures, particularly when dealing with border management processes. This includes personal data. For example, an XR system might collect information on where a person moves, but not on eye movement.

Content standardisation: This is a challenge that arises with different teams or instructors involved in content creation. Consistency in the level of quality and realism across various training modules becomes crucial.

Data security and GDPR compliance: Ensuring that XR training platforms comply with requirements for processing and protection of personal data is crucial. This would involve the implementation of encryption, access controls and secure storage mechanisms to safeguard sensitive information collected during training sessions.

Outcome/future potential

With significant advances on all fronts (comfort, size, computing power, etc.), these technologies are expected to be implemented quickly by companies. XR headset adoption is projected to show a ten-fold

increase, estimated at 105 million units by 2025 (from 11 million in 2021).[29] Having a headset available will be as common as having a notebook on your desk. To optimally use these devices, shorter and more focused learning experiences could be translated into an immersive format. As a result, current learning methods could be enriched by technology that allows new skills to be tested in a practical and immersive manner. The next stage could be haptic feedback technology: providing users with tactile sensations, allowing them to feel virtual objects or environments. In XR, this could simulate the sensation of touching or interacting with virtual objects, adding a new dimension to the immersive experience.

Customisation and personalisation: Extended reality technologies are expected to evolve to include higher degrees of personalisation and customisation. With future evolution, training programmes can be better tailored to individual learning styles, preferences, and skill levels, providing a more adaptive and effective experience.

Integration of artificial intelligence and machine learning: Integrating artificial intelligence (AI) and machine learning (ML) algorithms into extended reality training solutions could help analyse user performance data to dynamically adjust training content and challenges to suit each learner. AI could also help introduce dynamic elements, adaptive scenarios and intelligent virtual entities. This integration will not only make training more engaging but will also facilitate the development of AI-powered virtual mentors and tutors, providing personalised guidance to learners.

Cloud-based XR training: Cloud-based XR training solutions are likely to gain prominence, enabling seamless access to training content and simulations from various devices. This shift towards cloud-based solutions could help enhance collaboration, accessibility, and scalability, allowing organisations to deploy extended reality training programmes more efficiently across different locations.

Regulatory compliance and standards: With the increasing adoption of extended reality, there will likely be a focus on establishing regulatory frameworks and industry standards. Ensuring compliance with data protection regulations, ethical considerations and safety standards will be essential for widespread acceptance and integration of extended reality in training.

Continuous innovation and collaboration: The evolution of extended reality in training will be characterised by a culture of continuous innovation and collaboration. Industry players, technology developers, and educational institutions will need to work together to push the boundaries of XR capabilities in training functions.

Collaboration: Establishing strategic partnerships between border security agencies, technology developers and experts in immersive training will be crucial for the implementation of extended reality in the training curriculum. This collaboration will allow for the sharing of insights and resources to create comprehensive XR training scenarios tailored to the specific challenges faced in border security.

Incentivisation: Implementing incentivisation programmes within border management organisations to recognise and reward personnel actively engaged in extended reality training can aid in successful adoption.

Technology development and accessibility: Investing in the continual development of extended reality technologies tailored to the unique needs of border management training would need to become a key focus for the EBCG community. This includes advances in virtual environments, realistic simulations, and integration with specific equipment unique to border management organisations.

User-centricity: The successful adoption of XR solutions in training will depend on designing training modules with a focus on customised scenarios that closely mimic real-world border security challenges. Personalising the learning experience to match the specific and unique roles and responsibilities of border management personnel can help enhance engagement and knowledge retention.

[29] *https://www.counterpointresearch.com/insights/xr-vrar-headset-shipments-grow-10-times-cross-100-million-units-2025/*

Technology 6: Learning management systems

Learning management systems offer a central hub for training and education.

Training technology overview

Learning management system (LMS) technology empowers organisations to develop and roll out standardised training programmes that encompass up-to-date and consistent learning materials, covering procedures, protocols, legal aspects and more. An LMS enables organisations to centralise their learning resources and enhance the structure of learning content through modules, learning paths, journeys, etc. This enhances workforce training and development, making the learning experience more adaptable and versatile. All employees, agents or members of an organisation's network will be able to engage in continuous learning regardless of their location. LMSs range from specialised 'virtual academies' to comprehensive platforms with expansive libraries of diverse learning content. They also enhance workforce allocation through real-time updates, mobile accessibility and robust analytics. These platforms even provide blended learning options, complemented by a sophisticated learning experience layer, all while retaining their 'academy' essence.

Main benefits

LMSs can be highly useful for workforce training in border management personnel. They provide a centralised platform for creating, managing and delivering training content, allowing for standardisation and consistency in training programmes. They also enable the tracking of training progress and compliance, which is crucial in high-stakes environments such as border control. Additionally, LMSs facilitate the delivery of online training, which can be particularly beneficial for geographically dispersed border management personnel. This allows for cost-effective and efficient training, reducing the need for in-person training and minimising the time personnel spend away from their duties. Furthermore, LMSs can support the ongoing compliance training required for border management personnel, ensuring that all personnel are trained on laws, regulations and security protocols. A technology-enabled academy, backed by a comprehensive LMS that supports cutting-edge learning technologies, holds immense promise for the EBCG community to achieve better interoperability at various levels:

Centralised training hub: An LMS acts as a centralised platform, aggregating all training materials. The LMS software backing up a virtual technology-enabled academy can serve as a central platform for sharing and collaboration. For example, it facilitates the centralised deployment of initiatives like immersive virtual reality training, benefiting all stakeholders. Additionally, it simplifies resource decision-making for L&D teams, ensuring efficient maintenance, updates and tailoring to specific regions and actors. For example, existing content in a Member State can be translated for another rather than creating a new piece of content. Furthermore, the virtual academy can seamlessly integrate with other platforms, content libraries and universities, offering a unified hub for training while accommodating each actor's preferred approach. This approach optimises resource use and fosters a strategic, collaborative training ecosystem.

Management of training and content distribution from a single point: An LMS efficiently automates learning administration tasks, and managing events, certifications and subscriptions.

Cost and time efficiency: A learning management system significantly reduces the time and costs associated with traditional training methods. By eliminating the need for physical materials, travel and extensive instructor time, it maximises resource allocation for optimal organisational efficiency. It also handles reporting and data management from a central point with the ability to integrate with other HR and operational tools.

Real-time progress tracking: The advanced analytics embedded in an LMS allow real-time tracking of individual and collective progress. This feature empowers administrators with actionable insights, facilitating data-driven decision-making for continuous improvement in training programmes.

Integration with emerging technologies: A learning management system offers integration options with emerging technologies like virtual reality, augmented reality and artificial intelligence. This ensures that border management personnel receive training experiences enriched with innovative tools, enhancing their readiness for evolving operational challenges.

Implementation insights

This section explores implementation aspects that could help border management organisations better understand and decide on the deployment of emerging technologies within their learning and development process. It provides key insights into different areas affected when implementing LMS technologies for training and education.

Centralised access for learners: For learners, this technology offers a seamless experience by providing centralised access to diverse learning resources. Whether it's training materials, courses, or interactive modules, everything is conveniently available under one roof. They can effortlessly manage their learning journey, customising it to fit their needs and preferences. With personalised learning paths, users can tailor their educational experience to align with their skill development goals and career aspirations. The platform goes a step further by enhancing user engagement through gamification elements and leveraging cutting-edge technologies like extended reality to make learning immersive and dynamic, all while ensuring a user interface that reflects the organisation's branding, fostering a sense of belonging and familiarity.

Streamlined management for administrators: For administrators, this technology streamlines management tasks by centralising control over training and content distribution. They can efficiently handle inputs across multiple languages, catering to a diverse workforce seamlessly. Integration with HR data enables a deeper understanding of staff skill levels and career pathways, facilitating targeted training initiatives. Automated administrative features significantly reduce the burden of ICT tasks, freeing up time for more strategic endeavours. Moreover, the platform provides robust reporting and data management capabilities, enabling administrators to glean insights and track progress effectively, with the added advantage of seamless integration with other HR and operational tools for a comprehensive organisational overview.

Box 11. Case studies on the adoption of LMS technology in education

The Science and Technology Directorate of the US Department of Homeland Security recently partnered with the US Border Patrol and the Federal Law Enforcement Training Centers to provide a comprehensive solution to improve agents' knowledge and skills.

Network Rail trained 100,000+ learners with Moodle Workplace, increasing efficiency through data-driven decisions, multi-tenancy and integration for custom reports. The implementation provided a scalable, unified learning platform, reducing the technical burden and improving training efficiency.

The future of LMS integration is dynamic, incorporating trends like microlearning, extended reality platforms and seamless integration with advanced technologies. Artificial intelligence and machine learning will be central, transforming LMS into a hub for immersive learning experiences across various tech platforms.

Major integration opportunities

This section explores the integration opportunities offered by LMS technology. By identifying key areas where the technology can be seamlessly integrated, organisations can harness its full capabilities to enhance efficiency and drive innovation.

Artificial intelligence integration: Artificial intelligence augments LMS by personalising learning paths, optimising content delivery and providing data-driven insights. This enhances skill development for border management personnel through tailored and efficient training programmes.

Extended reality and metaverse solutions: Integration with extended reality and the metaverse brings immersive experiences to border management training. Simulations and real-world scenarios in virtual environments enhance practical skills, ensuring personnel are prepared for dynamic challenges.

5G/6G connectivity: 5G/6G integration enhances LMS accessibility, enabling seamless mobile access even in remote border areas. This ensures continuous learning opportunities, overcoming geographical constraints.

Gamification frameworks: Gamification within LMS transforms training into engaging experiences. Points, badges and interactive elements motivate learners, fostering a competitive yet collaborative environment that enhances learning outcomes.

Microlearning and mobile integration: The integration of microlearning modules and mobile-friendly content delivery within the LMS has provided significant benefits. This includes facilitating on-the-go learning, improving accessibility and aligning with the preferences of contemporary learners, who prefer concise, focused learning experiences.

Integration with talent management systems: the integration of talent management platforms with LMSs has streamlined staff development, succession planning and performance management. This integration aligns

learning initiatives with organisational goals, fosters staff growth and optimises talent acquisition and retention strategies.

API: The adoption of API-first approaches in the LMS has simplified integration with various tools, applications and systems. This integration has led to increased flexibility, scalability, and the ability to connect with a diverse range of external applications, fostering a more interconnected and efficient learning ecosystem.

Artificial intelligence: The integration of artificial intelligence features into the LMS has resulted in AI-driven capabilities for customised learning paths, adaptive assessments and intelligent content suggestions. This has led to better user engagement and learning outcomes, and the ability to tailor learning experiences to individual needs.

Augmented and virtual reality: The integration of augmented/virtual reality elements into the LMS has facilitated immersive learning experiences, virtual simulations and interactive training modules. This has proven beneficial by providing realistic training scenarios, increasing engagement and enabling the simulation of complex tasks in a controlled environment.

Blockchain technology: The incorporation of blockchain technology for secure and transparent record-keeping of certifications, credentials and achievements has yielded significant advantages. This implementation ensures the integrity of learner credentials, reduces fraud and establishes a decentralised, tamper-proof verification system.

Data analytics and predictive insights: Integration of advanced analytics tools into the LMS enables the collection and analysis of learner data to derive predictive insights. This allows for proactive identification of learning gaps, personalised learning recommendations and data-driven decision-making for educators and administrators.

Major challenges

Integration and implementation of LMS technology into the training and education of EBCG personnel will require various gaps and challenges to be addressed.

Ensuring content accuracy: The efficacy of an LMS hinges on the quality and relevance of the training content. Ensuring accurate, up-to-date and pertinent learning materials (learning curation) is paramount to the success of LMS in fostering effective staff training and development.

Focusing on user experience: The success of the LMS lies in its user-centric design. Ensuring that the interface is intuitive, well-designed and accessible, and accommodates different learning styles, increases user engagement and promotes an enriching learning experience.

The rapid pace of innovation: The dynamic nature of workforce training requires the LMS to evolve continuously. Regular updates to content, technology integrations and adaptability to emerging learning trends ensure that the LMS remains a forward-looking and effective tool in staff training and development. The LMS needs to be scalable and flexible enough to cope with a variety of present and future needs, both in terms of integrations as in terms of broader learning ecosystems.

Compatibility within the technological organisational ecosystem: An LMS needs to communicate with other technologies used within the organisational tech landscape (HR tools, planning tools, etc.), which often limits scalability and leads to complexities linked to training and onboarding. Adhering to various national, European and international compliance standards is critical for border management operational staff. Implementing an LMS that ensures all staff are trained on the applicable laws, regulations and policies is essential, but it can be challenging to achieve and maintain.

Scalability & security: The integration of an LMS within the EBCG community necessitates addressing the complexities of establishing a common architecture. Balancing agility, centralisation and diverse data protection and data quality rules and regulations poses significant challenges. This challenge is exacerbated in collaboration with third countries, especially in cross-border data transfers. The need to allow external user access for skill bridging can further complicate LMS implementation. Additionally, the intricacies of EU decision-making processes require a strategic approach, emphasising supporting infrastructure, streamlined decision-making and resource management. Change management strategies are vital to mitigate resistance, and the architectural design must balance technological readiness and adherence to standardisation protocols, ensuring a seamless transition from legacy models within the EBCG community.

Stakeholder alignment: Aligning the agendas of various stakeholders poses a complex challenge. Different entities may have divergent interests and priorities, requiring a concerted effort to ensure unified progress. Establishing common ground and fostering collaboration among stakeholders is essential for the successful development and implementation of the academy.

Resource management: Resource availability, both in terms of financial investment and dedicated commitment from various entities, is a critical factor influencing

the academy's development. Managing change and transitioning to a virtual academy with numerous layers requires careful consideration to ensure smooth adoption and integration. This underscores the importance of effective resource management and strategic investments.

Technological readiness and standardisation: The technological readiness of existing systems plays a crucial role in the successful implementation of a technology enabled academy. Standardising and sanitising data and ensuring its long-term maintenance are prerequisites for enabling advanced technologies such as artificial intelligence. Maturity levels across entities may vary, presenting a challenge in establishing common references and benchmarks.

Ethical and legal considerations: The use of LMSs in any professional context, including border management, raises ethical and legal considerations related to data privacy, security and the fair and transparent use of training and assessment data. Organisations using LMS systems for workforce training are required to ensure that they adhere to the highest ethical standards and legal requirements to protect the rights and privacy of their employees and to maintain the integrity of their training and development programmes.

Outcome/future potential

In the coming years, LMS technologies will continue their centralising role, facilitating cost-effective and accessible training for organisations of all sizes. Their crucial function in ensuring compliance with industry regulations through certifications and documentation will remain unchanged. They will also evolve further, leveraging new technologies to support skills-based learning.

Integration with emerging technologies: The evolving LMS industry expects transforming education delivery to align with a dynamic working culture. Future LMSs will prioritise integration, adopt a mobile-first approach and leverage artificial intelligence/machine learning for streamlined administrative tasks. Key trends include user-centric design, personalised learning journeys, micro-learning, integration, a mobile-first approach and data-driven decision-making.

Leveraging advanced analytics: Although LMSs commonly integrate with current business intelligence applications like PowerBI, there is an anticipation that LMSs will incorporate advanced analytics independently to enhance insights into training programmes. This potential enhancement could optimise learning content, pinpoint skills gaps and customise training to address specific needs.

Future learning opportunities: The future of LMS and virtual academies will prominently feature personalisation. Using AI and machine learning, learner data will be analysed to formulate tailor-made learning paths. This will result in personalised content recommendations, assessments and pacing that adjust to the unique needs of each learner, ultimately enhancing engagement and optimising knowledge retention.

Microlearning and bite-sized content: In the era of diminishing attention spans in the digital age, LMSs will increasingly adopt microlearning and bite-sized content. Learners will engage with brief, targeted modules designed to deliver specific learning outcomes. This methodology seamlessly integrates into busy schedules and aligns with contemporary learning preferences.

Gamification and engagement: To enhance learner engagement and motivation, elements of gamification, such as badges, leaderboards and rewards, will be seamlessly integrated. LMS will harness game mechanics to craft interactive and enjoyable learning experiences.

Continued professional development: An LMS will address the needs of lifelong learners and professionals pursuing ongoing development. Micro-credentials, badges and stackable certificates will become more prominent, giving individuals opportunities to augment their skills and advance their career prospects.

Evolving with educational technology, LMSs will remain a crucial element in education, delivering knowledge globally. Institutions and educators adapting to this changing landscape will excel in offering high-quality, engaging learning experiences, preparing learners for success in the digital age.

Cross-technologies: feedback from the Frontex Conference on Future Training Technologies

The following section provides an overview of preliminary feedback from leading training officials in the EBCG community, as well as industry technology experts, on the key technologies researched. It was collected during the Frontex Conference on Future Training Technologies.[30]

Learning management system technology enables a centralised, technology-enabled training hub: A centralised, technology-enabled training hub emerged as a valuable concept for future training in the EBCG community. Participants explored how LMS technology could support it. Notably, about 52% of the addressed attendees ranked LMS as the number one priority for the future of EBCG training. When considering collaborative learning, the largest group (28%) indicated that this particular use case held the most promise.

Emerging training technologies prompt the question of training equipment sharing: During discussions at the conference, augmented/virtual reality emerged as key technologies to investigate for future EBCG training (with about 50% and 32% of respondents, respectively, ranking these as their number one priority). Adopting these technologies requires specific equipment, like simulators and headsets. Therefore, participants explored ways to streamline procuring and sharing training equipment among stakeholders across dispersed locations. This collaborative approach is deemed to not only optimise resource use, but also cultivate a sense of community and mutual support among training centres within the EBCG community.

Emerging training technologies and concepts provide an opportunity for tech-enabled approval processes: Emerging technologies, like UAV deployment through augmented/virtual reality, are associated with complex approval processes, due to the diversity of regulations across different regions and the potential risks associated with UAVs usage. Therefore, technologies that can simplify these processes, ensure compliance and enhance decision-making efficiency are valuable. By leveraging technology-driven solutions, border management authorities can streamline approval workflows, ensuring compliance with regulatory requirements while mitigating operational complexities. The integration of advanced technology makes decision-making more agile and efficient, facilitating smoother border operations.

Trainers need to be empowered to adopt emerging training technologies and concepts: Based on the insights gained from engaging with participants at the conference, the contribution of trainers and instructional designers in adopting technology emerged as paramount. As frontline users of training technology, participants discussed ways to ensure trainers have the right skills to harness the full potential of emerging tools.

Collaboration can be enabled through different technologies: Along with the potential value of technologies in future EBCG training, conference attendees highlighted the critical importance of collaboration. They mentioned that effective collaboration not only enhances the learning experience but can also benefit document and agreement management. Some attendees mentioned they are already successfully sharing resources with partners in the EBCG community.

A pragmatic approach to technology adoption is favoured: Conference attendees favoured a pragmatic approach to adopting emerging and future training technologies and concepts, emphasising the value of incremental progress and experimentation. By piloting new technologies and concepts on a smaller scale, attendees deemed they could assess and identify potential challenges early on and make informed decisions about scaling up.

3.3.2 Summary of the workshop for brainstorming on the main use cases

The objectives of the workshop were to inspire participants with relevant academy cases and facilitate alignment on specific goals for a technology-enabled academy. Other objectives included identifying key components related to learning, collaboration and administration, and collectively pinpointing obstacles and success drivers to conceptualise and define a technology-enabled academy. Annex 2 provides more details on the workshop set-up. The summary below recaps its main outcomes.

[30] The Conference on Future Training Technologies, hosted by Frontex in Krakow on 28-29 February 2024, aimed to establish a long-lasting multi-stakeholder dialogue on future training technologies for the EBCG community; discuss existing best practices; and gauge the current and emerging needs of learners and trainers to ensure an abundant EBCG training landscape. For more information on the Conference, please refer to the report on the Frontex Conference on Future Training Technologies.

Objectives of the technology-enabled academy

The objective of outlining a technology-enabled academy is to inspire Agency stakeholders regarding the potential of existing and emerging technologies and concepts for training and education. Such an academy aims to transform learning in the EBCG community into an integral part of daily work routines, seamlessly incorporating snackable content into the flow of work, and to make skill acquisition more efficient by ensuring that learning materials are readily available anywhere: physical office spaces, field deployments, remote training sites, etc. The incorporation of artificial intelligence at various levels, for example, could further amplify the academy's progressive nature. AI could be not only leveraged for traditional training but extended to skills assessment and addressing specific gaps in expertise. By embracing AI in these capacities, the academy would position itself as an **adaptive and anticipatory learning hub**, ready to evolve alongside the dynamic needs of the EBCG community.

Such a platform would go beyond immediate training requirements; it aspires to be a proactive solution, **anticipating future challenges and developments**. This strategic foresight underscores the academy's commitment to continuous professional development, ensuring that the EBCG community would be well-equipped to navigate evolving operational and technological landscapes. This commitment extends to supporting the EBCG community in establishing, by 2027, an EBCG standing corps of 10,000 people.[31] Among them, 3,000 will be category 1 officers representing Frontex staff, while the remaining 7,000 will be seconded from EU Member States.

Academy components

The overarching goal would be not only to meet current training needs, but also to cultivate a culture of continuous learning and collaboration within the EBCG community. This is also aligned with the Technical and Operational Strategy for European Integrated Border Management, which mentions that "the capabilities needed to control the EU's external borders in an integrated manner should revolve around trained personnel using technical equipment, drawing on the full potential of technological developments and necessary logistic support."[32] By fostering this culture, the academy aspires to create a dynamic environment where individuals are motivated to actively engage in their professional development, contributing to the overall resilience and adaptability of the EBCG in the face of emerging challenges and advances in the field.

The landscape of border management and security is rapidly evolving, driven by the integration of cutting-edge technologies. The following section explores the fusion of emerging technologies — artificial intelligence, immersive experiences enabled by the metaverse and extended reality solutions, gamification principles, 5G/6G connectivity, etc. — within three academy pillars: learning and teaching; collaboration; and education. AI makes learning more efficient by automating administrative tasks and tailoring content recommendations to individual learners, thus providing valuable insights through learning analytics. For example, the incorporation of immersive experiences facilitates hands-on training in virtual environments, overcoming physical constraints and fostering collaboration. Gamification principles promote engagement and motivation, while 5G/6G connectivity enables high-quality streaming and real-time collaboration, creating a globally connected learning community. Furthermore, robust data protection protocols ensure secure personalised learning.

This holistic integration propels the EBCG community into an era of personalised, efficient and globally connected educational experiences, setting a new standard for educational excellence in border management and security based on three pillars:

- Learning & teaching: The integration of cutting-edge technologies like AI and immersive experiences revolutionises learning, creating a dynamic and personalised educational environment for border management personnel, for both learners and trainers.
- Collaboration: Transformative technologies such as learning management systems and high-speed connectivity foster seamless interaction among stakeholders, promoting a culture of shared knowledge and innovation within the border management community, empowering collaboration without boundaries.
- Communication: Through AI-driven analytics and centralised platforms, administrative processes become more efficient and interconnected, facilitating streamlined, fast and relevant communication, and enhancing the overall learning experience for all involved in border management.

[31] Frontex - Single Programming 2022-2024

[32] Frontex – Technical and Operational Strategy for European Integrated Border Management (2019)

Use cases elaborated based on the input collected: learning and teaching

The landscape of border management and security being transformed by the infusion of cutting-edge technologies. This section examines the amalgamation of emerging technologies such as AI; immersive experiences enabled by the metaverse and extended reality solutions; gamification principles; 5G/6G connectivity; and advanced data protection protocols. This comprehensive integration is reshaping the learning and teaching environment within the EBCG, fostering a dynamic, personalised and data-driven educational experience.

Newly integrated into learning tools, AI is an additional layer on the learning ecosystem, enabling higher efficiency across all learning tasks: from content creation to learning administration. It amplifies L&D teams' ability to embrace technological evolution, through automating administrative tasks such as training approvals, or tailoring content recommendations to learners. Artificial intelligence not only addresses unique needs and preferences but also provides invaluable business intelligence through insights into learning analytics. These analytics provide high visibility across various parameters such as individual performance data, in-depth insight into staff engagement levels and tracking and monitoring of knowledge and skill gaps at individual learner levels. This creates a large dataset for driving informed decision-making within the technology=enabled academy.

- **Automated content creation:** AI-driven tools accelerate training material creation, automating processes faster than traditional methods. This involves extracting insights from diverse sources and streamlining development. These algorithms enhance relevance, recommending supplementary materials based on progress and preferences, and fostering quicker, deeper staff understanding.
- **Individualised study curricula:** AI allows the creation and curation of individualised study curriculums tailored to the unique needs and preferences of each learner. Through sophisticated algorithms, artificial intelligence analyses historical data on individual learning patterns, performance metrics and preferences, generating personalised learning based on gaps in skill levels and areas for improvement. This not only optimises the educational experience for each participant, but also ensures that the training recommendations adapt dynamically to their required skills or next job level, fostering a strategically tailored learning environment. Considering the Artificial Intelligence Act, this feature of an AI-enabled digital academy falls under the definition of High-Risk AI Systems pursuant to Article 6(2) and Article 6(3) of Annex III,[33] which will require careful consideration and assessment of these technologies' risk and privacy impact.
- **Personalised learning content recommendations:** In the technology-enabled academy, AI functions as an intelligent guide, not only curating curricula but also tailoring content recommendations to learners. By evaluating the learner's historical interactions, preferences and performance, AI algorithms present content that is not only relevant to the specific needs of the individual but also delivered in a manner that aligns with their preferred learning styles. This personalised content delivery significantly enhances engagement and comprehension, contributing to a more effective learning experience.
- **Analytics for an efficient digital platform:** AI is a transformative force in education, prioritising learning analytics over traditional business intelligence to revolutionise teaching and learning practices. Predictive analytics anticipates learner performance, allowing personalised interventions. Adaptive assessments dynamically tailor evaluations, providing real-time feedback for a customised learning environment. Learning platforms enriched with AI capture detailed engagement insights, aiding nuanced understanding of student participation and driving targeted interventions. AI's real-time data analysis becomes the cornerstone for continuous improvement, scrutinising performance metrics, knowledge gaps and skill deficiencies. Educators gain actionable insights to refine curricula and optimise teaching methodologies, ensuring a dynamic and effective educational experience. This shift places AI at the forefront of learning, reshaping practices and maximising the potential of both educators and learners. This enables learning administrators to manage their content and platform efficiently for overall improved learning outcomes.

Immersive learning takes centre stage with **metaverse and extended reality** solutions through integration and combination of various types of content, and offers a centralised, personalised and collaborative platform within the academy. The content library is enriched with virtual scenarios, accessible both on and off the job, across multiple locations and facilities. Immersive experiences, facilitated by metaverse and extended reality solutions, enable realistic virtual environments for simulations and seamless collaboration. This integration allows border security personnel to refine their skills in a digital realm that mirrors real-world situations, overcoming physical limitations and fostering hands-on team interactions in a digital space without geographical constraints.

[33] https://www.euaiact.com/annex/3

- **Centralised learning hub:** In the technology enabled academy, the metaverse takes centre stage as a centralised, personalised, and collaborative platform. Learners from different geographical locations converge in a shared digital space, creating a hub where educational content is not only centralised, but also tailored to individual needs. The AI factors in the users' preferences, learning styles and collaboration dynamics, leading to an adaptive virtual environment offering a unique user experience and enhancing the overall immersive learning experience.
- **Content library with virtual scenarios:** The content library within the metaverse is a rich repository of virtual scenarios, offering a diverse range of immersive experiences. Accessible both on and off the job, across multiple locations and facilities, this content library allows learners to engage with realistic simulations that mirror real-world situations. These scenarios are designed to challenge and refine the skills of border management personnel, providing hands-on experiences that transcend the limitations of traditional training methods.
- **Real-time collaboration and virtual teaching in digital space:** The immersive experiences facilitated by the metaverse and XR solutions go beyond static content consumption: learners can engage in real-time collaboration within a dynamic digital space. Through avatars and interactive interfaces, border management personnel from different locations can connect and interact seamlessly. It also provides AI-powered virtual assistants within extended reality and metaverse environments, delivering on-the-spot guidance and support. These assistants leverage natural language processing and computer vision to understand user queries, offering information, explanations or additional resources within the immersive learning context. This collaborative approach fosters a sense of teamwork and camaraderie, essential qualities in the field of border management, even in a digital environment.
- **Physical constraints and geographic limitations:** Immersive learning in the metaverse transcends physical limitations and geographical constraints. Regardless of their location, learners can participate in training sessions, simulations and collaborative activities. This flexibility not only accommodates the diverse schedules of border management personnel but also enables the EBCG to tap into a wider pool of expertise and experience. The metaverse becomes a borderless learning space where knowledge is shared without the hindrance of geographical boundaries.

The academy leverages **gamification** principles to increase engagement and intrinsic motivation among learners. A reward and progress system, featuring points or levelling up, acknowledges accomplishments and reinforces positive behaviours. The narrative flow of gamified content, presented in a story-like manner, captures learners' interest, providing a cohesive and immersive learning experience. The adoption of a portioned learning-content approach further enables users to consume information at their own pace, promoting a flexible and adaptive learning environment.

- **Increases engagement and user experience:** Gamification enhances engagement in training, creating immersive experiences that positively influence the training process and underscore the organisation's commitment to staff development. These modules often include real-time feedback mechanisms, allowing trainees to receive instant responses to their actions.
- **Better retention of content:** Gamification, using techniques like repetition, challenges, and rewards, increases trainees' knowledge retention. Gamification incorporates neuroscientific principles in its design, considering how the brain processes information and learns. By aligning training content with cognitive principles, gamification maximises learning efficiency, fostering better information retention.
- **State-of-the-art approach to learning:** Gamification incorporates mobile learning strategies and microlearning modules, catering to the preferences of a mobile-first generation. This approach allows border security personnel to access training content on the go, fostering a culture of continuous learning and adaptability within the organisation.

The integration of **5G** and the anticipated advent of 6G revolutionise online learning experiences by providing high-quality video streaming and enabling seamless real-time collaboration. This transformative capability bridges geographical gaps, making education accessible to a wider audience. The technology also supports the integration of AR/virtual reality solutions into training scenarios, enabling mobile learning that connects UAVs, mobile devices and other hardware for on-the-go simulation exercises. Additionally, the promise of global connectivity at unprecedented speeds with 5G/6G facilitates a borderless exchange of knowledge and expertise, allowing for global access to experts and fostering a truly interconnected learning community.

- **Low latency for responsive control:** Decreased latency enables new scenarios of asset usage, such as responsive unmanned vehicle control. Training exercises involving responsive control benefit from minimal delays, providing learners with realistic simulations and hands-on experiences, enhancing their operational readiness.
- **Maximising IoT sensor usage:** Sensors can be deployed along borders, fences and in terrain to detect breaches, general movement, heat signatures, vibrations, etc. By maximising IoT sensor usage, training programmes can simulate real-world scenarios where sensors detect breaches, monitor movement, identify heat signatures

and capture vibrations. This hands-on experience enhances the practical skills of border management personnel, preparing them for diverse challenges they may encounter in the field.

- **Immersive learning experiences:** The high bandwidth and low latency of 5G and 6G enable immersive technologies like virtual reality (VR) and augmented reality (AR) to be integrated into education. Students can explore complex concepts through immersive simulations, enhancing comprehension and retention.

The technology-enabled academy ensures robust personal data protection protocols and learning profile management, creating a secure and tailored learning experience. Advanced training feedback is instantaneous, creating an ecosystem where both teachers and learners continuously adapt and thrive. Valuable insights derived from learning feedback serve as a robust data source, guiding informed decision-making and contributing to the continuous improvement of the educational experience.

The convergence of AI-driven personalisation, immersive experiences through metaverse and extended reality solutions, gamification principles and 5G/6G connectivity — underpinned by robust data protection protocols and fundamental rights safeguards — is propelling the EBCG community into an era of personalised, efficient, and globally connected educational experiences. This holistic integration not only enhances the skills and adaptability of border management personnel but also creates a collaborative and secure learning ecosystem that sets a new standard for educational excellence in the realm of border management and security.

Use cases elaborated based on the input collected: Collaboration

In the vision of the future academy, the implementation of emerging technologies serves as a catalyst for transcending conventional boundaries and fostering seamless collaboration among diverse stakeholders. This section explores the transformative role played by technologies such as **learning management systems (LMS), 5G/6G** connectivity and **gamification** in enhancing collaboration among learners, trainers, course managers, global expert communities and beyond.

The adoption of **LMS technology** emerges as a cornerstone in enabling collaboration within the future academy. LMS provides a centralised learning platform, acting as a collaborative hub where diverse teams — learners, trainers, course managers and global expert communities — interact seamlessly. Through LMS, the academy becomes a plug-and-play learning environment, allowing for the centralised deployment of learning initiatives that benefit all stakeholders. The collaborative nature of LMS facilitates the sharing of diverse training resources, software, tools and content across the ecosystem, breaking down silos and promoting a culture of shared knowledge.

The integration of **5G** and the anticipated advent of **6G** connectivity mark a transformative leap in the collaborative capabilities of the future academy. High-quality video streaming and seamless real-time collaboration become the norm, transcending geographical constraints. Learners, teachers, team members, course managers and global expert communities can connect effortlessly, fostering real-time knowledge exchange and collaborative problem-solving. The promise of global connectivity at unprecedented speeds facilitates a borderless exchange of expertise, ensuring that collaboration is not hindered by physical distance.

Gamification principles are woven into the fabric of the future academy, amplifying collaboration by fostering social learning and motivation. Badges, earned through collaborative efforts, become symbols of achievement and expertise, making the acquired skills in real-world scenarios more portable. The narrative flow of gamified content, presented in a story-like manner, captures learners' interest, creating a cohesive and immersive learning experience. Gamification elements, such as collaboration incentives and interactive challenges, motivate learners to actively participate, promoting a collaborative mindset that extends beyond traditional boundaries.

- Gamification principles extend beyond the learning environment to the recruitment processes, making them more interactive and personalised. This not only enhances engagement, but also ensures that potential candidates are evaluated in scenarios that mirror the challenges of the border management profession. The gamified approach in training and recruitment fosters collaboration by promoting a culture of interactive learning and shared experiences, creating a cohesive workforce that is aligned with the dynamic needs of border security and management.
- The use of gamification in workforce learning and development can significantly enhance an organisation's branding by projecting an innovative image that resonates with both employees and external stakeholders, reinforcing an employee-centric culture within the organisation, and demonstrating a visible investment in training of their resources.

Emerging technologies facilitate an inclusive learning environment by integrating diverse learning material and approaches from Member States and third countries. This collaborative ecosystem ensures that the academy's

content is enriched with a multitude of perspectives, enhancing innovation and knowledge exchange. Technologies like LMS enable the seamless sharing and adaptation of content between Member States, fostering a collaborative environment where existing resources can be used and customised without the need to create entirely new pieces of content. This adaptability promotes a culture of collaboration and shared resources, contributing to the overall quality and relevance of the educational content.

The adoption of emerging technology enables the transformation of the future digital academy into a centralised platform that can be easily leveraged by stakeholders, accessible and adaptable to individual team needs. It ensures existing content is seamlessly used across Member States, promoting efficient resource sharing, reducing duplication of efforts and enhancing overall effectiveness.

The implementation of emerging technologies transforms collaboration into an integrated reality. Major technologies, such as LMS, 5G/6G connectivity and gamification, create a collaborative ecosystem where stakeholders seamlessly interact and contribute to innovation. This would allow the future academy to remain dynamic, inclusive, and globally connected, enhancing the learning experience for all.

Use cases elaborated based on the input collected: Administration

Administration within border security and management organisations is undergoing a transformative evolution, driven by the intentional integration of emerging technologies. This section explores how automated processes, AI-driven analytics, digitalisation and a commitment to interoperability combine, emphasising their collective impact on shaping an efficient, responsive and connected administrative landscape.

The envisioned transformation places a significant emphasis on interoperability, aligning with various organisations and involving key stakeholders from internal and external EU parties. This commitment extends to the management of trainers/expert pools, fostering a dynamic and responsive team. The academy's dedication to interoperability also encompasses analytics, enabling the integration of insights from diverse sources for a comprehensive understanding of the technology-enabled academy's functioning and impact.

Automated administrative processes within the technology-enabled academy extend beyond routine tasks to encompass certification subscriptions, compliance reporting and scheduling tools. This strategic approach ensures seamless integration with diverse entities, alleviating administrative burdens across the spectrum. AI takes a central role in learning content administration, reducing the manual workload for tagging, categorising and resource management. The dynamic nature of this approach makes the learning process more efficient, allowing users to concentrate on the substantive aspects of their education. AI-driven content administration becomes a valuable ally in maximising the academy's effectiveness, converging accessibility, organisation and efficiency for an optimised and enriched educational experience.

AI's role extends beyond content administration to providing first-line technical support, ensuring an uninterrupted learning experience and establishing links to career paths. Such capabilities are classified as 'high-risk' technology under the EU AI act, highlighting the need for a comprehensive impact assessment before adoption. This multifaceted application of AI enables organisations to focus more on automation and limit investment in administrative processes, leading to the holistic development of learners and an efficient management process for the border security and management organisation.

The centralised nature of LMS technology serves as a consolidated hub, facilitating prompt implementation of changes to ensure that learning content remains current and relevant. This agility in content management minimises disruptions and makes educational offerings more adaptable. The centralised LMS approach simplifies administrative tasks associated with resource management, enabling L&D teams to efficiently allocate and monitor resources, track learner progress and assess the effectiveness of different learning modules. The LMS becomes a comprehensive tool generating valuable insights through analytics, aiding L&D teams in making data-driven decisions to enhance the overall quality and effectiveness of the learning experience.

The integrated vision for transforming the administration of border management related training is one that harmonises automated processes, interoperability and advanced AI applications. By aligning with various organisations, engaging key stakeholders, and committing to interoperability, administration becomes not only efficient but also seamlessly connected. The strategic use of AI in content administration, technical support and career-path linkage further amplifies the positive impact, fostering an environment where technology propels administrative efficiency and enriches the learning experience within the border security and management organisation. This comprehensive approach positions the technology-enabled academy at the forefront of the digital evolution in the broader context of the EU border control ecosystem.

Hurdles and drivers for success

The following hurdles for the establishment and adoption of a technology-enabled academy have emerged.

- **Strategic decision-making**: Strategic decision-making by the EBCG community and stakeholders presents a distinctive challenge tied to the intricacies of EU decision-making processes and the involvement of Frontex stakeholders. The complexity lies in delineating decision-making responsibilities between Member States, Agencies/organisations and Frontex itself. Establishing effective infrastructure and processes becomes crucial in articulating and streamlining these decisions. It is key to ensure seamless decision-making aligned with the overall objectives of the academy, all while efficiently managing available resources.
- **Mindset shift**: A specific challenge lies in cultivating a cultural and mindset shift within the EBCG community towards the integrated technology-enabled academy. This transformation requires a deliberate change in individuals' thinking and operational approaches, emphasising a proactive embrace of technological advances, continuous learning and a collaborative mindset. Effectively addressing resistance, especially from experienced professionals, calls for targeted interventions in change management strategies to promote an ideal mindset characterised by openness to innovation, adaptability and a commitment to lifelong learning.
- **Interoperability and communication**: Effective communication, both technologically and operationally, is crucial for the smooth functioning of the academy. Achieving interoperability, where different Member States and parties can seamlessly communicate and work together, complying with each's specific context and regulations, requires standardised protocols. This further heightens the importance of mitigating communication barriers.
- **Stakeholder alignment**: Aligning the agendas of various stakeholders poses a complex challenge. Different entities may have divergent interests and priorities, requiring a concerted effort to ensure unified progress. Establishing common ground and fostering collaboration among stakeholders is essential for the successful development and implementation of the academy.
- **Resource management**: Resource availability, in terms of both financial investment and dedicated commitment from various entities, is a critical factor influencing the academy's development. Managing change and transitioning to a virtual academy with numerous layers requires careful consideration to ensure smooth adoption and integration. This underscores the importance of effective resource management and strategic investments.
- **Comprehensive staff training**: Training staff in artificial intelligence, extended reality, learning management systems, 5G/6G, gamification and metaverse technology presents a challenge, as the diverse range of technologies demands versatile skill sets. Staff proficiency requires tailored programmes that cover both technical and strategic aspects across technologies. This demands a holistic approach for aligning staff training with academy objectives for effective technology use.
- **Regulatory challenges:** There are often delays in adopting advanced technologies due to the intricate process of conducting privacy impact assessments and formulating adoption strategies at border management organisations. The challenge lies in harmonising these assessments and strategies amid varying national approaches and legislation across Member States. Navigating and aligning with diverse privacy standards and legal nuances within the border management organisation's jurisdiction demands a nuanced, adaptable approach.
- **Technological readiness and standardisation**: The technological readiness of existing systems plays a crucial role in the successful implementation of a technology-enabled academy. Standardising data and ensuring its long-term maintenance are prerequisites for enabling advanced technologies such as AI. Maturity levels may vary across entities, presenting a challenge in establishing common references and benchmarks.
- **Equipment availability**: Ensuring the availability of equipment across multiple centres is imperative to enable effective training delivery. This underscores the importance of establishing foundational elements for the smooth functioning of the academy.

Establishing a technology-enabled academy faces challenges such as strategic decision-making complexities, cultural shifts, effective communication, stakeholder alignment, resource management, technological readiness, logistical and operational hurdles, cultural adoption resistance and ensuring equipment availability. These underscore the multifaceted nature of navigating the path towards successful integration of a technology-driven learning environment.

The following drivers for the successful establishment and adoption of a technology enabled academy have emerged.

- **Technology taxonomy for training and education**: The establishment of a technology taxonomy contributes to the systematic organisation of knowledge related to existing and emerging education and training technologies. This taxonomy creates a structured framework that categorises key research topics,

technologies and concepts relevant to the EBCG community's training areas of interest. By mapping these elements logically, by category, the taxonomy promotes a shared understanding and homogeneity of interpretation in discussions of educational technologies. This standardised classification facilitates communication and collaboration among stakeholders, ensuring a common language for discussing and implementing technology-enabled solutions within the academy.

- **Strategic collaboration**: One pivotal driver lies in strategic thinking towards digitalisation, emphasising collaboration not only at the policy level, but also fostering cooperation among entities and individuals. Sponsorship is crucial to create, adopt and operate the academy.
- **Regulatory frameworks and business needs**: Aligning with EU regulations, particularly those promoting digitalisation, becomes a driver for success. Recognising key moments and needs for change, such as the growing staff population and transforming staff capabilities, positions the academy as an adaptable and responsive solution.
- **Continuous professional development and incentivisation**: By offering digital career paths that provide recognition and visibility, the academy becomes a catalyst for individual growth. Learners steering their own learning journeys; personalised learning; and clear links to career paths all create an environment conducive to continuous learning.
- **Technology development and accessibility**: The academy's integration into daily routines (ensuring accessibility within the workflow) and user-friendly interfaces enhance its practical utility.
- **User-centric learning experience**: A user-centric approach, where learners steer their own learning journeys, lies at the core of success. Offering a personalised learning experience through self-paced learning, micro-learning and practical tools in the field ensures that the academy aligns seamlessly with the needs of the trainees.
- **Ambassadorship**: The presence of pioneers, change champions and ambassadors (internal and external) serves as a driving force. These individuals play a crucial role in advocating for the academy, fostering enthusiasm and navigating change with a positive impact.

The success of the technology-enabled academy centres on a strategic, collaborative approach, technological advances, user-centric learning experiences and adaptability to regulatory frameworks and business needs. Continuous professional development, efficient resource use and the presence of ambassadors contribute to a holistic environment that fosters growth, learning and sustained success.

3.3.3 High-level use cases identified for the EBCG community

This section outlines a proposed architecture for an EBCG technology-enabled academy. The design is informed by industry standards, desk research and discussions with the EBCG community at the Frontex Conference on Future Training Technologies. It is important to note that neither the academy nor its architecture are definitively established at this stage. The intent is to offer a glimpse into what the academy could become and create a shared understanding to stimulate and structure discussions.

In this section each aspect of the design is elaborated, including role-based access, interconnected building blocks, underlying infrastructure and the involvement of internal and external users. Additionally, the section examines how these components could synergise with each other. As such, this section supports understanding of the various use cases explored and further detailed in this report. At a later stage, concrete use cases will be linked to this provisional architecture.

Architecture of a technology-enabled academy

For an overview of the various elements and sub-elements, and how they are positioned in the architecture, please refer to the image at the end of this section.

Role-based access rights: Access rights in a digital academy are usually role-based, determining what resources or sections of the platform a user can access. This can include access to the training portal, research materials, knowledgebase, engagement hub, communication channels, content library and analytics centre. This system makes the platform more secure and efficient by ensuring users can only access resources or sections relevant to their roles.

Interconnected building blocks: The architecture of a digital academy includes several interconnected building blocks, described below.

Learning, teaching, and research

- **Facilities and work order management:** managing rooms, scheduling activities, supporting teacher training and initiating academic development programmes.
- **Learning systems:** a digital library, providing a vast range of resources for learners.
- **Curriculum management:** analysing and reviewing learning needs, ensuring alignment with educational standards.
- **Course management:** generating and curating educational content to meet learners' needs.
- **Exam and certification management:** assessing the impact of learning, grading and providing accreditation.
- **Feedback and evaluation management:** maintaining regulatory compliance and ensures continuous improvement based on feedback.
- R**esearch management:** publishing research facilitating effective knowledge sharing; nurturing a culture of interdisciplinary collaboration among researchers and learners; and continuously improving teaching methodologies to enhance research outcomes.
- **Content management:** the entire lifecycle of content — from creation, curation, review and approval, to distribution, integration and localisation.
- **Learning resource administration:** inventory, storage, organisation and archiving of all learning content.

Administration

- **Document management:** from the creation, storage and version control of documents to their retrieval, sharing and collaborative editing. Also document review, approval, security, archiving and backups.
- **Contract management:** contract renewals, managing vendor relationships, controlling costs, managing risks and overseeing procurement.
- **User administration management:** user enrolment, application processing and HR management tasks, including payroll and management of staff/external experts.

Communication & collaboration

- **Student and alumni relationship management** are key, focusing on programmes for alumni engagement, provision of career services and initiatives for mentorship and coaching.
- **Communication and community management** forms another significant element, taking on tasks such as planning and coordinating events, moderating activities and providing support to members of the community.
- **Virtual collaboration management** plays a critical role in this digital age, facilitating hybrid and online collaboration, enabling real-time interaction and creating collaborative learning spaces.
- **Marketing management** is crucial in shaping the academy's image, managing the brand, overseeing advertising and promotional activities and handling public relations.

Governance & performance

- **Enterprise or academy architecture management:** efficient functioning of the academy's structure.
- **Business intelligence:** reporting and dashboards, learning and performance analytics, search engine capabilities and metadata analytics.
- **Quality management:** upholding quality standards, promoting continuous improvement and quality control, ensuring compliance and accreditation, managing risks and overseeing stakeholder relations.
- **Project, programme, and portfolio management:** covering the efficient allocation of resources and project scope management.

Underlying infrastructure

- **Integration management:** seamless communication between software systems and tools, enhancing operational efficiency.
- **Security management:** safeguarding sensitive data through authentication, encryption and access controls.
- **Database management:** organising and retrieving data efficiently, supporting essential functions such as student records and course scheduling.

Users

Within the architecture of a technology-enabled EBCG academy, a diverse array of internal and external users plays an integral role in using, managing, developing and maintaining the institution's capabilities. Positioned alongside the interconnected building blocks (as detailed in sub-section 3.3), these users have varying degrees of engagement with the academy's functionalities. It's essential to recognise that they represent user groups, each comprising numerous individuals, profiles, roles, interests and needs. Internal users encompass individuals directly associated with the academy, including students, alumni and staff members.

Students engage with the academy's learning resources, participate in courses and various academic services. Alumni maintain connections with the academy, accessing resources, networking opportunities and continuing education offerings. Staff and faculty contribute to

the academic community through teaching, research, administrative tasks and student support services.

External users include government agencies, international organisations, media, the public, suppliers and service providers.

- Government entities may interact with the academy for regulatory compliance, funding or collaborative educational initiatives. International organisations may exchange know-how and best practices or conduct joint learning initiatives.
- Media may seek information, partnerships, or coverage of academy-related events. The public may engage with the academy through outreach programmes, public events or access to educational resources.
- Suppliers and service providers offer goods and services to support the academy's operations, ranging from technology infrastructure to catering services.

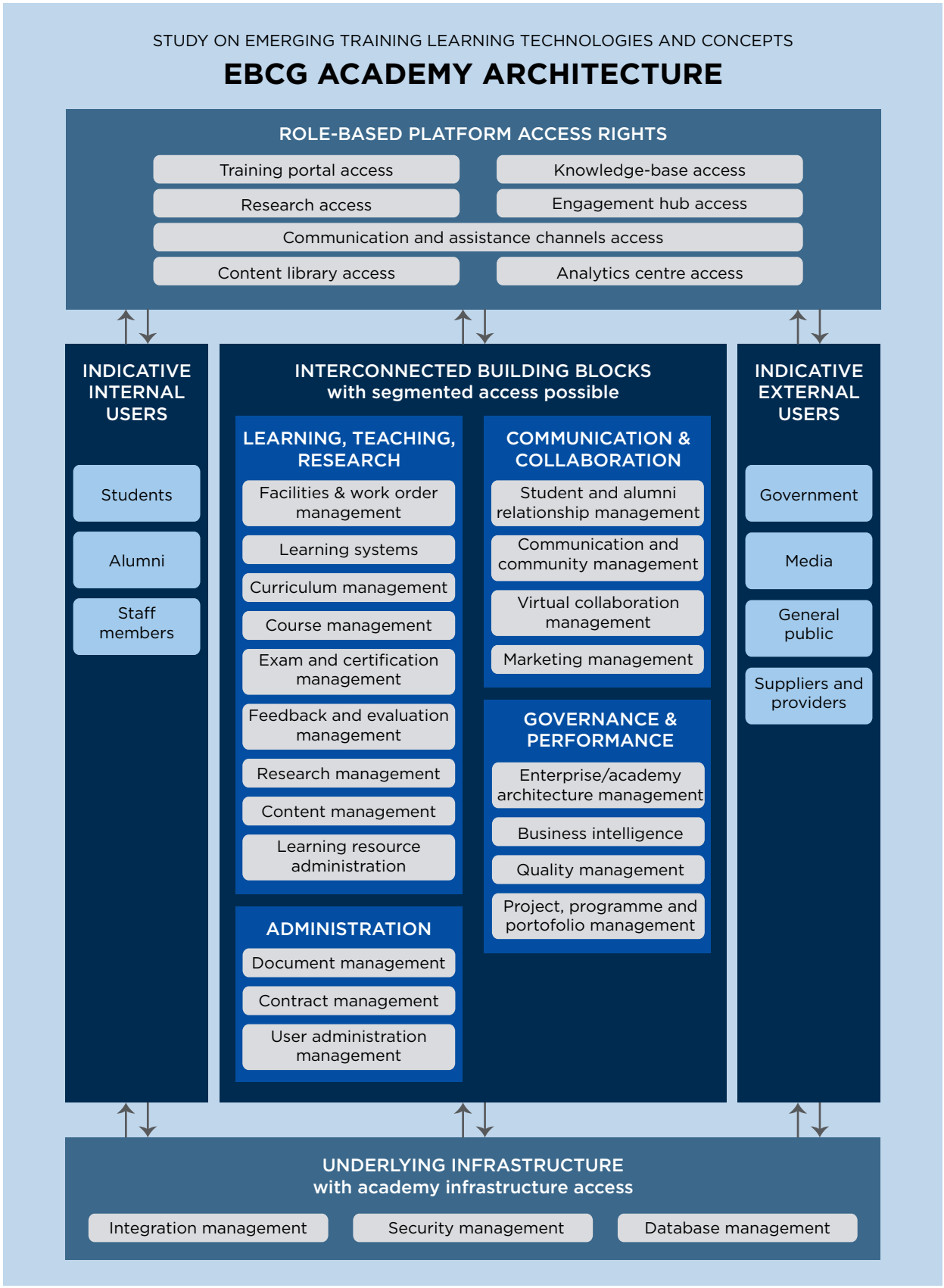
Additionally, there is a final distinct category of users composed of operational entities that would like to monitor the performance of their operational staff and consult the training schedule for planning purposes, for example. These users play a crucial role in ensuring the smooth functioning of day-to-day operations within their respective entity.

Synergies among the elements of a technology-enabled academy

This section has outlined the various elements of the EBCG academy individually, portraying the academy as a consolidation of elements. It is important to note synergies within the EBCG academy architecture, as it demonstrates the holistic nature and underscores the interconnectedness of its various elements. The discussion of synergies within the EBCG academy architecture relates to interoperability, namely the ability of different systems or components to communicate, exchange data and work together seamlessly. In the context of the academy, highlighting synergies underscores how different building blocks and functionalities interact to create a cohesive ecosystem, facilitating interoperability. Synergies manifest distinctly from both learner and administrative perspectives, yet they are not exhaustive and extend to other facets of the academy's operation.

From the learner's standpoint, the interconnected building blocks harmonise to provide a cohesive learning experience. For instance, a student's journey begins with streamlined access to courses through the curriculum management block. This access extends to a comprehensive digital library, enabling learners to supplement their studies with extensive resources. As students engage with course materials, the interconnectedness of systems ensures seamless navigation and interaction, enhancing the overall learning experience. Furthermore, the analytics centre empowers learners by providing insights into their performance and progress, facilitating personalised learning paths and continuous improvement. Thus, from a learner perspective, the synergy among these components amplifies accessibility, engagement and professional goal achievement.

From the administrative viewpoint, synergies within the architecture bolster operational efficiency and strategic decision-making. Access to analytics equips administrators with valuable insights into learner behaviours and performance trends, enabling proactive interventions and targeted support mechanisms. Administrative personnel can leverage this capability to disseminate information, promote initiatives and foster partnerships, strengthening the academy's reputation and outreach efforts. Additionally, the integration of marketing strategies and engagement hubs empowers administrators to tailor communications, aligning messaging with the needs and preferences of diverse user groups. This strategic alignment ensures cohesive branding, amplifying the academy's visibility and impact. In essence, the synergies among administrative components streamline operations, fortify stakeholder relations and drive strategic growth initiatives.


STUDY ON EMERGING TRAINING LEARNING TECHNOLOGIES AND CONCEPTS
EBCG ACADEMY ARCHITECTURE
ROLE-BASED PLATFORM ACCESS RIGHTS
Training portal access
Knowledge-base access
Research access
Engagement hub access
Communication and assistance channels access
Content library access
Analytics centre access
INDICATIVE INTERNAL USERS
Students
Alumni
Staff members
INTERCONNECTED BUILDING BLOCKS with segmented access possible
LEARNING, TEACHING, RESEARCH
Facilities & work order management
Learning systems
Curriculum management
Course management
Exam and certification management
Feedback and evaluation management
Research management
Content management
Learning resource administration
COMMUNICATION & COLLABORATION
Student and alumni relationship management
Communication and community management
Virtual collaboration management
Marketing management
GOVERNANCE & PERFORMANCE
Enterprise/academy architecture management
Business intelligence
Quality management
Project, programme and portofolio management
ADMINISTRATION
Document management
Contract management
User administration management
INDICATIVE EXTERNAL USERS
Government
Media
General public
Suppliers and providers
UNDERLYING INFRASTRUCTURE with academy infrastructure access
Integration management
Security management
Database management

4. High-level technology taxonomy for training and education

4.1 Introduction

This chapter translates the research insights gained in the preceding chapters into a practical taxonomy of training technologies for EBCG. The objective is to enable a common understanding of training technology concepts, use cases and their relationship to one another across different stakeholder groups, facilitating conversations and decision-making on training and development initiatives. In addition, the taxonomy aims to guide the understanding of elements in Frontex's ongoing research on emerging training technologies for EBCG. It is a holistic framework, including access management; academia and industry partnerships; key management areas such as course and curriculum; student and alumni relationships; and content. The framework links these elements to the highlighted use cases and shows how they are being implemented across the industry.

4.2 Methods

The research involved comprehensive secondary desk research, which allowed us to gather and analyse reports, case studies, academic papers and other relevant literature. Information was gathered and mapped based on its relevance for the EBCG community, including the most relevant points. To extract the most relevant use cases and align the final taxonomy structure, the research also leveraged publicly available sources (open web, vendor websites, blogs, whitepapers, etc.), our internal tools and databases and third-party research tools (Factiva, Gartner, EMIS, WSJ, Deloitte Marketplace intelligence reports, etc.).

4.3 Results

4.3.1 High-level groupings and sub-categories identified and listed

The taxonomy is organised into distinct categories that directly correspond to the design of the EBCG academy architecture. Each category is broken down into sub-categories aligned with the research on use cases and emerging technologies.

Sources of information

EBCG academy architecture: This outlines the structural design of a potential EBCG academy. The following aspects, each comprising more granular design elements, are included in the design.

- Interconnected building blocks: learning, teaching, research; administration; communication and collaboration; and governance and performance.
- Underlying infrastructure: integration management, security management and database management.
- Role-based access: access to the training portal, research materials, knowledgebase, engagement hub, communication channels, content library and analytics centre.
- Involvement of internal and external users: students, alumni, staff members, government agencies, international organisations, media, the public, suppliers and service providers.

Study on emerging training technologies for EBCG: This is a comprehensive exploration of emerging technologies and their potential applications in the training and development domain. The study includes the following emerging technologies:

- Artificial intelligence (AI)
- Extended reality: augmented reality (AR), virtual reality (VR), mixed reality (MR)
- Metaverse
- Learning management system (LMS)
- Gamification
- 5G/6G

Study on use cases of emerging training technologies for EBCG: This study researched and shortlisted the most relevant global law enforcement and defence initiatives and projects that are transforming education and training systems through process redesign and digitalisation, using emerging technologies. We provide insights along with initial conclusions regarding their significance to the EBCG community.

- United States Border Patrol: leveraging mixed reality for training
- Network Rail: eLearning solution powered by enterprise LMS technology
- Interpol: building a digital academy powered by metaverse technology
- Deloitte: digital academy powered by enterprise LMS technology
- BSH: web-based virtual campus powered by metaverse technology

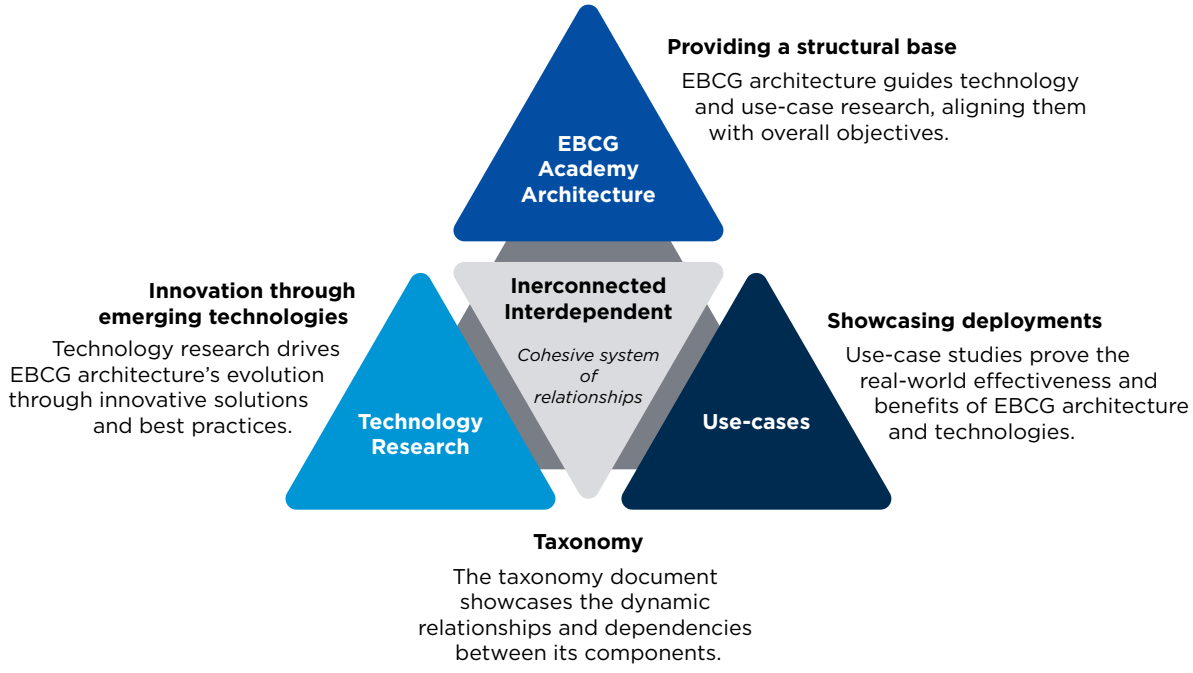


Figure 1. Conceptualisation of the taxonomy for EBCG training technologies

Interrelated character

The taxonomy interrelates different sources of information, creating a unified network of relationships. The study on technology, the EBCG academy architecture and the study on use cases inform one another. Concretely, the relations among these sources of information can be understood as follows:

- By identifying innovative solutions and best practices, the technology study informs the development and evolution of the EBCG academy architecture. This link ensures that the architecture is fit to embed the latest technological advancements, enhancing effectiveness and efficiency. For example, the study found that AI will

be a key training technology for EBCG, but regulations and ethical considerations need to be addressed. This informed the decision to include quality management (referring to issues including compliance and managing risk) in the EBCG academy architecture.

- The study on use cases validates the technology research by demonstrating practical applications and benefits in real-world scenarios. It provides empirical evidence of the technologies' effectiveness, facilitating their adoption in the EBCG architecture. For example, the US Border Patrol use cases validate the study outcome that mixed reality is advantageous to train border guards.
- The EBCG academy architecture provides the framework for use case research. This ensures that use cases are relevant and applicable to the EBCG context, and provide practical examples for the academy, enhancing the learning experience by connecting theory with practice. Visualising these connections as a triangle emphasises the interrelated nature of the taxonomy, highlighting the dynamic interdependence between its components. This holistic approach enables comprehensive analysis, planning, discussion and implementation within the context of EBCG initiatives.

EXISTING TAXONOMIES

Existing taxonomies in the field of learning and training (technology) provide a solid baseline for establishing a training technology taxonomy for the EBCG community. This section explores their benefits and potential drawbacks.

Consistency: One benefit of employing established taxonomies lies in promoting consistency across educational practices and initiatives within the EBCG community. This not only enhances the coherence and continuity of educational programmes, but also facilitates communication and collaboration among educators, administrators and other stakeholders. Additionally, consistency helps ensure that learning outcomes are transparent and measurable, enabling stakeholders to assess the effectiveness of teaching and learning initiatives and make informed decisions to improve educational quality and outcomes over time.

- **Validity and reliability:** Bolstered by their grounding in extensive research and theoretical foundations, the taxonomies have been rigorously developed and validated. Empirical studies and scholarly discourse lend credibility to their utility as frameworks for informing instructional decisions. Users can trust the robustness of these taxonomies, knowing that they have been tested and refined over time to accurately capture the complexities of teaching and learning processes. By relying on such credible frameworks, users can make informed decisions supported by evidence-based practices, ultimately improving their approaches.
- **Oversimplification and rigidity:** These taxonomies may inadvertently oversimplify the complexity of the learning and training (technology) landscape by categorising information into discrete categories or domains, potentially leading to a reductionist view. Moreover, the perceived rigidity of existing taxonomies could limit creativity and innovation, hindering the user's ability to adapt to diverse learning needs and contexts.
- **Applicability:** Existing taxonomies often lack consideration for context-specific standards, particularly those relevant to the EBCG community. These generic taxonomies may have been developed without accounting for the requirements, challenges and operational contexts specific to the EBCG community (e.g. the EBCG academy architecture). As a result, they may fail to capture all the intricacies of technology adoption and implementation within EBCG. Regulatory frameworks, security concerns, operational requirements and other factors paramount in the EBCG context may not be adequately addressed by generic taxonomies.

Taken together, existing taxonomies offer valuable guidance and structure for developing a taxonomy around emerging training technologies for EBCG, but should be approached critically, acknowledging their limitations and potential drawbacks. Hence, the taxonomy was developed in close consultation with a series of experts in border and coast guard control; learning and development; and training technology, as well as key stakeholders involved in the establishment of an EBCG academy and its use cases.

4.3.2 High-level technology taxonomy for training and education

This section provides a systematic description of the objects, concepts and information within the taxonomy for training technologies, following the structure outlined in the EBCG academy architecture. Beginning with this architecture provides clarity as a comprehensive source of information. It is worth noting that the study on emerging training technologies (and their use cases) could also serve as a starting point.

Category 1: Interconnected building blocks

LEARNING, TEACHING AND RESEARCH		
Facilities and work order management	Learning systems	Curriculum management
Course management	Exam and certificate management	Feedback and evaluation management
Research management	Content management	Learning resource administration
COMMNUCATION AND COLLABORATION		
Student and alumni relationship management	Communication and community management	
Virtual collaboration management	Marketing management	
ADMINISTRATION		
Document management	Contract management	User administration management
GOVERNANCE AND PERFORMANCE		
Enterprise/academy architecture management	Business intelligence	
Quality management	Project, programme and portfolio management	

Facilities and work order management

EBCG academy architecture component: This ensures the management and maintenance of physical and digital facilities. In an academy, it could include the management of various digital resources, platforms and technologies.

Use case: The management of a virtual classroom setup in an academy is a practical example. Here, administrators ensure all virtual resources are available and functional for smooth learning experiences.

The implementation of the LMS-enabled eLearning solution by Deloitte provided significant flexibility for the course administrators to manage multiple learning initiatives through the platform, enabling efficient management of faculty and training initiatives across the organisation.

Technology: AI technology can help predict the need for maintenance and help use resources efficiently. 5G/6G technologies can support real-time resource tracking and monitoring.

Product/practical impact: This component involves setting up a robust system for tracking and managing resources, training staff to handle them and setting up protocols for their use.

Learning systems

EBCG academy architecture component: Learning systems form the backbone of the academy, providing the platforms and technologies through which learning content is delivered.

Use case: LMS technology that delivers online courses, tracks students' progress and manages student data.

The 'Major Programmes' solution deployed by Deloitte through enterprise LMS technology provided personalised learning modules for users; microlearning content enabling an easy learning journey; and other resources to support the user learning process, such as gamified modules.

Technology: AI and LMS technology are crucial here, providing personalised learning paths and tracking student progress. 5G/6G technologies ensure seamless delivery of learning materials.

Product/practical impact: Implementation would require the selection and setup of suitable LMS technology, integration with other systems in the academy and training for both educators and students on how to use the system.

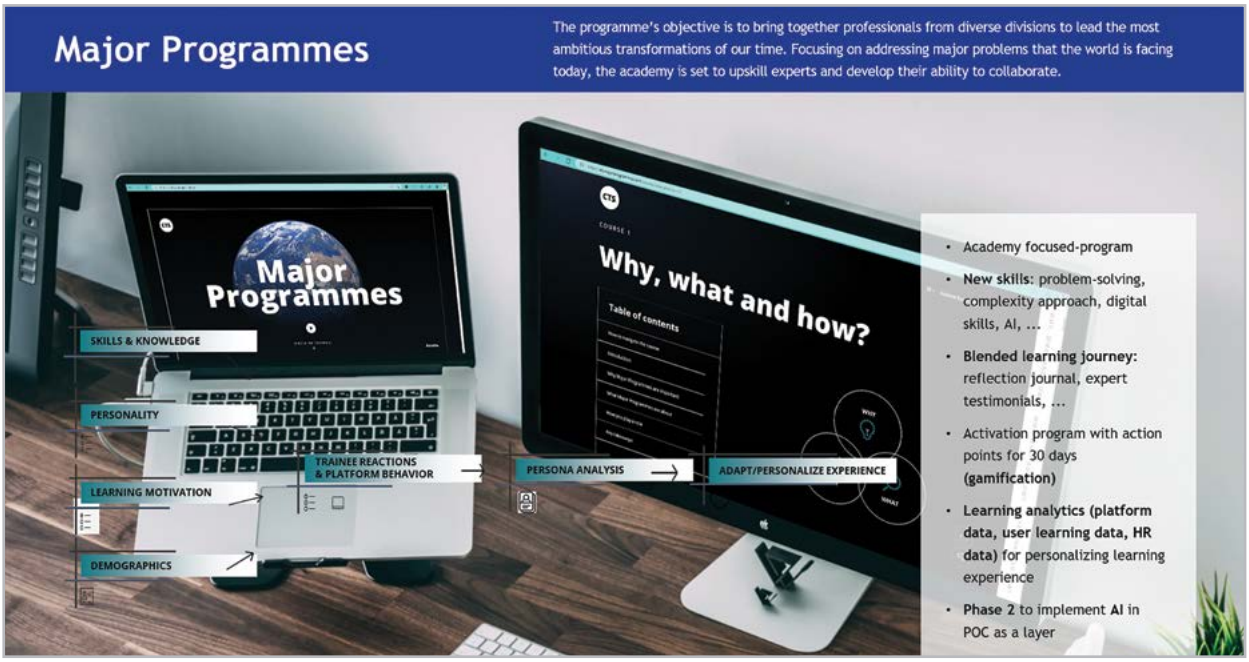


Figure 2. Visualisation of the use case at Deloitte

Curriculum management

EBCG academy architecture component: This involves managing the lifecycle of the Common Core Curriculum (CCC), the educational framework designed by Frontex and adopted by all Member States and some third countries. It also includes developing and managing various course curricula aligned with the CCC, encompassing the sequence of courses, learning objectives and assessment methods.

Use case: A course-planning tool that helps in structuring the curriculum, defining course prerequisites and outlining learning outcomes.

The implementation of 'Major Programmes' by Deloitte through enterprise LMS offered flexibility for the administrator group to understand user learning gaps, improve engagement with users and update learning content aligned with organisational requirements.

Technology: AI technology can assist in course curriculum and CCC management by analysing student data and providing insights into courses; identifying trends and gaps in learning; and suggesting improvements to the curriculum structure. The use of LMS technology can also assist in organising and delivering the curriculum by providing a centralised platform for managing course content, tracking curriculum updates, ensuring consistency and facilitating the alignment of national education programmes with the CCC.

Product/practical impact: This would involve detailed planning of the CCC and course curriculum, defining learning objectives and setting up systems for managing and delivering the curriculum.

Course management

EBCG academy architecture component: Course management involves the creation, delivery and evaluation of individual courses within the course curriculum aligned with the CCC.

Use case: An online course creation tool that helps educators design, deliver and assess courses.

The web-based metaverse virtual campus deployed by BSH leveraged the expertise of a third-party content development firm to create learning content using AI, making it efficient and leading to significant cost savings in the long run.

Technology: learning management systems are crucial for course management, and AI can assist in personalising course delivery based on student needs.

Product/practical impact: Setting up this component would involve training educators to create and manage online courses, setting up systems for course delivery and evaluation, and continuous updating of courses based on feedback, evaluation and changes to the course curriculum aligned with the CCC.

Exam and certificate management

EBCG academy architecture component: This entails the management of assessments and completion certificates, providing students with recognition for their achievements.

Use case: A platform that delivers secure online exams, automatically grades them and generates digital certificates.

The 'Major Programmes' solution deployed using enterprise LMS technology by Deloitte enabled administrators to evaluate users' learning journey and award certificates at the end of the course based on the fulfilment of pre-defined parameters and standards.

Technology: AI technology can provide analytics and insights on certificate evolution in groups of students. It can also assist in grading assessments and identifying areas of improvement. Learning management systems can enable the management of exam delivery and issuance of certificates.

Product/practical impact: This involves establishing a secure and reliable system for online assessments, creating protocols for issuing certificates and ensuring that they are acknowledged and accepted for their validity and credibility by other authorities, organisations and institutions. Training certificates are official documents that validate the completion of a specific course or training programme and signify that the recipient has acquired certain skills or knowledge. Unlike exams, which are assessments meant to test a candidate's understanding and proficiency at a particular moment, training certificates represent the culmination of an entire learning process. Managing these certificates involves several key steps: issuing them through a verified and tamper-proof digital platform; maintaining a secure database to track and store issued certificates; and establishing clear guidelines for renewal or continuing education requirements.

Feedback and evaluation management

EBCG academy architecture component: This enables the collection and analysis of feedback from students and educators, feeding into the quality assurance process and aiding in the continuous improvement of the academy.

Use case: An online survey tool that collects student feedback at the end of each course, and analytical tools that analyse this feedback to generate insights.

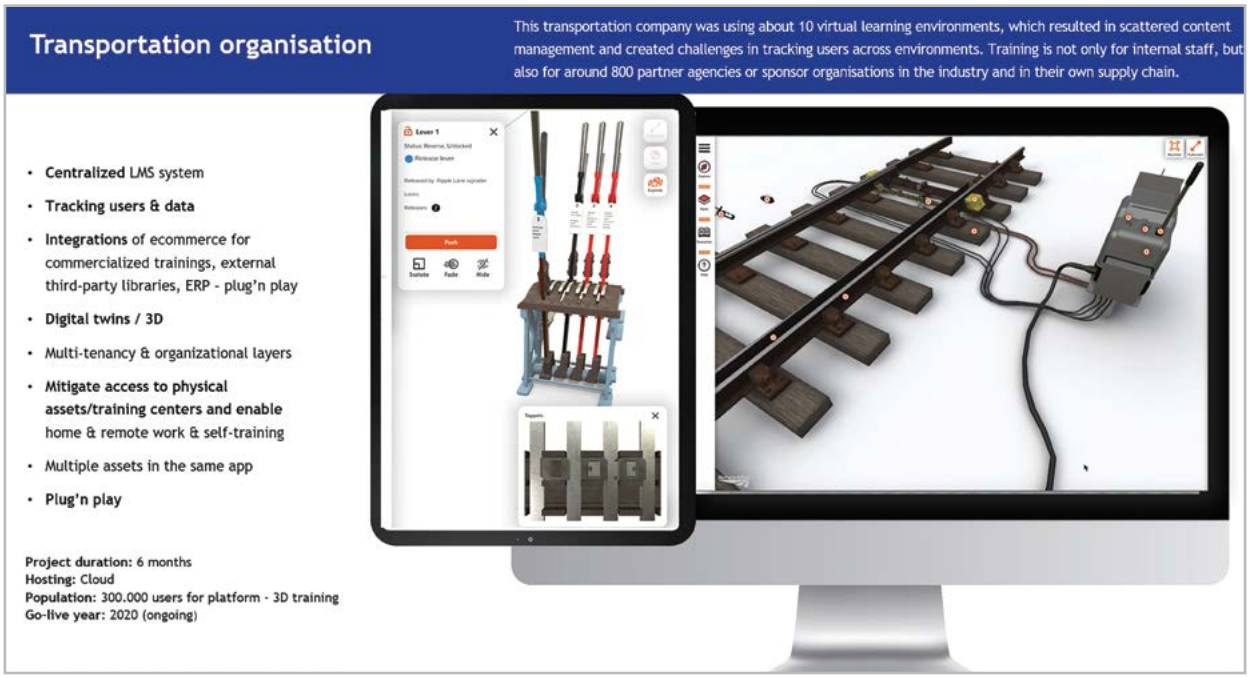


Figure 3. Visualisation of the use case at Network Rail

The eLearning solution developed by Network Rail using Moodle LMS technology offered increased visibility on parameters such as users accessing the system and courses being taken, leading to improved regulatory compliance, and updates based on user feedback.

Technology: LMS technology often comes with built-in analytics tools that can help analyse feedback data effectively. Moreover, it can help implement protocols for using feedback to make the academy more effective. It can also provide features such as discussion forums, surveys and assessment tools that enable continuous feedback loops between instructors and students, allowing for timely adjustments to the curriculum and teaching methods. AI technology can analyse feedback and generate insights, helping improve the learning experience. 5G/6G technologies can support real-time feedback systems.

Product/practical impact: This entails establishing comprehensive ICT tools for collecting and analysing feedback. It also requires training staff to use these systems and establishing protocols for using feedback to improve effectiveness. Furthermore, it helps ensure compliance with existing ICT frameworks, and robust protocols to safeguard privacy and protect personal data.

Research management

EBCG academy architecture component: This involves the management of research activities within the academy, including the collection, analysis and dissemination of research findings.

Use case: An online research portal where educators and students can share their research findings, collaborate on research projects and access research resources.

Technology: AI can enhance research management by automating tasks. It can streamline data collection by automatically extracting information from sources such as academic journals, databases and online repositories. AI-powered analytics tools can then analyse this data to identify trends, patterns and insights, accelerating the research process and enabling researchers to make more informed decisions. Additionally, AI can facilitate the dissemination of research findings by personalising content recommendations to target audiences based on interests and preferences. This ensures that research outputs reach the right audience effectively. 5G/6G technologies can enhance the efficiency of research by enabling faster and more reliable data transmission. Researchers can collaborate in real time, share large datasets seamlessly and access cloud-based research tools and resources from anywhere with high-speed connectivity. This enables faster, more agile, more accurate and collaborative research processes.

Product/practical impact: Implementation would involve setting up systems for managing educational research, providing training on research methods and creating protocols for ethical research practices.

Content management

EBCG academy architecture component: This involves the creation, organisation and delivery of content within the academy.

Use case: A content management system that helps educators create, organise and deliver multimedia content.

Network Rail leveraged the multi-tenancy structure for course administrators that enabled the L&D team to create multiple sub-platforms for each entity or partner within its supply chain network, offering its improved management structure centrally for its LMS platform.

Technology: AI can assist in tailoring content delivery based on individual stakeholder needs (e.g. general content created to serve the entire community can be further tailored with national specifics). Learning management systems can then facilitate its organisation and delivery.

Product/practical impact: Implementation would involve selecting and setting up a content management system; training educators to create and manage content; and creating protocols for content quality and accessibility. The availability and relevance of data and information that would support content development are crucial.

Learning resource administration

EBCG academy architecture component: This component focuses on handling training registrations to ensure students are linked to the appropriate learning content, in line with their registration.

Use case: This approach is used to ensure students complete mandatory courses or certifications required by regulatory bodies or organisational policies. It involves assigning specific groups of students (audience cohorts) to tailored learning journeys based on their roles, responsibilities and compliance requirements. The requirements ensure individuals are knowledgeable about the regulations and standards pertinent to their roles, thereby maintaining legal and ethical standards within the organisation. Administrators can streamline the process of enrolling students into the right modules and broader training programmes, monitor their progress and track compliance requirements through these capabilities.

The eLearning solution deployed by Network Rail using Moodle LMS technology provided significant options for the administrative team to control learning content and grant access to users based on their learning pathways. The team also leveraged the solution to create a data backup for efficiently managing their workforce, onboarding new hires and keeping track of employees leaving the organisation.

Technology: LMS technology supports learning-resource administration by providing a centralised platform for managing training registrations, tracking compliance and assigning audience cohorts to tailored learning journeys.

Product/practical impact: Developing this component involves setting up learning-resource databases and implementing processes to manage training resources efficiently.

Student and alumni relationship engagement

EBCG academy architecture component: This focuses on maintaining and enhancing relationships with students and alumni. It involves regular communication, providing ongoing support and fostering a sense of belonging to the academy.

Use case: This component is used to keep students and alumni engaged, through initiatives like alumni mentorship programmes, regular newsletters and exclusive events.

Technology: AI can personalise communication based on the recipient's profile, and 5G/6G technologies can ensure timely and efficient communication. Gamification technology can encourage engagement through rewards and recognition.

Product/practical impact: Developing this component involves creating a student and alumni engagement strategy, setting up communication channels and regularly monitoring and improving engagement levels.

Communication and community management

EBCG academy architecture component: This involves managing the academy's online and offline communities. It's responsible for facilitating communication, moderating discussions and ensuring a positive and respectful community environment.

Use case: Academies often have online forums or social media groups where students can discuss course material, share resources and support each other.

Technology: AI can help with managing large volumes of communication and moderate online discussions. 5G/6G technologies can enhance connectivity and enable real-time communication. Metaverse and extended reality technology can create immersive virtual communities.

Product/practical impact: Setting up this component involves creating community guidelines, setting up community platforms and training community managers.

Virtual collaboration management

EBCG academy architecture component: This manages the academy's virtual collaboration spaces. It ensures that students and academy members can collaborate on projects and discussions, regardless of their physical location.

Use case: Academies use virtual collaboration tools to allow students to work together on group projects, participate in virtual labs and engage in peer review.

At the United States Border Patrol, the implementation of the Sign Cutting and Tracking Training solution showcased improved engagement in a virtual environment. This was accomplished by extending ARES technology in a virtual reality environment, which allowed the staff to operate without interruptions, enabling coordinated network coverage in remote areas where the physical sand table could not be accessed.

Technology: Metaverse and extended reality technology can create immersive virtual collaboration spaces. 5G/6G can ensure seamless connectivity, and AI can help manage and organise collaboration.

Product/practical impact: To implement this component, the academy needs to select virtual collaboration tools, set up virtual collaboration spaces and train users.

Marketing management

EBCG academy architecture component: This component is responsible for promoting the academy and its courses to potential students. It involves creating and implementing marketing strategies, managing marketing campaigns and monitoring and improving marketing effectiveness.

Use case: Organisations attract potential students using online marketing techniques, such as search engine optimisation, social media marketing and content marketing.

The metaverse academy developed by Interpol is a collaboration between global technological organisations including Microsoft, Meta and Decentraland. Such initiatives allow the implementing organisation to position itself as a pioneer and early adopter, improving its market brand value and credibility in the eyes of the workforce.

Technology: AI can help personalise marketing messages and analyse marketing data. 5G/6G can enhance the reach and speed of marketing campaigns.

Product/practical impact: Setting up the marketing management component involves creating a marketing strategy, selecting marketing channels and setting up systems to monitor and improve marketing effectiveness.

Document management

EBCG academy architecture component: Document management is a sub-component of the administration function. It involves the systematic handling of the academy's information documents throughout their lifecycle. This includes the storage, retrieval and sharing of documents, ensuring information is readily available and secure.

Use case: Organisations use document management systems to store and manage educational resources, administrative documents and student records.

Technology: AI can be used to classify and organise documents, making retrieval easier. 5G/6G can speed up document transfer and access.

Product/practical impact: A robust document management system is essential for the efficient operation of the academy. It requires careful planning and implementation, including determining the types of documents to be managed, establishing a system for document classification and storage, and setting up protocols for document retrieval and sharing.

Contract management

EBCG academy architecture component: Contract management, another sub-component of administration, involves the management of contracts with vendors, partners and other stakeholders. It entails the creation, review, execution and analysis of contracts to maximise operational and financial performance.

Use case: Organisations use contract management systems to handle contracts with vendors for services such as technology provision, content creation and facility management.

Technology: AI can be used to automate contract creation, review and analysis, while 5G/6G can make contract management processes faster and more reliable.



Figure 4. Visualisation of the use case at Interpol

Product/practical impact: Effective contract management is essential for ensuring the academy's smooth operation and financial health. Implementing a contract management system involves identifying needs, selecting a suitable system and training staff to use it.

User administration

EBCG academy architecture component: User administration involves managing the academy's user database, including students, staff and other stakeholders. It includes tasks such as creating and managing user profiles, granting and managing access rights, and handling user inquiries and issues.

Use case: Organisations use these systems to manage their user databases, ensuring that users can access appropriate resources and services.

The eLearning solution deployed by Network Rail using Moodle LMS technology provided significant options for the administrative team in terms of control over learning content and granting access to users based on their learning pathways. The team also used the solution to create a data backup for managing their workforce, onboarding new hires and keeping track of employees leaving the organisation.

Technology: AI can automate tasks such as profile creation and access management, while 5G/6G can make user administration processes faster and more reliable.

Product/practical impact: A robust user administration system is essential for a good user experience. Implementing it requires careful planning and execution, including determining the types of users to be managed, setting up a system for user administration and training staff to use the system.

Enterprise/academy architecture management

EBCG academy architecture component: This sub-component ensures that the academy's structure and operations align with its strategic goals. It plays a pivotal role in ensuring that the academy's overarching strategic objectives are not just outlined, but also effectively implemented and sustained. It involves planning, designing, implementing and managing the academy's structure and processes.

Use case: Academies typically use this component to design and manage their course structure, administrative processes and other operational aspects. In a use case such as designing and managing course structures and administrative processes, this component would encompass all these steps. It would involve carefully planning the courses to be offered, designing the curriculum and learning materials, implementing the courses according to schedule and managing the administrative tasks associated with enrolment, grading and certificates. Throughout this process, the component would ensure that all activities align with the academy's strategic goals and contribute to its overall success.

Technology: Some of the technologies that can be used are LMS, authoring tools, data analytics, virtual classrooms, workflow management and mobile learning applications. It is important to note that all applications and technologies need to be compliant with the organisational ICT environment and stipulations, as well as aligned with the ICT strategy. The solutions' sustainability needs to be considered to ensure they are future-proofed. This means that not only should the technologies integrate seamlessly with existing systems, but they should also be adaptable to future advancements and shifts in educational paradigms. Regular updates and maintenance will be crucial to preserve their effectiveness and security. Additionally, staff training and support are vital to maximise the potential of these technologies, ensuring that educators and administrators are fully equipped to leverage them to enhance the learning experience. By prioritising compliance, sustainability and adaptability, the EBCG academy can establish a robust, innovative and resilient architectural framework.

Product/practical impact: Establishing this sub-component involves defining the academy's structure, designing its processes and implementing systems for managing these processes. By leveraging these technologies, the EBCG academy can enhance the effectiveness and efficiency of its operations, aligning them more closely with its strategic goals. Additionally, technology enables the academy to adapt to evolving educational trends and preferences, ensuring its relevance and competitiveness in the digital age.

Business intelligence

EBCG academy architecture component: This sub-component involves collecting and analysing data to provide actionable insights for decision-making. It enables the academy to monitor its performance, identify trends and patterns and make data-driven decisions.

Use case: Academies often use business intelligence tools to monitor key performance indicators such as student enrolment, course completion rates and student satisfaction levels.

Deloitte's 'Major Programmes' solution, which leveraged enterprise LMS technology for its deployment, enables effective analysis for the administrative team. The use of learning analytics was achieved through a combination of survey data, learning platform data, and HR data, with individual data to personalise the learning journey. This highlighted a clear difference in the learning patterns of individuals and/or groups of students.

Technology: AI is used extensively in business intelligence to analyse large volumes of data and provide actionable insights. 5G/6G can facilitate real-time data collection and analysis.

Product/practical impact: Implementation involves selecting suitable business intelligence tools, setting up systems for data collection and analysis and training staff to use these tools.

Quality management

EBCG academy architecture component: This sub-component ensures the academy's services meet the desired quality standards. It involves setting quality standards, monitoring performance and implementing improvements when necessary.

Use case: Academies typically use quality management systems to ensure the quality of their courses, teaching methods and administrative processes.

Technology: AI can monitor performance against quality standards and suggest improvements. A learning management system can help manage course quality.

Product/practical impact: Establishing this sub-component involves defining quality standards, setting up a system for quality monitoring and implementing a process for continuous improvement. Furthermore, it requires collaboration with the Member States to set and review quality benchmarks. This will also facilitate the sharing of best practices in quality management, enhancing the academy's overall effectiveness.

Project, programme, and portfolio management

EBCG academy architecture component: This sub-component oversees the planning, execution and management of projects and programmes within the academy. It ensures these initiatives align with the academy's strategic goals and are completed on time and within budget.

Use case: Academies often use project management tools to manage initiatives such as the development of new courses, implementation of new systems and execution of marketing campaigns.

Technology: AI can streamline project management processes and provide real-time updates on project status. 5G/6G can facilitate real-time communication among project team members.

Product/practical impact: Implementation involves defining the academy's project management methodology, selecting suitable project management tools and training staff to use them. Furthermore, it requires collaboration with Member States to ensure that the projects and programmes not only meet internal objectives, but also align with broader regional and international goals.

Category 2: Underlying infrastructure

INTEGRATION MANAGEMENT
SECURITY MANAGEMENT
DATABASE MANAGEMENT

Integration management

EBCG academy architecture component: Integration management ensures all components of the academy's infrastructure work together seamlessly. It involves the coordination of various ICT systems and applications, enabling them to operate as a unified whole.

Use case: Academies typically use integration management to connect their LMS with other systems, such as student information systems, content management systems and administrative systems.

Technology: AI can automate the integration process and optimise the performance of the integrated systems. 5G/6G can enhance the speed and reliability of data exchange between integrated systems.

Product/practical impact: Implementation involves identifying the systems to be integrated, designing the integration architecture and monitoring and improving the integrated system's performance.

Security management

EBCG academy architecture component: Security management protects the academy's infrastructure from threats and breaches. It involves implementing security measures to protect data, systems and networks, and managing the response to any security incidents.

Use case: Academies typically use security management to protect sensitive data, such as student records, course materials and administrative data.

Technology: AI can enhance security management by detecting unusual activity and potential threats. 5G/6G can provide the speed and bandwidth needed for real-time security monitoring.

Product/practical impact: Establishing this component involves designing a security policy, implementing security measures and setting up a system for incident response.

Database management

EBCG academy architecture component: Database management involves the storage, retrieval, update and management of data in the academy's databases. It ensures that data is organised, accessible and protected.

Use case: Academies typically use database management systems to store and manage data such as student records, course materials and performance data.

Technology: AI can automate many database management tasks, such as data classification and query processing. 5G/6G can enhance the speed and reliability of data access.

Product/practical impact: Implementation involves designing the database structure, selecting a suitable database management system and setting up systems for data entry, retrieval and protection.

Category 3: Role-based access rights

ENGAGEMENT HUB
TRAINING PORTAL
CONTENT LIBRARY
RESEARCH MATERIALS
ANALYTICS CENTRE
COMMUNICATION CHANNELS
KNOWLEDGE BASE

Engagement hub

EBCG academy architecture component: The engagement hub fosters interaction and collaboration among users. It is a vital component of the EBCG academy architecture, promoting a sense of community and enhancing the learning experience.

Use case: The hub could be used to facilitate discussions and collaborations on training topics, provide a platform for peer-to-peer learning and foster a sense of community among students and staff.

Technology: virtual reality and augmented reality could be used to create immersive, engaging experiences in the hub. Gamification technology could be used to motivate user participation.

Product/practical impact: Establishing an engagement hub would involve designing and implementing a platform that supports interaction and collaboration, fostering a community of users and regularly monitoring and moderating the hub to ensure it remains a positive, productive space.

Training portal

EBCG academy architecture component: The training portal is the primary gateway where students and staff access training resources and learning materials. It is a central component of the EBCG academy architecture, providing a user-friendly interface and streamlining the learning process.

Use case: The portal could be used to deliver a variety of training programmes, including those for onboarding new recruits, advanced training for specialised roles and continuous professional development courses. A well-designed portal could provide personalised learning paths, allowing users to progress at their own pace and according to their learning preferences.

Technology: The portal could leverage AI and LMS technology to deliver personalised learning experiences and track student progress. Extended reality technology could also be used to provide immersive learning experiences.

Product/practical impact: Establishing the training portal would involve identifying user requirements; designing the user interface; integrating with other components of the academy architecture such as the content library and analytics centre; and regularly updating and maintaining the portal to ensure it continues to meet user needs.

Content library

EBCG academy architecture component: The content library is a repository of learning materials and resources. A crucial component of the EBCG academy architecture, it provides users with access to the resources they need to support their learning.

Use case: The library could store and organise a variety of learning resources, including course materials, multimedia resources, and supplementary materials.

Technology: AI could be used to deliver personalised recommendations from the content library. Extended reality could be used to create immersive, engaging learning resources.

Product/practical impact: Building a comprehensive content library would involve collecting and organising high-quality learning resources; creating a user-friendly and efficient system for accessing them; and regularly updating the library to ensure it remains relevant and valuable.

Research materials

EBCG academy architecture component: Research materials encompass a variety of resources that support innovative and evidence-based training practices. They provide valuable insights into emerging trends, best practices and effective strategies in training and education.

Use case: These materials could be used by instructional designers to design training programmes, by trainers to stay up to date with the latest methodologies and by students to deepen their understanding of the topics they are studying.

Technology: AI can be used to analyse and categorise the materials, making them easily searchable and accessible. 5G/6G can support quick and seamless access to them.

Product/practical impact: Developing a robust collection of research materials would involve collecting and curating high-quality resources; creating a user-friendly and efficient system for accessing them; and regularly updating the collection to ensure it remains relevant and valuable.

Analytics centre

EBCG academy architecture component: The analytics centre provides insights into user behaviour, learning outcomes and platform performance. It is a key component of the EBCG academy architecture, supporting decision-making and continuous improvement.

Use case: The centre could be used to track user engagement, assess learning outcomes, identify areas for improvement and support decision-making.

Technology: AI could be used to analyse data and generate insights in the centre. 5G/6G could support real-time data collection and analysis.

Product/practical impact: Establishing an analytics centre would involve selecting and implementing appropriate data collection and analysis tools, ensuring compliance with data privacy regulations and training users to interpret and use analytics.

Communication channels

EBCG academy architecture component: Communication channels support collaboration, feedback and information sharing.

Use case: These channels could be used for a variety of purposes, including delivering announcements, facilitating discussions, providing feedback and supporting collaboration on projects.

Technology: AI could be used to personalise communication and ensure users receive relevant information. 5G/6G could support real-time, high-quality communication.

Product/practical impact: Establishing effective communication channels would involve selecting appropriate technologies, establishing communication protocols and guidelines and providing support and training to users.

Knowledge base

EBCG academy architecture component: The knowledge base is a repository of information and resources that supports learning and performance at the EBCG academy. It provides users with self-service access to information, reducing the need for direct support and enabling users to find answers to their questions quickly and efficiently.

Use case: The knowledge base could be used to provide answers to frequently asked questions; instructional guides and tutorials; and other resources that support learning and performance.

Technology: AI could be used to deliver personalised recommendations from the knowledge base, suiting the user's role, learning preferences and past behaviour. 5G/6G would ensure fast, reliable access.

Product/practical impact: Building a comprehensive knowledge base would involve collecting and organising information, creating a user-friendly and intuitive interface and maintaining and updating the collection to ensure it remains accurate and helpful.

Category 4: Involvement of internal and external users

INTERNAL USERS
Students
Alumni
Staff members
EXTERNAL USERS
EU Agencies and Government Institutions
Media
General public
Suppliers and providers

Students

EBCG academy architecture component: Students are the primary users of the EBCG academy. They use all its resources and tools for learning and development. Their activity and interaction within the academy contribute data that helps refine and personalise the learning experience.

Use case: Students can access the training portal to take their courses, participate in the engagement hub for collaborative learning, use the knowledge base for self-service learning and get their queries resolved through communication channels.

Technology: AI and LMS would be crucial for personalised learning paths. Metaverse and extended reality would provide immersive learning experiences, and 5G/6G would ensure seamless and instant access to resources.

Product/practical impact: The steps to be followed for operationalizing a technology enabled academy would involve defining student roles and access rights, creating student profiles, enrolling them in courses and familiarising them with the academy features.

Alumni

EBCG academy architecture component: As former students, alumni play an important role in the EBCG academy. They can contribute by sharing their experiences, participating in discussions and even mentoring current students. They might also participate in continuous learning programmes.

Use case: Alumni can engage with current students and staff in the engagement hub, contribute insights from field experience to the knowledge base and participate in ongoing training programmes for continuous learning.

Technology: AI can be used to recommend interactions and learning materials to alumni. Communication and collaboration can be enhanced using 5G/6G.

Product/practical impact: Inclusion of alumni in the academy would involve defining their roles and access rights, creating their profiles and integrating them into the academy community. Their inclusion can provide valuable insights for current students and staff members.

Staff members

EBCG academy architecture component: Staff members, including administrators and instructors, manage and operate the EBCG academy. They create and manage content, interact with students and use analytics to improve academy operations.

Use case: Staff members would use the analytics centre to track student progress and course effectiveness, the content library to store and manage course materials and the communication channels to interact with students.

Technology: AI can be used for data analytics, course management and personalised communication. LMS can be used for managing courses and student data.

Product/practical impact: Staff would need clearly defined roles and access rights, as well as training in using the academy's systems. Their inclusion is crucial for the academy's operations, from course creation to student management.

EU Agencies and government organisations

EBCG academy architecture component: EU agencies/government organisations may need access to the academy for collaboration/information purposes. This would likely be limited to certain platform areas.

Use case: Officials of other EU agencies/government organisations may use the research area to understand the latest trends in training and education.

Technology: AI can be used to provide comprehensive analytics and reports. 5G/6G would ensure seamless access.

Product/practical impact: To include other EU agencies/government entities, their roles and access rights would need to be defined. Their inclusion can help ensure regulatory compliance, secure funding and guide policymaking.

Media

EBCG academy architecture component: Media access might be required for reporting and public relations purposes. They could have limited access to general information and academy news.

Use case: Media entities might receive updates through academy communication channels or interact with the academy community via the engagement hub.

Technology: AI can be used to provide personalised news updates. 5G/6G would ensure seamless access.

Product/practical impact: Including media would require defining roles and access rights. Their inclusion can help in public relations and outreach efforts.

General public

EBCG academy architecture component: The public might have access to certain academy areas, such as public resources, news updates or open courses.

Use case: The public could use the academy's resources for self-learning, or to stay updated about academy activities.

Technology: AI can be used to provide personalised learning recommendations. 5G/6G would ensure seamless access.

Product/practical impact: Including the public would require defining roles and access rights. Their inclusion can contribute to the academy's reputation and outreach.

Suppliers and providers

EBCG academy architecture component: Suppliers and providers might need access to the academy for maintenance, support and system upgrades.

Technology: AI could be used for predictive maintenance and support. 5G/6G would ensure seamless access.

Product/practical impact: Including suppliers and providers would require defining roles and access rights. Their inclusion is crucial for the smooth operation and maintenance of the academy's systems.

Cross-categories: Process for handling uncategorised terms

The taxonomy for EBCG training technologies offers a comprehensive structure to understand different terms and concepts in the EBCG training technology landscape. Given the rapidly evolving landscape, new terms and concepts are likely to emerge that might not fit neatly into the predefined categories. This section provides a structured approach for integrating emerging terms and concepts. By doing so, the taxonomy remains flexible, up-to-date and capable of incorporating innovations. This process involves several steps.

Step 1: Understand the meaning. Begin by thoroughly understanding the meaning of the term. Clarify its definition and conceptual boundaries to ensure a precise understanding within the context of training technologies for the EBCG community. This foundational step establishes a clear understanding of the term's essence and facilitates its effective integration into the taxonomy.

Step 2: Evaluate the relevance. Assess the term's relevance, examining its frequency across various sources. Terms that appear frequently in reputable literature, industry reports and academic studies are likely to hold greater significance and recognition. Moreover, delve into the scope of the term to comprehend the breadth and depth of its applicability within EBCG training. Analyse the range of applications the term covers, considering its potential impact on multiple aspects of training for the EBCG community. Understanding the scope of the term ensures that its categorisation within the taxonomy accurately reflects its multifaceted implications for EBCG training. This evaluation can be done through a comprehensive literature review and by engaging subject matter experts in the field of training technologies for the EBCG community.

Step 3: Decide on the category. If the term (along with related concepts) appears frequently and is highly relevant to EBCG training, it may justify a new category or sub-category. If it represents a significant new area of study or practice, creating a new category may be beneficial. Assess whether the term can be integrated into an existing category without diluting its unique importance. If integration would obscure the term's significance, a new category is warranted.

Step 4: Integration and documentation. Once the decision is made, integrate the term into the taxonomy, either within an existing category or as a new category. Document the term's definition and relevance, and its relationships within the taxonomy. If a new category is created, outline the rationale and criteria used for this decision.

By following these steps, terms that do not initially fit into the predefined categories can be systematically assessed, integrated and documented, ensuring the taxonomy remains dynamic and inclusive of all relevant training technologies for the EBCG community.

5. Proposal of relevant research topics

5.1 Introduction

As we navigate the challenges and opportunities of the 21st century, it is crucial to continually develop our training and education methods to ensure optimal preparedness of our border management organisations. **This chapter provides a practical starting point, proposing potential avenues to inform the piloting, adoption and further development[1] of innovative solutions for EBCG**. It provides a flexible framework to guide the definition and implementation of tailored initiatives from 2025 to 2027. They may be implemented on either a planned or an ad-hoc basis, ensuring the EBCG's ability to adapt to evolving needs and circumstances. A structured approach identifies and prioritises focus areas, while maintaining a degree of generality that ensures applicability across a variety of contexts.

5.2 Methods

The first step was to extract common, cross-applicable denominators that should be reflected in all stages of potential endeavours, both in technical implementations and in methodological structuring. While some principles could be universally applicable, others are tied to the nature of the ECBG community and its way of working. These principles are based on previous deliverables (and their possible commonalities), with an emphasis on success factors. They would be applicable during general technical adoption where recommended practices are already established, and where they could logically and feasibly transfer to the new implementation.

These principles were then compared with use cases where they correlated with successful functionality and benefits. Not only do these common denominators provide the foundation for practical approaches, but their detailed breakdowns also propose specific steps to ensure their application. These steps were considered when narrowing down the following directions of future research and actions.

Second, considering the vast landscape of the researched areas, the need for structure is addressed. The output is constructed to reflect the complexity of the EBCG environment and allow additional options for the future, and thus is granular. It is not recommended to explore all these options together, or to fully explore all of them. This is rather a summary of the researched areas, with practical information if the area becomes relevant over time. It serves as a point of reference for the future, where needs and priorities may evolve or shift completely. Based on the full scope of the original research, it offers a short format that can be referred to comfortably and intuitively, with a practical approach.

5.3 Results

5.3.1 Common denominators for relevant research topics

This section explores common denominators essential for technology management within the EBCG community. From seamless data sharing to safeguarding operations, it covers interoperability, standardisation, futureproofing, scalability, modernisation, clarity and security. These denominators form the baseline to understand the research topics that will be explored hereafter.

[1] Further research in context of Horizon Europe: This is a huge global research and innovation programme with €93.5 billion in funding from 2021 to 2027. It's open to participants worldwide, offering opportunities for collaboration and funding. The programme focuses on major research projects and international partnerships to address global challenges. *https://research-and-innovation.ec.europa.eu/funding/funding-opportunities/funding-programmes-and-open-calls/horizon-europe_en*

Interoperability

Comprehensive compatibility analysis: To ensure seamless operation across diverse technological platforms, it is critical to implement solutions that are inherently compatible. This involves creating a framework that supports functional interoperability and efficient data-sharing across different systems and components within the organisation.

Unified communication protocols: Developing standardised protocols that facilitate communication between various technological solutions enables them to operate cohesively and share crucial operational data without barriers.

Standardisation

Organisational standards: Standardise solutions and approaches to align with diverse use cases throughout the organisation. Standardisation should encompass terminology, function and structure, and aim to set or update organisational standards to streamline operations and reduce inefficiencies.

Regular review and updates: A periodic review process ensures that the standards remain relevant and reflect the latest industry practices and technological advances.

Future-proof

Strategic evaluation of current technologies: Identify and deploy technological solutions likely to remain relevant and functional in the foreseeable future. This involves selecting technologies at the forefront of innovation within their respective categories.

Continuous market and trend analysis: Keep abreast of emerging technologies and industry trends to continuously update and adapt the organisation's technology strategy and pre-empt obsolescence.

Scalability

Adaptive solution frameworks: Ensure that all solutions and approaches are designed with scalability in mind. They should be capable of accommodating future growth in user numbers, data volume and transaction capacity, without significant reconfiguration.

Provisions for enhancements and upgrades: Establish mechanisms to easily implement enhancements and add functionalities or content as needed, ensuring the solutions evolve in line with organisational growth and technological progress.

Modernisation

Implementation of cutting-edge technologies: Actively pursue the adoption of advanced technologies that contribute to the modernisation of the organisational landscape, both technologically and culturally. This entails embracing innovations that streamline processes, enhance productivity and foster a progressive work environment.

Balancing ideals and practicalities: Balance idealistic visions of technological progress with the practical realities of the organisation's operational needs.

Clarity

Clear documentation and user education: Ensure all technological implementations are accompanied by comprehensive documentation and educational resources that explain their functionality and underlying principles to all stakeholders.

Avoidance of vendor lock-in: Make conscious choices to adopt technologies and platforms that do not lead to dependence on specific vendors, thus maintaining organisational autonomy and flexibility in technological development.

Security

Implementing robust security protocols: Ensure all solutions adhere to the highest standards of security, particularly considering the sensitive nature of the data and the needs of the EBCG community.

Continuous security assessments: Regularly evaluate and update security measures to protect against evolving threats and vulnerabilities, ensuring the integrity and confidentiality of organisational data.

5.3.2 Research categories and topics

The research topics are divided into two categories: innovative solutions, and underlying methodologies and frameworks. Each category represents a diverse spectrum of research topics worth considering, as they are crucial to the integration and application of emerging training technologies within the EBCG community.

The first category entails a technology-driven approach aimed at exploring and implementing cutting-edge methodologies and tools to address specific challenges within the training landscape of the EBCG community.

The second adopts a more holistic perspective, delving into the foundational structures, governance principles, and ethical frameworks necessary to ensure the integration, implementation and sustainable use of emerging training technologies within the broader organisational context of the EBCG community. It is important to note that the research topics in this category are rather foundational, as they explore broader concepts and principles within training and education, rather than focusing on specific applications or tools.

#	Categories	Research topics	Indicative research questions
1.	**Innovative solutions**	Innovative solutions for human-AI collaboration in training	In what ways do AI platforms, through automated content delivery and adaptive learning algorithms, improve the retention of training materials and reduce the time required for skill acquisition compared to traditional training methods? What tools are available and emerging for preparing law enforcement personnel to work collaboratively with AI systems in real-world scenarios? How can AI be integrated into border control training programmes to enhance decision-making and situational awareness while maintaining the human-centric approach? What are the specific challenges faced by the EBCG community in adapting to AI-driven training methods and how can these challenges be addressed?
		Innovative solutions to advance the training environment of border and coast guards	How can an integrated a simulated environment enhance the training of border guards to use a range of border surveillance technologies, including surveillance cameras, seismic sensors, mobile vehicles and UAVs, to enhance border and coast security and response capabilities? What design elements are essential for developing an advanced training environment that enables border management operational staff to master the operation and maintenance of various border security technologies such as passport readers, e-gates, video spectral comparators and heart rate detectors? How does the cognitive load, or mental demand, placed on border guards within an integrated simulation environment affect their ability to apply learned skills from training to real-world situations involving border surveillance technologies, thereby enhancing their capability to respond to security challenges?

#	Categories	Research topics	Indicative research questions
		Innovative solutions to enable train-the-trainer in dispersed locations	What available and emerging technologies and platforms can be leveraged to enable train-the-trainer programmes in dispersed locations, particularly for law enforcement agencies operating across diverse geographic regions? What technologies, platforms and/or tools can facilitate remote train-the-trainer sessions for EBCG personnel stationed in dispersed locations?
		Innovative solutions for adaptive learning models and personalisation of learning content	How do the real-time interaction capabilities of generative AI and chatbots facilitate immediate feedback and personalised learning experiences in collaborative settings? What are the latest technological innovations driving advances in adaptive learning, such as machine learning, natural language processing and affective computing? What are the key criteria for developing adaptive learning algorithms tailored to the specific needs and skill levels of EBCG personnel? How can personalised learning pathways be created to accommodate the diverse backgrounds, experiences and learning styles within the border control community? How can adaptive learning technologies accommodate individual differences, while promoting inclusive and equitable learning experiences? What strategies and technologies can be implemented to ensure that training programmes are inclusive and accessible for border control community members from diverse backgrounds? How can user-centred design principles be applied to develop training materials and tools that cater to unique needs and preferences?
		Innovative solutions for integrating disparate training data sources to create a holistic view of training activities and outcomes	What are the major challenges and barriers to integrating disparate training data sources, such as learning management systems (LMS), virtual reality (VR) simulations and performance tracking tools, into a unified and cohesive training data ecosystem, and how can these challenges be overcome? What frameworks or methodologies can be employed to integrate data from various training resources, including simulations, field exercises and classroom instructions, to provide a comprehensive analysis of training transfer? How can data analytics and machine learning techniques be used to identify patterns, trends and areas for improvement in border control training programmes?
		Innovative solutions for collaborative training delivery that are environmentally sustainable	How can AI platforms track and analyse data on energy consumption, resource use and carbon emissions associated with training programmes, and how can this information be used to optimise training processes for greater environmental sustainability and continuous improvement?

#	Categories	Research topics	Indicative research questions
2.	**Underlying methodologies and frameworks**	Governance frameworks for cross-organisation collaboration on technology-driven training initiatives	What are the key components of a governance framework that fosters trust, transparency, and accountability in cross-organisational collaboration on technology-driven training? How can governance structures be designed to ensure accountability, transparency and knowledge sharing while respecting the sovereignty and operational autonomy of each participating organisation?
		Methodologies for measuring the impact of rapidly evolving technology-driven law enforcement training	How do different types of emerging training technology (e.g. virtual reality simulations, AI-driven analytics) impact the effectiveness and efficiency of law enforcement training programmes, and how can their impact be assessed and compared? What methodologies and performance metrics can be used to assess the training impact, especially in the context of rapidly evolving technologies? How can qualitative and quantitative data be collected, analysed and synthesised to provide insights into the impact of emerging technologies on skill acquisition, decision-making and operational readiness within border control agencies?
		Methodologies for strategically aligning the development and deployment of future training technology with the overarching strategic objectives and priorities of EBCG	How can technology on road mapping, scenario planning and foresight methodologies be used to anticipate future training needs and align technology investments with organisational goals? What strategic planning frameworks and methodologies can be employed to align the development and deployment of training technologies with the long-term vision, mission and strategic priorities of the EBCG community? How can stakeholder engagement processes, scenario planning exercises and risk assessments be integrated into strategic decision-making to ensure that training technology investments align with the evolving needs and challenges faced by EBCG?
		Privacy-preserving methodologies to protect training data while still allowing for meaningful analysis and insights	What state-of-the-art privacy-preserving methodologies, such as differential privacy and homomorphic encryption, are applicable to training data collected through innovative training technologies? What privacy-enhancing technologies, encryption techniques and data anonymisation methods can be used to protect sensitive training data, including personal information and operational procedures, while still enabling effective analysis and knowledge extraction? How can differential privacy principles and federated learning approaches be applied to training datasets distributed across multiple border-control agencies and jurisdictions, ensuring data privacy and security without compromising analytical validity?
		Frameworks for measuring the impact of the AI Act on future AI-driven training solutions	What are the potential implications of the AI Act on the design, development and adoption of AI-driven training solutions, and how can organisations proactively address these implications to ensure alignment with regulatory requirements and best practices? How can interdisciplinary collaborations between legal experts, ethicists, technologists and law enforcement professionals inform the development of ethical AI guidelines, auditing mechanisms and compliance procedures tailored to the specific requirements and constraints of border-control training applications?

6. Conclusions and considerations

This study examined emerging technologies and innovative educational concepts, evaluating their adaptability to EBCG needs and challenges, identifying gaps and facilitating informed decision-making for strategic improvements. Indeed, the following key findings emphasise the pivotal role of technologies within the learning ecosystem. This section of the report presents aggregated conclusions based on the information gathered, analysed and conceptualised regarding the study objective. Next to this, considerations are presented on leveraging this study to improve EBCG education and training.

6.1 Conclusions

Conclusion 1: Learning management system technology has become a pivotal and increasingly an indispensable enabler of training and education in the EBCG community. LMSs support collaborative learning initiatives, as they enable dynamic interaction, learning asset sharing, learning data centralisation and real-time feedback, making them an invaluable tool in cultivating a collaborative and interactive learning experience.

The desk research highlighted that LMS supports organisations in developing and implementing standardised training programmes with up-to-date, consistent learning materials, ensuring that all personnel receive high-quality training regardless of location. This centralised approach is increasingly valuable for the EBCG community, as organisations operate in various Member States, necessitating standardised training programmes to ensure consistent approaches.

Moreover, LMS supports continuous learning regardless of location, through online training, reducing the need for in-person sessions and minimising time away from duties. Mobile-friendly access and advanced connectivity ensure seamless training opportunities even in remote areas. This flexibility is complemented by real-time data on training progress and compliance, which is crucial for high-stakes environments like border control. Advanced analytics in LMS allow for real-time tracking of individual and collective progress, providing insights that facilitate data-driven decision-making for continuous improvement.

Furthermore, the EBCG community relies on a range of technological tools and platforms for operations and training, and an LMS integrates seamlessly with HR tools, content libraries and other platforms, creating a unified training ecosystem that enhances the overall training infrastructure and operational efficiency. Adhering to regional and international compliance standards is vital for border management forces, and LMS supports ongoing compliance training, ensuring that all personnel are well-versed in laws, regulations and security protocols.

The value of LMS for the EBCG community was underscored at the Frontex Conference on Future Training Technologies, where approximately 52% of attendees ranked LMS as the top priority for the future of EBCG training. Moreover, when considering collaborative learning, the majority of these (approximately 28%) indicated that this particular use case held the most promise. This feedback highlights the value of a centralised technology-enabled training hub for the EBCG community's future training needs.

Organisations with similar needs to the EBCG community have implemented LMS technology for training and education. For instance, the Science and Technology Directorate of the Department of Homeland Security recently partnered with the US Border Patrol and the Federal Law Enforcement Training Centers to train agents. Network Rail, a public sector organisation with partner agencies in dispersed locations, trained over 100,000 employees with Moodle Workplace, improving efficiency through data-driven decisions, multi-tenancy and custom report integration. These implementations provided scalable, unified learning platforms, reducing technical burden and enhancing training efficiency, further demonstrating the potential and effectiveness of LMS for the EBCG community.

Conclusion 2: Integrating AI technology into the EBCG community offers opportunities to enhance efficiency, personalise learning experience and augment decision-making capabilities. AI in EBCG education and training is also associated with challenges, such as navigating high-risk AI applications in critical border management areas and adhering to stringent data privacy regulations like the GDPR. Balancing regulatory compliance with technological innovation is important for EBCG organisations to effectively deploy AI, ensuring secure, unbiased, and effective AI-driven solutions that meet the dynamic demands and unique context of EU border management and security.

This study highlighted the promising pathway of AI technology to transform training and education globally, offering opportunities to enhance efficiency, personalise learning experiences and augment decision-making capabilities. For instance, AI-driven analytics can provide deep insights into individual performance, facilitating tailored learning paths that align with professional development goals and organisational objectives. Moreover, AI can automate administrative tasks, freeing up valuable time for training administrators to focus on strategic initiatives and innovation.

As concluded during the conference, the integration of AI technology in EBCG education and training is associated with specific challenges. In line with the AI Act, AI applications used in critical areas of border management, such as real-time decision support or autonomous operations, may fall into high-risk categories, requiring comprehensive risk assessments and regulatory approvals before deployment. AI applications that simulate real-world scenarios for border security personnel must adhere to strict data privacy regulations such as the GDPR, ensuring that sensitive operational data remains secure and compliant.

Ethical considerations were also central to the conference discussions. These emphasised the importance of deploying AI technologies responsibly within the EBCG community, for example ensuring AI algorithms are free from biases and capable of transparently managing sensitive data. By ensuring adherence to ethical principles and regulatory frameworks, organisations can build trust and confidence in AI systems among stakeholders while advancing training and development initiatives to meet evolving challenges in border security.

Consultations with the EBCG community showed that addressing the unique work context of border and coast guards is important in implementing AI solutions effectively. These professionals operate in dynamic, high-stakes environments, where real-time decision-making and situational awareness are paramount. Seamlessly integrating AI technologies into their operational workflows could enhance real-time decision support, while upholding stringent regulatory standards such as the GDPR. By addressing these operational needs, AI can augment border and coast guards' capabilities, ultimately bolstering efficiency, effectiveness and overall safety

The conference prompted a nuanced approach that integrates regulatory compliance with technological innovation. By learning from successful implementations and continuously adapting to a changing regulatory landscape, EBCG organisations can harness AI to not only meet current training needs, but also prepare for future challenges in border management and security.

Conclusion 3: Collaboration has emerged as fundamental and supported in shaping the future of EBCG training and education, highlighting its role in adopting innovative technologies and enhancing learning outcomes. Consultations with the EBCG community underscored the shared perception of collaboration's value in improving the learning experience and document management, and notable interest within the community in further collaboration in training and education initiatives.

This study emphasised that collaboration is a key factor in shaping the future of EBCG training and education, stressing its pivotal role in adopting innovative technological advances and enhancing learning outcomes across the EBCG community. The desk research and consultation with the community showed that effective collaboration enriches the learning experience and enhances document and agreement management. Participants in the Frontex Conference on Future Training Technologies highlighted successful resource-sharing initiatives among EBCG organisations and demonstrated interest in further collaboration in training and education initiatives. Moreover, the conference concluded that successful and modernised border control and coast guard training and education relies on the collaborative efforts of academia, industry solution experts, EU Member States and third countries.

One way to operationalise this collaborative space is through the concept of an EBCG technology-enabled academy. It is conceptualised as a dynamic educational platform leveraging advanced technological infrastructure

and integrated functionalities. Synergies within the academy architecture emphasise interoperability and cohesive operation, enhancing both learner experiences and administrative efficiency. Learners benefit from streamlined access to educational resources, personalised learning paths and insightful analytics to support their academic journey. Administrators leverage integrated functionalities to optimise operations, promote initiatives and strengthen stakeholder relationships, thereby advancing the academy's strategic objectives and impact within the EBCG community and beyond. By nurturing a collaborative community environment, the academy aims to foster innovation and continuous improvement in EBCG practices.

Continued dialogue and strategic considerations are essential to effectively shape this technology-enabled academy, ensuring it meets the evolving needs of the EBCG community and supports ongoing advances in training methodologies and operational readiness.

One organisation and that has already established a virtual collaborative environment for its community and dispersed workforce is Interpol. Driven by metaverse technology, Interpol has fostered global collaboration and knowledge exchange, enabling its worldwide workforce to collaborate effectively in solving cases and enhancing skills relevant to virtual border management and digital investigations. This approach not only improved operational efficiency, but also provided insights into regulatory frameworks and potential risks associated with virtual environments.

6.2 Considerations

Consideration 1: A potential next phase of this research could involve exploring practical applications through a proof of concept, requiring a nuanced and thorough examination of how these findings could be integrated into different EBCG organisations and their training and education practices.

This phase would involve comprehensive discussions with key stakeholders to understand their specific needs and challenges and examine the current use of training technologies within these organisations. Additionally, it would be crucial to gather insights from various actors within the community, including trainers, trainees, and technology providers, to ensure a well-rounded perspective. This exploration could include pilot programmes or small-scale implementations of the identified technologies to assess their effectiveness in real-world scenarios. For example, pilot programmes could develop the concept of an EBCG community or test virtual reality border-patrol simulations for consistent training on situational awareness and decision-making skills. This deeper and more practical exploration is anticipated to unveil actionable insights that will inform strategic decisions, driving investments in the most promising technologies. It will also provide a clearer roadmap for scaling these solutions across the EBCG community, optimising the training ecosystem for greater effectiveness, adaptability and responsiveness to the dynamic challenges of border security.

Consideration 2: As the EBCG community explores the implementation of emerging technologies and innovative training concepts, solutions to ensure scalability and interoperability of an EBCG technology-enabled academy could be explored.

Achieving scalability and interoperability of an EBCG technology-enabled academy and similar concepts could involve establishing standards and protocols enabling seamless communication and resource-sharing among various EBCG organisations. Some emerging technologies, like advanced simulators and immersive headsets, require specialised equipment that may not be universally available across all EBCG organisations. To mitigate this, exploring streamlined procurement processes and establishing frameworks for equipment-sharing among centres in different locations could be considered. This collaborative approach would not only optimise resource use, but also foster a sense of community and mutual support among EBCG training facilities. Implementing these advanced technologies effectively requires robust infrastructure support and strategic partnerships with technology providers. Ensuring seamless integration and ongoing technical support are crucial to sustain long-term benefits.

Scalability extends beyond technical considerations to encompass organisational readiness and human resource capabilities. Investing in training programmes that empower personnel to use these technologies is

crucial. Initiatives such as train-the-trainer programmes and continuous professional development ensure that educators and trainers have the necessary skills to leverage emerging technologies for maximum impact in training and operational contexts. Furthermore, fostering a culture of collaboration and knowledge-sharing among trainers, stakeholders and learners remains central to navigating the complexities of integrating emerging technologies into EBCG training programmes. This collaborative effort would not only enhance the quality and relevance of training delivery, but also drive continuous innovation and improvement within the EBCG training landscape, strengthening operational readiness and effectiveness.

As concluded during the conference, a pragmatic approach is favoured to adopt emerging and future training technologies and concepts, emphasising the value of incremental progress and experimentation. By piloting new technologies and concepts on a smaller scale, attendees deemed they could assess and identify potential challenges early on and make informed decisions about scaling up.

Consideration 3: Establishing a collaborative space, like a technology-enabled EBCG academy, could be a valuable collective effort, looking into an inclusive governance structure involving stakeholders like Member States and academia to oversee its strategic direction, operational management and long-term sustainability, fostering collaboration and shared ownership to ensure effective resource mobilisation and ongoing adaptation to educational and technological advancements.

Establishing a collaborative space, such as a technology-enabled EBCG academy, raises considerations regarding governance and ownership to ensure its successful launch, setup and maintenance. The governance structure should be robust, transparent and inclusive, involving key stakeholders such as Member States and academia.

Any governance structure should define clear roles and responsibilities among stakeholders to oversee the academy's strategic direction, operational management and financial sustainability. This may involve creating a steering committee or board that ensures diverse perspectives in decision-making. Ownership of the academy could be shared among participating stakeholders to promote collaboration and collective responsibility. This shared ownership can facilitate resource mobilisation, including the funding, expertise and technological infrastructure necessary for the academy's establishment and ongoing operations.

Regarding setup, a phased approach may be prudent, starting with pilot projects or partnerships to test and refine technological solutions, curriculum development and operational processes. This iterative approach allows for adjustments based on initial outcomes and stakeholder feedback before scaling up.

Maintenance of the academy requires a long-term commitment to continuous improvement, and adaptation to evolving educational and technological landscapes. Governance mechanisms should include regular evaluations and feedback loops to assess effectiveness, address challenges and incorporate innovations in training methodologies and AI technologies.

Consideration 4: When implementing emerging technologies in the education and training of EBCG personnel, it is valuable to stay closely informed about regulatory changes and their implications, such as those introduced by the AI Act. Consider looking into approaches to ensure that training and education support comply with evolving regulations to secure sensitive information. This includes not only the learning assets, but also the infrastructure supporting EBCG training.

This study underscores the role and ongoing evolution of regulations and standards governing emerging technologies. As technology progresses, regulatory frameworks will continue to evolve to address emerging challenges and ensure responsible innovation. For instance, the EU AI Act, introduced during the study period, establishes a comprehensive framework to regulate the development, deployment and use of AI systems across the Member States.

Some regulations and standards linked to emerging technologies are expected to be within the scope of EBCG training and education. At the same time, the design and implementation of training programmes

themselves must adhere to regulations and standards. For example, compliance with foundational laws such as the EU General Data Protection Regulation (GDPR) is crucial, ensuring that training platforms securely manage personal data and uphold individuals' rights regarding automated decision-making and profiling. Beyond curriculum content, the infrastructure supporting training delivery must also comply with these regulations.

As concluded during the Frontex Conference on Future Training Technologies, adhering to these standards and frameworks not only mitigates risk, but also supports public trust in these technologies. By ensuring compliance and accountability across all facets of training and education, EBCG organisations can foster a secure and trustworthy environment that supports continuous improvement and innovation in their operational practices.

7. Annexes

7.1 Annex 1: Existing taxonomies in the field of training and education (technology)

The following taxonomies are most used in the field of learning and training technology. It is important to note this list is indicative and by no means exhaustive.

Taxonomy of educational objectives

Bloom's taxonomy[1] classifies learning objectives into six hierarchical categories of cognitive complexity: from lowest to highest, they are remembering, understanding, applying, analysing, evaluating and creating. Bloom's taxonomy has been widely used in curriculum design, instructional planning and assessment. Within each category, some sub-categories delineate specific cognitive activities and skills. This hierarchical structure provides educators with a framework for designing instructional objectives, assessments and activities that target different categories of cognitive complexity, ultimately facilitating deeper learning and understanding.

Although Bloom's taxonomy is not inherently a technological concept, it can certainly inspire the development of learning technologies. The taxonomy underscores the importance of structuring learning experiences in a manner that promotes retention and understanding. Therefore, when building learning technologies, it's crucial to integrate elements that align with Bloom's taxonomy to ensure that students engage with content in ways that facilitate deep learning and memory retention.

Structure of Bloom's taxonomy of educational objectives:

- **Remembering:** Students demonstrate their ability to recall basic facts, concepts or information from memory, such as historical dates or mathematical formulas.
- **Understanding:** Students are expected to demonstrate comprehension by explaining ideas or concepts in their own words, interpreting information or summarising key points.
- **Applying:** Students apply their understanding by using acquired knowledge in new or familiar situations, solving problems or executing procedures.
- Analysing: Students break down information into its constituent parts, identify patterns or relationships and examine the underlying structure of concepts or arguments.
- **Evaluating:** Students assess the validity, credibility or effectiveness of information, arguments or solutions based on specific criteria or standards.
- **Creating:** Students demonstrate their ability to synthesise knowledge and ideas to generate original products, designs or solutions, fostering creativity and innovation.

Building upon Bloom's original taxonomy, a revised taxonomy[2] was published. The revision presents a reconfiguration of the original framework, emphasising a more dynamic and student-centred approach to educational objectives. The structure shifts from a hierarchical model to a more flexible and interactive framework, allowing for the integration and interaction of cognitive, affective and psychomotor dimensions. This revised taxonomy encourages educators to consider students' holistic development, fostering deeper engagement and meaningful learning experiences.

1 Seddon, G. M. (1978). The properties of Bloom's taxonomy of educational objectives for the cognitive domain. Review of educational research, 48(2), 303-323.

2 Krathwohl, D. R. (2002). A revision of Bloom's taxonomy: An overview. Theory into practice, 41(4), 212-218.

Structure of Krathwohl's taxonomy of educational objectives (i.e. revised version of Bloom's taxonomy):

- **Cognitive**: Remembering, understanding, applying, analysing, evaluating, creating.
- **Affective**: Receiving, responding, valuing, organising, characterising.
- **Psychomotor**: Perceiving, guided response, mechanism, complex overt response, adaptation, origination.

Around the same time as the revised Bloom's taxonomy, Fink's Taxonomy of Educational Objectives was launched,[3] which incorporates six categories to guide course design. These categories include foundational knowledge, application, integration, human dimension, caring and learning how to learn. Within each category, sub-categories delineate the progression of learning objectives, ranging from basic understanding to more complex application and synthesis. By emphasising the integration of multiple dimensions and sub-categories, Fink's taxonomy encourages educators to design courses that engage students in deep, transformative learning across a variety of cognitive, affective and interpersonal domains.

Structure of Fink's taxonomy of educational objectives:

- **Foundational knowledge**: Basic understanding of essential concepts and principles. Mastery of foundational knowledge through application and analysis.
- **Application**: Applying knowledge and skills to solve problems or address real-world challenges. Transferring knowledge and skills to new contexts or situations.
- **Integration**: Connecting disparate ideas, concepts or experiences to form a coherent understanding. Synthesising and integrating knowledge from multiple sources or disciplines.
- **Human dimension**: Understanding the impact of course content on personal values, beliefs and identity. Developing empathy, cultural competence and interpersonal skills through course activities.
- **Caring**: Recognising and valuing the significance of course content in personal and societal contexts. Demonstrating compassion, ethical reasoning and social responsibility in course-related activities.
- **Learning how to learn**: Acquiring metacognitive skills and strategies to monitor and regulate one's learning process. Developing lifelong learning skills, such as self-reflection, goal setting and seeking feedback.

Marzano's taxonomy of significant learning

This taxonomy offers a framework for classroom instruction. At its core are nine instructional strategies, each designed to enhance student learning and achievement. They encompass key elements such as providing feedback, reinforcing effort and providing recognition; cooperative learning; setting objectives and providing feedback; generating and testing hypotheses; cueing and questioning; non-linguistic representation; summarising and note-taking; and homework and practice. Within each strategy, sub-categories outline specific techniques and approaches for implementation in the classroom. By delineating these categories and sub-categories, the taxonomy provides educators with a comprehensive framework to discuss instructional practices.

[3] Fink, L. D. (2003). A self-directed guide to designing courses for significant learning. University of Oklahoma, 27(11), 1-33.

Structure of Marzano's taxonomy of significant learning:[4]

- **Feedback**: Various methods for providing specific, timely and actionable feedback to students.
- **Reinforcing effort and providing recognition**: Techniques for acknowledging and rewarding students' efforts and achievements to motivate continued progress.
- **Cooperative learning**: Strategies for organising and facilitating collaborative activities that promote active engagement and peer interaction.
- **Setting objectives and providing feedback**: Approaches for establishing clear learning objectives and providing ongoing feedback to guide students' progress.
- **Generating and testing hypotheses**: Methods for encouraging critical thinking and hypothesis formation, as well as strategies for testing and evaluating hypotheses.
- **Cueing and questioning**: Techniques for using effective cues and questions to stimulate students' thinking and facilitate deeper understanding.
- **Non-linguistic representation**: Approaches for incorporating visual aids, graphic organisers and other nonverbal representations to enhance learning and retention.
- **Summarising and note-taking**: Strategies for summarising key information and teaching students effective note-taking techniques to aid comprehension and retention.
- **Homework and practic**e: Methods for assigning meaningful homework and providing opportunities for deliberate practice to reinforce learning and skill development.

SOLO Taxonomy

The Structured Observation of Learning Outcomes (SOLO) Taxonomy offers a structured framework for evaluating the quality of observed learning outcomes. It consists of five categories, each representing a progressively more sophisticated category of understanding: pre-structural, unistructural, multi-structural, relational and extended abstract. It focuses on the depth of understanding and complexity of thinking demonstrated by students, rather than simply the quantity of knowledge.

Structure of the SOLO taxonomy:[5]

- **Pre-structural**: Students have a minimal understanding or organisation of knowledge. Students demonstrate confusion or lack of comprehension regarding the topic or task.
- **Unistructural**: Students grasp isolated aspects of the concept or task. Understanding is limited to one relevant aspect or dimension.
- **Multi-structural**: Students can identify several relevant aspects or dimensions of the concept or task. However, these aspects are not integrated into a cohesive understanding.
- **Relational**: Students establish connections between different elements or dimensions. They begin to perceive relationships, patterns and interactions among various aspects.
- **Extended abstract**: The highest category of understanding. Students can generalise principles, apply them across contexts and demonstrate creativity and originality. They exhibit deep comprehension and can synthesise information to generate new insights or solutions.

4 Marzano, R. J., Gaddy, B. B., & Dean, C. (2000). What works in classroom instruction.

5 Biggs, J. B., & Collis, K. F. (2014). Evaluating the quality of learning: The SOLO taxonomy (Structure of the Observed Learning Outcome). Academic Press.

Technology integration taxonomy

The Technology Integration Taxonomy (also known as the Technology Integration Matrix) provides a structured framework for understanding the progression of technology integration in education, comprising five distinct categories: entry, adoption, adaptation, infusion and transformation. Each category is further delineated by sub-categories that offer nuanced insights into the extent and nature of technology integration within educational practices. From basic awareness and exploration to transformative and visionary use, the hierarchical structure enables educators to assess their current category of technology integration and identify pathways for advancing towards more innovative and effective teaching and learning practices.

Structure of the technology integration taxonomy:

- **Entry level:** Awareness: Technology is present in the learning environment, but its use is sporadic or limited. Exploration: Teachers and students are beginning to explore technology tools and resources, but integration is inconsistent.
- **Adoption level:** Mechanical Use: Technology is used primarily for basic tasks, such as word processing or accessing information. Routine Use: Teachers and students regularly use technology for specific tasks, but integration may lack depth or creativity.
- **Adaptation level:** Technical Proficiency: Teachers and students demonstrate proficiency in using technology tools and resources. Beyond Proficiency: Technology is used to facilitate student-centred learning experiences, with teachers adapting instructional methods to leverage technology effectively.
- **Infusion level:** Empowering Use: Technology is seamlessly integrated into teaching and learning, empowering students to take ownership of their learning. Transformed Use: Technology facilitates transformative learning experiences, enabling students to engage in authentic, real-world tasks and collaborative projects.
- **Transformation level:** Innovative Design: Technology is used to design innovative learning experiences that transcend traditional boundaries and foster creativity and critical thinking. Visionary Leadership: Technology integration is driven by visionary leadership, with a focus on continuous improvement and innovation in teaching and learning practices.

7.2 Annex 2: Set-up of the workshop for brainstorming on the main use cases

The workshop took place on 12 December 2023 at Frontex's headquarters in Warsaw, Poland, using a hybrid format to ensure the participation of key stakeholders, including representatives of the Frontex Research and Innovation Team and the Frontex Training Team, Frontex Project Team and Deloitte Project Team. The workshop's design and hybrid nature aimed to maximise productivity, engagement and collaboration among the diverse set of participants. In total, there were 11 participants, four of whom joined online.

The agenda for the workshop was:

- **Introduction** (13:00-13:05): A presentation to set the stage and establish the context.
- **Inspiration from academies** (13:05-13:20): A presentation aimed at providing insight into the learning technology landscape, as well as inspiration and insights into relevant cases.
- **Alignment on the objectives of the academy** (13:20-13:25): An open conversation to ensure a shared understanding of the academy's goals among all participants.
- **Interactive session on academy components** (13:25-14:20): A dedicated ideation phase, encouraging active participation and collaboration among attendees.
- **Break** (14:20-14:30)
- **Identification of hurdles and drivers for success** (14:30-15:15): Another ideation session focused on pinpointing obstacles and enablers for the project's success.
- **Final wrap-up** (15:15-15:30): A concluding presentation to summarise key takeaways and wrap up the workshop.

In the section dedicated to **'Cases: Inspiration from academies'**, an exploration of ecosystem thinking in learning technologies was presented to provide context for the participants. To facilitate this discussion, the framework used was the Bersin learning technology framework, which categorises learning technologies into three distinctive focuses: (1) experience, (2) data and (3) content. The presentation aimed to deliver a nuanced understanding of the interplay between different elements within the learning technology ecosystem. Within this context, eight impactful cases were presented. These cases served as tangible illustrations of how learning technologies have been instrumental in optimising operational efficiency across a diverse range of industries. The cases spanned both private and public sectors, offering a well-rounded perspective on the applicability of learning technologies across different organisational landscapes. Furthermore, they included examples from organisations operating in the field of law enforcement, providing a unique and critical lens into the specific challenges and opportunities within this sector. The technologies demonstrated a wide spectrum of applications, ranging from learning administration and collaboration to user experience. By presenting these diverse case studies, the aim was to equip participants with a comprehensive understanding of the versatility and adaptability of learning technologies. The intention is to prompt thoughtful discussions and reflections on how similar principles and technologies could be integrated into the specific context of Frontex. The following cases were presented and discussed:

- **Telecommunication organisation:** The organisation's learning needs prompted it to implement a comprehensive Learning Management System (LMS). This system was designed to cater to a wide range of requirements spanning across continents, aligning with business goals and accommodating individual learning preferences. The solution incorporated various learning formats, including online, blended and in-person approaches. Additionally, the organisation introduced "Create-your-own-journey" programs, providing a personalised experience for soft skills learning journeys, such as a leadership programme. The LMS also served as a portal connecting to other academies and external content libraries, facilitating a plug-and-play approach. Automated administrative processes and the incorporation of gamification elements increased the solution's effectiveness: learners can request external trainings from their Learning & Development team, and managers that can approve them in the system. The LMS works as a one-stop shop for employees to access an entire learning ecosystem.
- **Fin-learn:** The FinLearn academy is a resource for worldwide organisations seeking to understand the evolving role of finance in a digital landscape. This initiative aims to provide practical insights through interactive and immersive learning experiences. The academy caters to multiple companies and ensures secure access through five types of connections, including three Single Sign-On (SSO) options. The platform boasts an extensive analytics dashboard, offering insights into user engagement and performance. With a focus on adaptability, FinLearn supports mobile learning through micro-learning modules. To enhance user participation and motivation, the academy incorporates gamification elements, making the learning experience not only informative but also engaging and interactive.
- **Transport organisation:** The organisation faced challenges with its learning and development infrastructure, using around 10 scattered virtual learning environments. This decentralised approach led to

difficulties in content management and tracking users' behaviours and compliance across various platforms. The scope of training extended beyond internal staff to encompass approximately 800 partner agencies and sponsor organisations within the industry and supply chain. To address these issues, the organisation opted for a centralised LMS. The new system focused on tracking users and data, integrating e-commerce functionalities for commercialised training, third-party libraries and ERP systems through plug-and-play capabilities supporting multi-tenancy and organisational layers. Additionally, the implementation could support the integration of additional pilot projects like digital twins and 3D elements for technical training. The goal was to mitigate access challenges to physical assets and training centres, enabling flexibility for home, remote work and self-training. The solution embraced a user-friendly approach, allowing multiple assets to be managed within the same application, emphasising a plug-and-play architecture for seamless integration.

- **Consumer organisation**: The organisation has earned a commendable reputation for producing high-quality bicycle parts; a key audience critical for maintaining this reputation is the independent mechanics who work with their products. To address their training needs and provide certification, the organisation has established a digital academy. The platform offers a comprehensive system for reaching out to mechanics, providing them with training opportunities. The digital academy boasts an impressive array of offerings, with over 160 programmes and 250 courses available in 13 European languages. The administrative processes are decentralised, streamlining operations, and the system incorporates elaborate automated notifications. Additionally, the implementation of such a technology-enabled academy has facilitated effective quality follow-up in the training and certification processes. Today, more than 5,000 open badges have been earned on the platform. This strategic initiative not only ensures that the bike mechanics stay well-versed with the organisation's products but also highlights a commitment to continuous improvement and excellence within the community of partners.
- **Interpol**: The Interpol Metaverse is a platform for immersive learning and development initiatives. The primary goal is to deliver comprehensive training to employees and associated law enforcement organisations. This immersive approach extends to replicating real-world scenarios, including crime scenes and border control situations, fostering a dynamic and experiential learning environment. Moreover, the initiative is designed to facilitate the delivery of Interpol's services while promoting international cooperation, with a specific emphasis on addressing the transformative requirements arising from emerging digital threats and the application of law enforcement in digital realms. Additionally, the initiative emphasises the recovery of evidence from the Metaverse infrastructure, recognising the evolving nature of crime and the critical role of law enforcement in navigating the complexities of virtual spaces.
- **Telecommunication organisation:** The organisation relied heavily on SharePoint for training as its staff expanded. However, with the growing number of files on SharePoint, the organisation sought ways to streamline and enhance the learning process. This led to the adoption of a Learning Experience Platform (LXP) for onboarding multiple teams, centralising resources and providing training for various stakeholders. AI-driven learning content recommendations, social learning features and the implementation of more than 30 learning journeys were integrated to cater to diverse learning needs. The system was also designed to integrate seamlessly with SharePoint and provide Single Sign-On (SSO) access, ensuring coherence with existing systems and increasing accessibility. This strategic approach reflects the organisation's commitment to optimising learning processes and creating an efficient learning environment for its expanding workforce using a technology-enabled academy.
- **Proof-of-concept virtual reality connected to the digital academy:** The proof-of-concept virtual reality learning platform is a project dedicated to advancing immersive teamwork training through digital platform access. Beyond developing virtual reality learning experiences across diverse environments, integrating gamification, personalised scenarios and collaboration, virtual reality training is accessible directly from the centralised digital academy. The platform retrieves all results and information of the virtual reality training and facilitates skill development through consolidated views and dashboards. The immersive training involves multiple 3D environments accessible through goggles, leveraging Unity as a mid-layer for seamless integration. The platform is accessible for up to 10 co-workers to train together in VR. The Plug and Play feature enhance user-friendly interaction, making the technology-enabled academy a one-stop shop to centralise learners' information and progress on various types of training, virtual reality and beyond. Analytics embedded within the platform provides valuable insights, contributing to a data-driven approach to enable managers and Learning & Development teams to act upon them.
- **Deloitte Major Programmes:** The objective of these programmes is to bring together professionals from diverse divisions to spearhead ambitious transformations addressing the world's most pressing challenges. The focus is on upskilling experts and enhancing their collaboration abilities. The academy's curriculum centres on developing new skills such as problem-solving, adopting a complex-system analysis approach and acquiring digital and AI skills. The learning journey is designed as a blended approach, incorporating reflection journals and expert testimonials. An activation

programme with action points for 30 days, incorporating gamification elements, ensures practical application of newly acquired skills. Learning Analytics, using platform data, user learning data and HR data, plays a crucial role in personalising the learning experience. Looking ahead, Phase 2 involves the implementation of AI in proof-of-concept scenarios as an additional layer, further enriching the learning environment.

In the section of the applied questionnaire dedicated to the **'objectives of the academy'**, participants were asked 'What is the objective of a technology-enabled academy?' This query served as a catalyst for participants to articulate their individual visions and aspirations. Leveraging a plenary open discussion format, each participant's response became a valuable piece of a collaborative puzzle. The open format not only allowed for diverse perspectives to emerge but also facilitated a dynamic exchange where answers could be compared, linked and clarified in real-time.

In the section focused on the **'academy components'**, a systematic approach was employed to foster collaboration and ensure clarity. The facilitation involved the use of both a virtual whiteboard and flipchart, both of which adhered to a consistent structure, making the process more coherent. First, participants individually brainstormed on key components within three distinct sections (i.e. Learning, Collaboration and Administration) on the virtual whiteboard or flipchart. This collaborative ideation process was followed by a sharing session, where participants presented their ideas. The facilitators guided brief discussions after each session, providing a platform for clarification and the expansion of presented components. This interactive exchange fostered a deeper understanding of the diverse perspectives and insights brought forth by the participants. A final summary aimed at distilling the wealth of ideas into a cohesive and meaningful framework. Participants engaged in discussions to identify the most crucial elements within each category and establish clear connections to the overarching objectives of the tech-enabled academy.

The section dedicated to the **hurdles and drivers for success** followed a structured and collaborative process. During this brainstorming phase, participants actively engaged in identifying potential obstacles or challenges that could impede the establishment of a technology-enabled academy and noted those on the virtual whiteboard or flipchart. Simultaneously, they explored factors that were deemed crucial for the academy's success. This initial brainstorming session allowed for a diverse range of perspectives to be captured. The subsequent plenary group discussion delved deeper into the identified hurdles and key drivers, merging the insights from participants, who shared their experiences and insights in a dynamic exchange of ideas. This discussion aimed to provide a comprehensive understanding of the challenges and opportunities associated with the envisioned academy. Finally, the workshop concluded with a summary of the key hurdles and drivers. This approach ensured a holistic exploration of potential challenges and success factors, allowing for a nuanced understanding and strategic planning to address these considerations in the development of the technology-enabled academy.

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